

Essays in Financial Economics

by
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Abstract

This thesis comprises three chapters on climate and international finance.

Recent pressure on publicly traded (public) firms to divest high greenhouse gas-emitting assets has raised concerns that these assets are flowing to more opaque, privately held (private) firms that may be operating them in more emissions-intensive ways and that sellers are being rewarded for such sales by increased valuations. Whether this is likely to be an important concern depends on the climate and valuation consequences of such asset transfers. These issues are explored in the first two chapters of the thesis.

The first chapter uses data from fossil fuel power plant operations in the United States and employs a difference-in-difference design to estimate the effects of sales on plant emissions outcomes 1998-2022. I find that eighteen months after sale, changes in power plant unit emissions were statistically indistinguishable at the 5% level from zero vis-a-vis comparable plant units that shared technological specifications and were in the same regional electricity area; this was true regardless of whether the buyer was a public or private firm. Then, using data on fossil fuel power plant sale announcements by public firms, I employ an event study methodology to estimate the effect of public-to-private sale announcements on sellers' market values. I find that, on average, the announcement of a sale to a private firm led to cumulative abnormal returns to the seller's stock; however, this average was not statistically different from the average return when the announcement was to a public firm.

The second chapter further investigates from a theoretical perspective. I present a general equilibrium model that predicts what will happen to asset ownership and emissions when public, but not private, firms experience a positive shock to their cost of emitting when there is trade in assets. I find three qualitatively distinct equilibria, one of which is a "greenwashing equilibrium." The public firm expresses the shock entirely through ownership decisions by selling to private firms and assets emit more than they would have if trade were suppressed. There are also equilibria with no trade and divestments of assets to public firms.

The third chapter studies exchange rate determination. I test a set of assumptions that imply the return parity of long-run, real bonds denominated in different currency numeraire. The joint hypothesis is rejected in our post-2009 sample of developing and developed market currencies; however, I document a strong relationship between changes in the log of bilateral, real exchange rate and real holding period bond returns in the direction of parity, contributing to the Meese-Rogoff puzzle on exchange rate determination.

Thesis Supervisor: Deborah J. Lucas

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Dedication

To Helena Jung and Deborah Lucas

And to my heavenly mother, Mary, the Mother of God

Ad Jesum per Mariam

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Chapter 1

Climate and Financial Effects of Fossil Fuel Power Plant Sales in the United States

1.1 Introduction

“If a corporation sells the dirtiest stuff to some private enterprise somewhere in the world and that private enterprise is doing exactly or even worse offenses to the environment, how do you define that? The company looks better...but the world is probably worse off.”

- Larry Fink, CEO of BlackRock

There has been accelerating interest in corporate contributions to climate change. Recent pressures on publicly traded firms to decarbonize and their decisions to sell dirty assets to privately held firms have sparked concerns that they are engaging in greenwashing - taking actions that appear superficially green but in reality have zero, or negative climate effects - and are being rewarded for doing so.

The argument for the hypothesis that these sales have zero, or negative climate effects is that such sales reshuffle asset ownership to private firms who may have weaker incentives to operate assets in environmentally responsible ways. That could be because private firms a.) are relative more opaque (Bernstein, 2022), b.) subject to fewer regulatory reporting requirements¹, and/or c.) because they're immune to investor activism through the acquisition of voting shares (Gozlugol and Ringe, 2022; Deshmukh, 2023). Thus, when asset ownership shifts to those who may be less concerned about their emissions impacts,

¹For example, as noted in Duchin et al. (2023), “in 2022, the SEC enforced ESG disclosure requirements for investment funds and other investment companies, whose portfolios largely comprise publicly traded firms. In contrast, no regulations impose such disclosure requirements on privately held firms.”

the argument is that assets will emit just as they would have before or more. These concerns gained renewed attention after a 2022 Environmental Defense Fund report which documented hundreds of cases in which assets were sold by operators with public commitments to cut methane emissions to those with no such obligations. It also reported alarming case studies of public to private transfers, one of which was of a Nigerian oil gas field after which there was a near four-fold increase in average methane flaring (Malek et al., 2022). If these sales have zero, or negative climate effects, this would suggest that ESG investment strategies that target publicly traded firms and stress divestment may be ineffective or even counterproductive.

One argument for the hypothesis that the stock market financially rewards such sales is that there are gains to trade when assets move from public to private ownership because private firms are more inclined to cut costs associated with maintaining or improving the environmental performance of assets, raising asset values under private firm ownership compared relative to those under public ownership. In a model in which gains to trade sometimes accrue to the seller, on average, there should exist a premium and thus incentive for selling to private firms over public firms.

While plausible, there are also arguments that sign the climate and financial effects in the other direction.

One the climate effects, one might expect climate effects are positive if one thinks public firms are myopic (Stein, 1989; Graham et al., 2005) and more so than private firms. In so, this would manifest in higher discounting of future environmental liabilities and could lead assets to be operated in more emissions intensive ways. Another is that private firms, because ownership is relatively more concentrated, have better managerial control (Bolton and von Thadden, 1998), leading them to excel at identifying and executing positive net present value projects that improve the productive capacities of firms. In a model in which emissions scale with inputs, this would reduce emissions.²

On the financial effects, one might expect a decrease in seller valuations if public firms are willing to sell assets at a discount and sacrifice shareholder value to quickly reduce their emissions associated with the firm, and assuage moral outrage over corporate contributions to climate change. In the 1980s, Wright and Ferris (1997) document negative cumulative abnormal returns to the announcement of sales of South African business units in response to

²For example, consider assets that combust fossil fuels, an activity that accounts for 73% of total anthropogenic greenhouse gas emissions. A project that increases how efficiently the heat generated from fuel combustion is converted into useful energy would directly reduce fuel costs, and emissions since emissions in this context is primarily determined by the quantity and carbon content of the fuel burned. Because of costs, not all such projects will be positive net present value projects, but industry periodicals have identified projects that are, and cover case studies of implementing such projects (Nowling, 2015; Korellis, 2014).

outrage over apartheid. Furthermore, one might think that sales to private firms particularly drive down valuations if private firms excel at bargaining [Liu \(2021\)](#).

Whether shifting asset ownership from public to private firms is “innocuous”—that is, have near zero impacts on assets’ emissions behavior—or “perverse”—that is, lead assets to become sharply more emitting, and rewards public sellers through increased valuations are empirical questions that have not yet been explored.

This paper investigates these questions in context of the fossil fuel power plant sector and with respect to short-horizon emissions outcomes. What happens in this sector is important because of its

1. Aggregate importance to US emissions: it is the second largest source of U.S. greenhouse gas emissions, accounting for 25% of the U.S. total;
2. Similarity to industrial sectors that generate 20% of additional US annual emissions by combusting fossil fuels to generate heat and/or mechanical energy study.

To answer questions about divestment effects on emissions, I combine power plant input, output, and emissions data from the Environmental Protection Agency (EPA), plant characteristics data from the Energy Information Administration (EIA), and ownership data from S&P Capital IQ Pro to construct a virtually complete characterization of ownership, emissions, and production from this sector from 2000-2023 at the weekly frequency. Two key advantages of this dataset are

1. The coverage, accuracy, granularity of the emissions and production data. Data cover 96% of US fossil fuel generation. Reporting is federally audited, and the ability of firms to misreport is sharply bounded by the physics and chemistry of fossil fuel combustion (the process that generates emissions), which tightly links emissions with observable input and output quantities. Furthermore, the data are reported at the level of the power plant unit, which is the level at which operators make decisions that determine the emissions intensity of production;
2. Significant volume of assets sold by publicly traded firms. I focus only on those sales that involved a transfer of a majority stake, in which the seller was a domestic, publicly traded company, and didn’t involve a repeat sales, allowing me to study 80 deals, 58% to public firms and 42% to private firms that transferred 548 power plant units. These sales transferred, in the years in which they were completed, a cumulative 420 million short tons of carbon dioxide, or 23% of annual 2022 sectoral emissions.

To estimate climate effects, I use a difference in difference design to estimate sale effects on four, primary operational decisions that determine power plant emissions (e.g., what kinds

of fuels to burn) eighteen months after sale, and by firm type: whether or not the firm was a public or private firm. I find that power plant units sold to private firms are not operated in more emissions intensive ways post-sale. In fact, they become less emissions intensive, but the effects are small and statistically insignificant at the 5% level. Thus, the narrative that private firms buy up dirty assets and ramp up emissions did not hold in this asset class and period. This suggests that “innocuous greenwashing” may be a concern in this setting, but not “perverse greenwashing”. I also find that these effects are statistically indistinguishable from the effects of sales to public firms. Thus, the narrative that private firms always operate dirty assets in more emissions intensive ways compared to public firms did not hold in this setting.

When breaking out divestment effects by the fuel type burned, I find statistically significant post-divestment declines in the emissions intensity of coal-fired (but not gas-fired) units of 4.2% when units were sold to private firms and 1.4% when units were sold to public firms, with the caveat that private firm effects were estimated on a small sample of seven power plants and fifteen units. Approximately 88-100% of these declines appear to be driven by improvements in productive efficiencies: how efficiently units convert the heat content of fuels to electricity. In this sector, as well as for industrial assets that produce an additional 20% of US annual emissions through fossil fuel combustion, productive efficiency is directly linked to economic and emissions efficiency: when less fuel is used to generate a MWh of electricity, firms save on fuel costs, which are 80% of plant marginal operating costs, and emit less, since emissions are primarily determined by the carbon content and quantity of fuel burned. This suggests that while these declines could be driven by different costs of emitting between buyer and seller, they could also be driven by buyers’ comparative skill in operating the asset, or implementing positive net present value projects that enhance assets’ productive efficiencies. However, while the magnitude of the effect is economically meaningful at the asset level, its implications for aggregate climate impacts are modest. Even if private firms were to acquire all coal units and make an additional reduction of emissions intensities by 4.2%, at the 2022 level of emissions by coal-fired units, the implied impact on aggregate emissions would be 2.4% of sectoral emissions or 0.6% of US emissions.

To estimate financial effects, I use an event study methodology to estimate the effect of public to private sale announcements on sellers’ market values. I find that on average, the stock market rewarded sales to private firms, but not above and beyond those to public firms in a statistically significant way in my sample.

In sum, this paper suggests that ESG investor strategies that stress divestment by public firms as a way to reduce aggregate emissions may have near zero impacts. A caveat is that this paper only focused on changes in the short-horizon emissions behavior of operating

units. Acquisitions may affect long-term decisions such as shut-down decisions or changes to the technology, which are important determinants of generators' long-term emissions, but low-frequency events not the focus of this paper and also explored by [Andonov and Rauh \(2023\)](#). Furthermore, this paper did not evaluate whether or not sales were motivated by a greenwashing motive.

The paper is organized as follows; Section [1.2](#) presents the literature review; Section [1.3](#) covers climate implications; Section [1.4](#) covers valuation implications. Section [1.5](#) concludes.

1.2 Literature Review

This paper is related to several strands of literature.

The first strand is the empirical literature on the environmental and financial effects of divestitures of pollutive assets. [Duchin et al. \(2023\)](#) find evidence consistent with greenwashing in assets that release chemical pollutants: following the divestiture of pollutive assets, changes in pollution are statistically indistinguishable from zero, but sellers earn positive announcement returns that are increasing in the pollutive level and intensity of divested assets. In the opposite direction, [Jacqz \(2021\)](#), using the same dataset, finds that toxic air releases, measured both in pounds of toxic releases and in the volume of releases adjusted for toxicity, fall dramatically in the ten years following divestment.³ This paper complements these studies by quantifying the real and financial impacts of divestitures on a kind of pollution not covered by their dataset, but with climate relevance: carbon dioxide, which accounted for 79% of all U.S. greenhouse gas emissions from human activities in 2021. In contemporaneous work, [Bai and Wu \(2023\)](#) examine the effects of power plant acquisitions on the emissions intensity of operations when the acquirer is a private equity firm and find that acquired plants operate with a lower heat rate, which translates into reduced fuel consumption and reduced output-scaled emissions. This paper's contribution is to focus on the absolute and relative effects of divesting to a private firm, where relative is defined vis-à-vis the effects of divesting to a public firm. I also examine other margins by which new operators can affect the environmental operation of power plants such as by changing the likelihood of being on in a given week, generation levels, and the fuel mix of generation, and also present a model. [Andonov and Rauh \(2023\)](#) study the effects of divestitures by domestically listed public companies to foreign publicly-listed companies, private equity, and institutional investors on operating decisions like capacity factor decisions and productive efficiency decisions. While not obvious, these decisions are primary determinants of assets' emissions. This paper complements this study by focusing on the short-horizon effects of divestitures on other decisions that determine plant emissions, such choice the turn on and the fuel mix to burn, and by studying these effects by whether or not the buyer was publicly traded or not.

The second strand is the empirical literature on how firm characteristics shape polluting behavior. [Akey and Appel \(2021\)](#) study the effects of limited liability on environmental outcomes and find that subsidiaries with parents that face strong limited liability protection are more likely to emit ground pollution. [Bellon \(2021\)](#) studies the effects of private-equity

³[Jacqz \(2021\)](#) focuses only on toxic air releases; [Duchin et al. \(2023\)](#) covers toxic releases without distinguishing by release type (e.g. air release, water release, land release, release to an off-site treatment or disposal facility).

acquisitions on the environmental operations of an asset in oil and gas well transactions. He finds that acquisitions are associated with a 70% reduction in the use of toxic chemicals and a 50% reduction in satellite-based measures of carbon dioxide emissions from gas flaring vis-à-vis nearby, observationally equivalent wells, but that private-equity backed firms increase pollution in locations and periods where environmental liability risk is low. This paper complements these studies by studying whether private firms operate acquired assets differently than publicly traded firms. [Andonov and Raub \(2023\)](#) study differences in plant retirement decisions between public and private firms. They find that public firms have the highest probability of decommissioning a power plant conditional on plant age and capacity, followed by private firms and that private firms operate plants at slightly lower capacity factors, particularly natural gas power plants. They find that domestic corporations have a 0.28 percentage points higher probability to retire coal and petroleum power plants, and 0.19 percentage points lower probability to retire natural gas power plants compared to other owners after controlling for plant capacity, plant age, and fuel-state-time fixed effects.

This third strand of literature is the theoretical and empirical literature on emissions leakage. When pressure is applied to only a subset of facilities contributing to the pollution problem, it has been long recognized that this pressure may be muted or wholly ineffective if it [prompts a shift in production from pressured to unpressured producers](#) (van Benthem et al., [2022](#); [Fowlie, 2009](#)). Existing studies, however, keep ownership fixed. This model focuses on emissions leakages mediated by changes in ownership.

The fourth strand is on the effect of divestitures on firm value. [Hite et al. \(1987\)](#) investigate the valuation consequences of announcements to sell part or all of a corporations' assets and find that successful sellers and buyers reap statistically significant abnormal returns. [Wright and Ferris \(1997\)](#) report significant and negative excess returns to the shares of companies announcing divestments of South African operations during a period in which there was a buildup of private and public pressure to divest US business interests in that country due to moral outrage over the treatment of black South Africans by the minority white South African government. Their results support the premise that noneconomic pressures may influence managerial strategies rather than value-enhancement goals that lead to a decline in firm values. This contributes by focusing on the divestment of fossil-fuel power plants.

The fifth strand is the empirical literature on how firm valuations and returns are related to the ownership of emitting assets. [Bolton and Kacperczyk \(2021\)](#) find that investors of publicly traded firms care about carbon risk: stocks of firms with higher total carbon dioxide emissions (and changes in emissions) earn higher excess returns, controlling for size, book-to-market, and other return predictors. The authors find that this premium cannot be

explained by differences in unexpected profitability or other known risk factors. Berk and van Binsbergen (2022) find that given current levels of socially conscious capital, the fraction of targeted firms in the economy and the return correlation between targeted firms and the rest of the stock market, the likely impact of divestiture strategies on the cost of capital is very small.

1.3 Climate Implications

In this section, I describe how I estimate the effect of divestitures on units' decisions to produce, how much to produce, and the emissions intensity at which to produce.

1.3.1 Data

This study utilizes a detailed panel dataset of weekly operations and emissions of, and ownership of the US fossil-fuel power plant sector from 2000 to 2022.

Data on plant operations and carbon dioxide emissions, characteristics, and regulatory regimes are from the EPA's Clean Air Markets Division database. The database covers 96% of total fossil fuel generation in the United States and reports the following variables: gross generation, defined as the total amount of electric energy; heat input, defined as the energy supplied to produce heat; carbon dioxide emissions; the primary and secondary fuels burned; unit type (i.e. whether or not the unit was a part of a combined cycle system); and participation in an emission trading program or other air quality program at the unit level. A power plant is typically composed of many units. I conduct my analysis at the unit level because emissions are primarily determined by the quantity and carbon content of the fuel burned, and the unit is the level at which the choice of the type and quantity of fossil fuel combusted is determined. Though fossil fuel-burning power plants emit greenhouse gases besides carbon dioxide, such as methane and nitrogen dioxide, only carbon dioxide is reported in the EPA's emissions dataset. However, because carbon dioxide is the primary greenhouse gas emitted, this is a small concern. In fact, where they can be compared, plant carbon dioxide emissions and total greenhouse gas emissions are close and have a correlation coefficient of 0.98 (Shiver and Forster, 2020). Thus, in this paper, I will use the terms carbon dioxide emissions and emissions interchangeably.

From this data, following Demirer and Karaduman (2022), I estimated units' nameplate capacities, defined as the maximum rated electricity output at full power, by using the 99th percentile the generation as the capacity for the unit every year. Data are based on federally mandated emissions reporting. Advantages of this dataset are a) the strict standards for and auditing of emissions estimations and b) the chemistry and physics of fossil fuel combustion, which imply a tight connection between emissions and observable inputs and outputs, and

sharply bounds the ability of plants to misreport.⁴⁵ The decision to burn fuels and thus, emit is unit specific (a unit typically consists of a boiler connected to a generator), which makes the unit the relevant level at which to conduct the analysis.

I then matched this data to the following plant characteristics data from the EIA’s Form 860 “Annual Electric Generator Report”: the plant’s longitude, latitude, carbon capture technology, and the North American Electric Reliability Corporation (NERC) region. The NERC region gives a boundary of the regional electricity market, which tends to but does not always fall along state lines. Over the sample, there are instances of NERC regions consolidating; in such instances, I used the consolidated region for my whole sample.

Ownership data are from Standard and Poor’s Capital IQ Pro. To my knowledge, this is the most complete and accurate dataset of power plant transactions in the US, where completeness is due to regulation of plant transactions (Demirer and Karaduman (2022)). Deals data include seller, buyer, and target identities and where available, stock tickers, the announcement date, the completion date, the deal type (e.g., involved a majority stake; percent owned by the buyer after the deal; percent acquired through the deal), deal summary, and the names and, for a subset, the EIA identifier known as the ORIS code of the power plants involved. For this dataset, I recognize a divestment by a publicly traded firm if (a.) the seller was listed on a major US trading exchange (i.e. NASDAQ, NYSE), (b.) the divestment involved an ownership transfer that changed the identity of the majority financial owner of the plant, defined as the owner of more than 50% of the asset’s financial interest, and (c.) the buyer was a for-profit entity. This excludes municipalities, co-ops, federal agencies, etc. I only consider transactions that changed the identity of the majority financial owner under the assumption that the primary controller of plant operations is the owner with the majority financial share. I also only include the first instance of a plant acquired in my sample and impose the restriction that the plant could not have been divested within the event window. Additionally, I exclude divested, oil-fired units because of insufficient deals.

⁴The federal Acid Rain Program required all units over 25 megawatts and new units under 25 megawatts that use fuel with a sulfur content greater than 0.05 percent by weight to measure and report emissions. For carbon dioxide, all units can either use a mass balance estimation, carbon dioxide CEMS, or oxygen CEMS. All CEMS used for compliance are certified by the EPA, and undergo daily calibration tests, and annual reference test audits. It is typically composed of a Data Acquisition and Handling System which collects and stores data on the emissions intensity. For compliance, it must be in continuous operation even if no process is on and for valid measurement, must record at least one reading every 15 minutes for 3 out of 4 quarters and the readings are averaged quarterly. See <https://web.archive.org/web/20090211082920/http://epa.gov/airmarkets/emissions/continuous-factsheet.html>.

⁵Carbon dioxide emissions are primarily determined by the carbon and heat content of the fuel burned, both of which are quantities that can be measured well.

Power plants are matched to the operations and emissions data using, where available, the plant’s ORIS code and with fuzzy matching on names, otherwise. I also used the deal summary and google search to algorithmically and manually identify and correct cases in which a majority transfer was incorrectly recorded or when a majority transfer was not identified though it had occurred.

Table [1.1](#) presents summary statistics of the divested units analyzed in this paper and the deals they were a part of. This study covers 80 deals that involved the divestment of 548 units from 215 power plants. In the year they were acquired, these units generated electricity that equaled 20% of annual 2022 fossil-fuel power plant generation and emitted 23% of annual 2022 fossil-fuel power plant emissions. Of these, 148 were coal-fired units from 58 plants.^{[6](#)} Though only 27% of divested units, these coal-fired units emitted, in the years in which they were acquired, the equivalent of 19% of annual 2022 fossil fuel power plant emissions, reflecting both the relative emissions intensity of coal generation, and the fact that some of the divested units were very large.^{[7](#)}

⁶For context, there are currently 217 operational coal power plants in the US.

⁷Coal units release 1.8x more carbon dioxide per MBtu, and 2.3x more carbon dioxide per Mwh vis-à-vis gas units.

Table 1.1: Summary Statistics

Panel A: Counts			
	Buyer Type		Combined
	Public	Private	
# of Units	314	234	548
Coal-fired	133	15	148
Gas-fired	181	219	400
# of Plants	124	91	215
Coal-fired	51	7	58
Gas-fired	74	84	158
# of Deals	47	33	80
# of Unique Sellers	32	24	44 ¹
# of Unique Buyers	34	23	57 ²

Panel B: Pre-Sale Characteristics			
	Buyer Type		Combined
	Public	Private	
Generation (bn. MWh, annual)	0.33	0.19	0.52
Coal-fired	0.25	0.08	0.33
Gas-fired	0.07	0.11	0.18
Emissions (bn. short tons, annual)	0.29	0.13	0.42
Coal-fired	0.25	0.08	0.33
Gas-fired	0.04	0.06	0.10
Emissions Intensity ($\frac{s.tons}{MWh}$)	0.80	0.62	0.72
Nameplate Capacity (MW)	69,993	48,083	118,076
Coal-fired	44,209	11,457	55,666
Gas-fired	25,784	36,626	62,410

Footnotes: (1) and (2) may not be sum of disaggregated numbers because of overlapping sets of entities.

Table [1.2](#) presents a tabulation of the industries of the seller and buyer. There was a total of 80 deals that spanned 2001-2019. Deals were evenly spread out across time. Public to public sales had a median deal announcement year of 2008; Public to private sales had a median deal announcement year of 2009. The majority of divestments to public buyers, around 89%, were by power firms to other power firms. The majority of divestments to private buyers, around 79%, were by power firms to financial firms, which includes private equity (PE), family offices, etc.

Table 1.2: Seller Buyer Matrix

(a) Deals with Public Buyer

Seller/Buyer	Finance	Power	Total
Finance	0 (0%)	1 (100%)	1 (100%)
Power	4 (9%)	42 (91%)	46 (100%)
	4	43	47

(b) Deals with Private Buyer

Seller/Buyer	Finance - Non PE	Finance - PE	Power	Total
Finance - Non PE	1 (100%)	0 (00%)	0 (00%)	1 (100%)
Power	18 (49%)	9 (31%)	6 (20%)	33 (100%)
	19	9	6	34

1.3.2 Empirical Strategy

To estimate the effect of divestitures on units' decisions about (a.) whether or not to engage in emissions generating production, and conditional on producing, (b.) how much to produce, and (c.) at what emissions intensity to produce, I estimate the following DD model using ordinary least squares:

$$\begin{aligned}
y_{it} = & \beta_1(Post \times Public\ Buyer)_{it} \\
& + \beta_2(Post \times Private\ Buyer)_{it} \\
& + X_{it} + \alpha_i + \eta_{it},
\end{aligned} \tag{1.1}$$

where outcome y_{it} is for each unit i and week t includes: weekly “starts”, defined as a dummy variable equal to one if gross generation was positive in week t ,⁸ the capacity factor, defined as the ratio of gross generation to nameplate capacity conditional on gross generation being positive; and the natural log of the emission rate, defined as the ratio of carbon dioxide

⁸This is an approximation to starts since some units will cycle on and off at a higher frequency.

emissions to gross generation conditional on gross generation being positive:

$$y_{it} = \begin{bmatrix} Starts \\ Capacity Factor \\ \ln(Emissions Intensity) \end{bmatrix}_{it} = \begin{bmatrix} Gross Generation > 0 \\ \frac{Gross Generation}{Nameplate Capacity} |_{Gross Generation > 0} \\ \ln(\frac{CO_2}{Gross Generation}) |_{Gross Generation > 0} \end{bmatrix}_{it} . \quad (1.2)$$

The dummy variable $(Post \times Public Buyer)_{it}$ is equal to one if unit i was divested by a public firm to a public firm, and t is in the post-divestment period and the dummy variable $(Post \times Private Buyer)_{it}$ is equal to one if unit i was divested by a public firm to a private firm, and t is in the post-divestment period. The post-divestment period is defined as the 18 months following divestment completion. I chose this horizon to balance the desire to capture short-term divestment effects, while respecting the fact that these effects may take more than a few months to materialize. Previous work has found divestment effects in the 8-month to 2-year horizon (Cicala, 2015; Demirer and Karaduman, 2022). Observations of treated units are dropped past this horizon so that the comparison for treated units is always with never-acquired or not yet acquired units. Time-varying trends or slow-moving processes within a state and NERC region are captured by X_{it} which is the interacted set of the following variables: state, NERC region, a combined cycle indicator equal to one if the unit was part of a combined cycle system, the types of fuel the unit can burn, the week, and year. Time-invariant unit fixed effects are denoted by α_i . Standard errors are clustered at the plant level. To ensure I am only capturing divested units for which I can obtain a DD treatment effect, I impose the condition that divested units included in the estimation must have observations in both the pre and post-divestment period and have had at least one control in both periods (i.e. another never-acquired or not-yet acquired unit that could be used to estimate X_{it}). The median number of controls is 13; the mean, 20.⁹

This strategy compares changes in the behavior of treated units with controls, which are never-treated or not yet treated units in the same state and NERC region with the ability to burn the same kind of fuels, and, if a unit that can burn natural gas, of the same system type (i.e. combined cycle or not combined cycle). I include the state, NERC region in the interaction set to control for regional trends in demand that can arise from local economic conditions and population growth, as well as to control for regional trends in generation supply that can arise from state regulation of carbon dioxide and non-carbon dioxide pollutants released by power plants, such as mercury, sulfur dioxide, and nitrous oxides. I include the fuel set in the interaction set for two reasons. The first reason is that the set of fuels the unit can burn is a technological constraint that determines the means by

⁹This led to the loss of 17% of units.

which owners can change the emissions intensity of generation. For example, an owner can reduce the emissions intensity of generation by burning a cleaner fuel, but only if burning multiple fuels is technologically feasible. The second reason is that without differences in private costs of emitting, the ranking of fuel prices for the major fuel categories: coal, oil, and gas would be the primary determinant of which fuels are burned to generate electricity and by extension, unit starts and generation. Furthermore, the ranks are a slow moving process.¹⁰ Thus, when estimating the divestment effect on units' emissions intensities, I want treated units to be compared with control units with the same ability to change their emissions intensities and subject to the same factors that would influence their propensities to change their emissions intensities like relative fuel prices. I control for the system type because combined cycle systems can be 50-70% more efficient at converting fuel energy to electricity than non-combined cycle systems, which significantly impacts marginal operating costs and thus start probabilities and generation.¹¹

One way to see whether the controls are good is to compare differences in means of the control and treated groups. Differences in the average annual generation, annual emissions, and emissions intensities of the two groups were statistically indistinguishable from zero. However, treated units were, on average, 178W and slightly smaller than the control group by 1MW; they were also likely to be a part of plants that were younger by nine years. I assume that the unit fixed effects absorb these differences.

A concern may be that divestment differentially affects coal and gas units and that aggregating divestment effects across unit types masks important heterogeneity in the divestment effect. For example, one may argue that new owners were unlikely to have changed coal-unit starts and generation because these units were, for a large fraction of the period covered, non-marginal electricity suppliers. Another argument is that new buyers may have greater opportunities to run coal-fired units in environmentally ruinous or beneficial ways vis-à-vis natural gas units because coal-fired operators have more room to alter emissions since they can switch from burning a dirtier to cleaner coal and vice versa. For example, coke and anthracite coal release 22% and 10% more carbon dioxide emissions per MMBtu,

¹⁰The relative rank of fuel prices that determine the unit's location in the dispatch curve, which is typically ordered from the least to greatest marginal cost unit, determines the order in which the unit is "dispatched" to meet electricity demand. Because the time varying changes in the relative rank of fuel prices is slow-moving, it is important to control for the fuel set in this setting when I have a sizable, but modest number of treated units. Not controlling for these time varying changes in relative rank can impact point estimates, whereas I wouldn't have had to if I could assume (a.) orderings of relative fuel prices were uniformly distributed and (b.) I had many treatment units.

¹¹In a combined cycle system, waste heat from combustion is often converted to steam, which is then used to drive an additional turbine to increase the plant's overall thermal efficiency.

respectively, than bituminous coal.¹²¹³ This is in contrast to gas units, the vast majority of which are made to burn pipeline natural gas, which is gas processed into a homogenous output according to standards fixed by gas transmission pipelines and local distribution companies, making substitution within natural gas less likely. Thus, I re-estimate the model with post-divestment dummy variables interacted with dummy variables of the divested unit's fuel type:

$$\begin{aligned} y_{it} = & \quad \Sigma_f \beta_1 (Post \times Public Buyer \times f Unit)_{it} + \\ & + \Sigma_f \beta_2 (Post \times Private Buyer \times f Unit)_{it} \\ & + X_{it} + \alpha_i + \eta_{it}, \end{aligned} \tag{1.3}$$

where the dummy variable $(Post \times Public Buyer \times f - Unit)_{it}$ is equal to one if unit i is in the post-divestment period at time t , was divested by a public buyer, and of type $f \in \{Coal, Gas\}$ and the dummy $(Post \times Private Buyer \times f - Unit)_{it}$ is equal to one if unit i is in the post-divestment period at time t , was divested by a private buyer, and of type f .

Identifying variation comes from average differences in the outcome variable from treated and never treated units and variation in treatment timing. Since the estimation strategy relies on comparing changes over time, it is important that pre-existing trends are not responsible for the subsequent differences between treatment and control units. A plot of pre-trends in the Appendix provides encouraging evidence that both treatment and control groups were following parallel paths before divestment.

1.3.3 Results

Table 1.3 presents estimated average treatment effects on the treated (ATT) that measure the average short-horizon impacts of divestments on starts, capacity factors, and emissions intensities, as well as the p-value of a Wald test of the null that effects are the same when units are sold to a public firm versus a private firm.

I find that following divestment to a private firm, starts decreased by 3.2 percentage points, or approximately two weeks a year, the capacity factor by 3.7 percentage points, and the log emissions intensity by 0.6 percent. The effects are both modest and statistically insignificant at the 5% level and the difference in divestment effects by buyer type is

¹²https://www.eia.gov/environment/emissions/co2_vol_mass.php

¹³Coal plants can burn blends of coal. Switching from one type of coal to another, entirely, however, can require large modification costs because different types of coal have different propensities to generate different amounts of slag on boiler and furnace surfaces (McCartney, 2006).

statistically insignificant. To get a sense of the magnitudes of estimated effects, back of the envelope calculations suggest a 3.2 percentage point decline in starts will lead to a 0.07% decline in annual emissions if all declines are replaced by zero-emissions clean generation and a 0.05% increase in emissions if all declines were by the relatively cleaner fossil fuel (gas) and replaced by the dirtier fossil fuel (coal). Similarly, a 3.7 percentage point decline in the capacity factor of all units acquired by the private buyer implies a 0.34% decline in annual US emissions if the decline is replaced by zero-emissions clean generation and a 0.26% increase in annual US emissions if declines are by the relatively cleaner fossil fuel and replaced by the dirtier fossil fuel. To put these numbers in context, between 2000 and 2022, one standard deviation of emissions as a percent of 2021 GHG emissions was 4 percentage points. Most of the time series variation in emissions over that time was due to business cycle fluctuations and a rise in the share of gas generation due to the decline in natural gas prices compared to coal prices.

Robustness As a robustness check, I re-estimate divestment effects using a longer ten-year symmetric window around divestment completion, and weighting observations by the nameplate capacity of units. In both specifications, I find that sale effects on starts, capacity factors, and emissions intensities were small and statistically indistinguishable from zero.

By fuel type Breaking effects out by fuel type, I find that no statistically significant divestment effects on coal-fired starts and statistically significant divestment effects on gas-fired starts to private but not public buyers. Table [1.3.2](#) reports. Specifically, starts of gas-fired units divested to private buyers decreased by an average 3.6 percentage points, which implies that these units were less likely to engage in emissions generating production.

Divestment effects on capacity factors at both coal-fired and gas-fired plants were near zero and statistically insignificant, implying that conditional on being on, on average, units did not change their levels of emissions generating production post-divestment relative to comparable units.

Turning to the divestiture effects on emissions intensities, I find that the divestiture of coal-fired units (but not gas-fired units) was associated with statistically significant effects. A caveat is that post-divestment effects when the unit was sold to a private buyer is estimated on fifteen units from seven unique plants. To put these numbers in context, there are currently 217 operational coal power plants. These units, however, were quite large, and in the year before sale completion had annual emissions that together were 4% of 2022 sectoral emissions. In divestitures of coal-fired units, units divested to public and private firms experienced an average decline in emissions intensities of 1.4% and 4.2%, respectively.

The estimates are statistically significant at the five percent level and their magnitudes are broadly in line with estimates of divestment effects on heat rates, a primary component of the emissions intensity (Bushnell and Wolfram, 2005; Demirer and Karaduman, 2022; Bai and Wu, 2023),¹⁴ as well as in the range of heat rate improvements estimated by the EIA at existing coal-fired plants (EIA, 2015). The implied impact on aggregate emissions, however, is modest: approximately 0.45% of sectoral emissions or about 0.11% of total US emissions in 2022. This does not support the extreme version of the greenwashing story in which transfers to private firms lead to dramatic increases in emissions, but it does support the version in which they have near zero emissions impacts.¹⁵ Even if private firms were to acquire all coal units and make an additional reduction of emissions intensities by 4.2%, at the 2022 level of emissions by coal-fired units, the implied impact on aggregate emissions would be 2.4% of sectoral emissions or 0.6% of US emissions.¹⁶ Furthermore, though the point estimate of the divestment effect when the buyer is private is nearly double the effect when the buyer is public, the difference is not statistically significant and I cannot reject the null that public and private firms decrease emissions intensities at coal-fired units by a similar amount.

¹⁴Since the choice to divest a particular asset is deliberate and unlikely random, one may be concerned that estimated effects on the emissions intensity is being driven by selection bias: sellers are divesting assets whose emissions intensities were on a trajectory to improve. This is unlikely to happen naturally as physical assets tend towards depreciation and increased inefficiency. The other possibility is that sellers divested assets with an existing pipeline of operational improvement project expected to come live upon acquisition. I cannot observe this and thus, cannot rule it out. I do, however, check to see if there were changes in the boiler type, which would change the heat rate, following the acquisition and find no change.

¹⁵At the average emissions intensity of coal units, which is $1.01 \frac{CO_2}{MWh}$, the implied effect of divestment to public and private buyers on the level of the emissions intensity at the means is a decline of $0.014 \frac{CO_2}{MWh}$ and $0.042 \frac{CO_2}{MWh}$, respectively. Coal-fired units divested to public and private firms generated approximately 0.29 bn MWh and 0.08 bn MWh of electricity. This implies a decline of $0.014 \times 1.01 \times 0.29 + .042 \times 1.01 \times 0.08 \approx .007$ bn short tons of carbon dioxide. In 2020, fossil-fuel units generated 1.65 billion short tons of carbon dioxide. Thus, the implied decline is about 0.45% of sectoral emissions or about 0.11% of total US emissions in 2022.

¹⁶Coal firing emitted 0.957 short tons of emissions. All fossil fuel plants emitted 1.650 billion short tons. A 4.2% improvement would generate a 2.4% decline in sectoral emissions. Since the sector is about a quarter of total US emissions this translates to a decline of 0.6% of US emissions.

Table 1.3: Divestment Effects on Unit Starts, Capacity Factors, and Emissions Intensities

	All		
	Starts (1)	Capacity Factor (2)	$\ln(\text{Emissions Intensity})$ (3)
<i>Post</i> $\times$ <i>Public Buyer</i>	-0.020 (0.014)	0.008 (0.033)	-0.012 (0.010)
<i>Post</i> $\times$ <i>Private Buyer</i>	-0.032* (0.017)	-0.037 (0.047)	0.006 (0.018)
Sample			
Observations	349,477	265,692	265,692
R-squared	0.39	0.87	0.93
Wald Test P-value			
All	0.54	0.43	0.42

Notes: This table uses OLS regressions to test the effect of divestment on starts, capacity factors, and the log of the emissions intensity by the primary fuel burned by the divested unit prior to divestment. Observations are at the unit level and weekly frequency. Starts is a dummy variable equal to one if gross generation is positive; the capacity factor is defined as the ratio of gross generation to nameplate capacity conditional on gross generation being positive; the log of the emission rate is defined as the ratio of carbon dioxide emissions to gross generation conditional on gross generation being positive. Columns (1)-(3) present the effects of divestment on all units without distinguishing by unit fuel type. The dummy variable (*Post* $\times$ *Public Buyer*) is equal to one if unit i was divested by a public firm to a public firm, and t is in the post-divestment period and the dummy variable (*Post* $\times$ *Private Buyer*) is equal to one if unit i was divested by a public firm to a private firm, and t is in the post-divestment period. Standard errors are clustered at the plant level and reported in parentheses. Fixed effects are unit fixed effects and interactions of the following variables: state, NERC region, a combined cycle indicator equal to one if the boiler was part of a combined cycle system, the fuels the unit is able to burn, week, year. The table reports also the number of relevant observations used to identify the coefficients reported in that column only, and the p-value of a Wald test of the null that the post-divestment dummies for public buyers and private buyers are equivalent. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Table 1.4: Divestment Effects on Unit Starts, Capacity Factors, and Emissions Intensities, By Fuel

	All		
	Starts (1)	Capacity Factor (2)	$\ln(Emissions\ Intensity)$ (3)
$Post \times Public\ Buyer \times Coal\ Unit$	-0.021 (0.020)	0.026 (0.038)	-0.014** (0.007)
$Post \times Private\ Buyer \times Coal\ Unit$	0.022 (0.033)	0.053 (0.070)	-0.042** (0.021)
$Post \times Public\ Buyer \times Gas\ Unit$	-0.02 (0.020)	-0.008 (0.051)	-0.010 (0.017)
$Post \times Private\ Buyer \times Gas\ Unit$	-0.036** (0.017)	-0.044 (0.050)	0.010 (0.019)
Sample			
Observations	349,477	265,692	265,692
R-squared	0.39	0.87	0.93
Wald Test P-value			
Coal	0.25	0.71	0.20
Gas	0.50	0.60	0.48

Notes: This table uses OLS regressions to test the effect of divestment on starts, capacity factors, and the log of the emissions intensity by the primary fuel burned by the divested unit prior to divestment. Observations are at the unit level and weekly frequency. Starts is a dummy variable equal to one if gross generation is positive; the capacity factor is defined as the ratio of gross generation to nameplate capacity conditional on gross generation being positive; the log of the emission rate is defined as the ratio of carbon dioxide emissions to gross generation conditional on gross generation being positive. Columns (1)-(3) present the effects of divestment on units by fuel type. The dummy variable $(Post \times Public\ Buyer \times f - Unit)$ is equal to one if unit i is in the post-divestment period at time t , was divested by a public buyer, and of type f and the dummy $(Post \times Private\ Buyer \times f - Unit)$ is equal to one if unit i is in the post-divestment period at time t , was divested by a private buyer, and of type f . Two types of units exist: coal and gas units. Standard errors are clustered at the plant level and reported in parentheses. Fixed effects are unit fixed effects and interactions of the following variables: state, NERC region, a combined cycle indicator equal to one if the boiler was part of a combined cycle system, the fuels the unit is able to burn, week, year. The table reports also the number of relevant observations used to identify the coefficients reported in that column only, and the p-value of a Wald test of the null that the post-divestment dummies for public buyers and private buyers are equivalent. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

So far, the discussion has largely been limited to the effect of divestments on emissions at the individual unit level. But what are the impacts on aggregate emissions? Declines in the emissions intensities unambiguously decrease aggregate emissions. However, while individually divested units did not increase emissions-generating production relative to comparable units, the aggregate implications are generally ambiguous. Since electricity demand is inelastic in the short run, any change in generation from a change in starts is replacing, or will be replaced by other units. Thus, the effect of divestment on aggregate emissions depends on the emissions profile of the replaced or replacement generation. However, when the divested unit burns the most or least emissions intensive fuel type, the divestment effects of individual assets can tell us something about aggregate effects.

Burning coal generates the most carbon dioxide per MWh of electricity generated of all fuel types. Thus, electricity generated from burning coal is most likely to displace or be replaced by generation that is similarly emissions intense (i.e. other coal plants) or less emissions intense (i.e. other gas, oil plants). In the results for divestment effects by unit fuel type, I find that coal-fired units did not contribute to worse aggregate climate outcomes in a robust, and detectable way: divestment effects on starts by, and the capacity factors of coal-fired units are statistically indistinguishable from zero, regardless of buyer type.

One may wonder why I do not test for aggregate effects more directly. For example, by testing for whether post-divestment, more/less emissions-intense units start more frequently or produce more. This would be like estimating post-divestment effects on the electricity region's non-acquired units, or the buyer's pre-existing generation units. I do not do this because in cases in which one or few units were sold, the divestment effect on the behavior of the firm's pre-existing generation units will likely be close to zero. This is because the estimated, average offsetting effect will be the change in the divested unit divided by many, non-acquired units in a given electricity region and/or the pre-existing units of the buyer, which will approach zero as the number of non-acquired units and/or pre existing units increases.

1.3.4 Decomposing Changes in the Emissions Intensity

One question is whether or not changes in the emissions intensity are due to genuine reductions, or changes in reporting.

There are four ways to change the reported emissions intensity of generation:

1. Change the unit's primary or secondary fuel category. A unit able to burn multiple fuels can decrease its emissions intensity by using more of the cleaner fuel to generate electricity. This behavior, however, is constrained to units that are technologically

equipped to burn multiple fuels. Table 1.5 reports on the emissions factors, defined as, emissions per unit of the fuel’s heat input, of the three, dominant fossil fuels used to generate electricity: coal, oil, and gas. Coal and oil are about 1.8 times and 1.4 times more emitting than natural gas, respectively.

Table 1.5: Emissions Factor by Fuel

Emissions		
Fuel Type	Scaled by heat input ($\frac{kg\ CO_2}{MBtu}$)	Relative to Gas
Coal	95.92	1.8
Oil	74.14	1.4
Gas	52.91	1.0

Source: [Carbon dioxide emissions coefficients, EPA](#)

2. Change the fuel subtype (e.g. shift from burning a dirtier coal to a cleaner coal). There are weaker technological constraints to doing this than changing the unit’s fuel mix, but substantial, possible emissions gains since the most emitting coal subtype coke emits 22% more emissions per heat input than the least emitting coal subtype bituminous coal.
3. Change the way emissions are estimated and reported.
4. Change the heat rate, which is the physical efficiency at which the energy in the fossil fuel is converted into electricity, known as the heat rate. Note that increasing physical efficiencies implies that less fuel is being combusted per unit of electricity generated. Since emissions are primarily determined by the quantity and carbon content of fuel burned, this mechanically implies a decline in the emissions intensity.

In this section, I show that the decline in the emissions intensity appears to be genuine and driven primarily by improvements in the heat rate.

Fuel-switching and Co-firing

There are units that are technologically able to fuel-switch (i.e. units running on one fuel can replace that fuel in its entirety with a substitute fuel) and, or co-fire (i.e. units can simultaneous use two or more fuels). In such units, new buyers can reduce the emissions intensity of electricity generation by switching completely to a cleaner fuel or by tilting the fuel mix towards a cleaner fuel.

I identify units with fuel-switching and co-firing abilities using their historical behavior¹⁷. In my estimation sample, none of the units were able to fuel switch¹⁸, but some were able to co-fire. As reported in Table 1.6, of divested units sold to public buyers, 33% of coal-fired units and 49% of gas-fired units were able to co-fire; of those divested to private buyers, 13% of coal-fired units and 29% of gas-fired units were able to co-fire.

Table 1.6: Type of Divested Units

Coal-firing Units		
Type	Public Buyer	Private Buyer
Coal Only	100	13
Coal, Oil	30	0
Coal, Gas	3	0
Coal, Gas, Oil	0	2
Total	133	15
Gas-firing Units		
Type	Public Buyer	Private Buyer
Gas Only	100	169
Gas, Oil	87	58
Gas, Coal	0	0
Gas, Coal, Oil	0	0
Total	187	227

To study whether divestment leads to changes in co-firing decisions, I estimate the effects of divestiture on coal and gas units separately to estimate linear probability models with intuitive, and easily interpretable coefficients.

¹⁷Identification is not perfect, but an approximation based on discussions with EIA staff. The Appendix goes into greater detail.

¹⁸Fuel switching is not technologically feasible in all units, especially the dirtiest: coal units. As of January 2021, 18% of US electric power generating capacity could functionally switch fuels, 27% of utility scale electric generation that could potentially use multiple energy sources could functionally switch fuels; less than 20% of coal capacity could fuel switch; 19% of natural gas-fired combined-cycle applications can fuel switch; 45% of natural gas combustion turbines can fuel switch (Source: [EIA, 2022](#)). Furthermore, transforming the asset to go from burning one fuel to many fuels is so costly that the EPA does not include fuel switching as part of the “best system of emissions reduction.” For example, “although the EPA acknowledged in the GHG Abatement Measures TSD that some coal plant owners are engaging in repowering projects, the agency concluded that this kind of fuel switching will be on average more expensive than other available options, such as constructing a new natural gas combined-cycle unit. Because gas and biomass cofiring options were found to be relatively expensive when national average cost data were used, the EPA declined to include fuel switching as part of the “best system of emissions reduction” in its proposed emissions guidelines” ([NACAA, 2015](#)).

Empirical Strategy To estimate the effect of divestitures on fuel switching in units able to burn coal, I estimate Eq. [L.3.2](#)

$$\begin{aligned} y_{it} = & \beta_1(Post \times Public Buyer)_{it} \\ & + \beta_2(Post \times Private Buyer)_{it} \\ & + X_{it} + \alpha_i + \eta_{it}, \end{aligned}$$

using observations of units able to co-fire with coal and using coal as the primary fuel that week. The dependent variable y_{it} is a dummy variable equal to one if the unit burned a secondary fuel:

$$y_{it} = Burned\ Secondary\ Fuel_{it}.$$

Time-varying trends or slow-moving processes within a state and NERC region are captured by X_{it} which is the interacted set of the following variables: state, NERC region, a combined cycle indicator equal to one if the unit was part of a combined cycle system, week, and year. Time-invariant unit fixed effects are equal to α_i . Standard errors are clustered at the plant level. Since for coal-fired units, alternative fuels (i.e. oil or gas) are cleaner than coal, a positive coefficient on the post-divestment dummy variables would imply that post-divestment, new buyers were co-firing units to make generation cleaner vis-à-vis comparable units.

To estimate the effect of divestitures on co-firing in units able to burn natural gas, I estimate Eq. [L.3.2](#) using observations of all units able to co-fire with natural gas and using natural gas as a primary fuel that week. The dependent variable is a dummy variable equal to one if the unit burned a secondary fuel. Everything is as before except that the interaction set X_{it} includes a combined cycle indicator equal to one if the unit was part of a combined cycle system. Since in our divested gas-fired units, the alternative fuels with which they could co-fire are dirtier, a positive coefficient on the post-divestment dummy variables would imply that new buyers are fuel-switching and co-firing units in ways that would make generation dirtier.

Results Table [L.7](#) presents the main results as ATT estimates to measure the short-horizon impact of divestments on units' co-firing behavior.

Table 1.7: Co-firing

Dependent Variable:	Coal	Gas
Burned Secondary Fuel	(1)	(2)
<i>Post</i> $\times$ <i>Public Buyer</i>	0.014 (0.014)	0.004 (0.003)
<i>Post</i> $\times$ <i>Private Buyer</i>	0.000 (0.000)	-0.044 (0.039)
Observations	14,573	89,251
R-squared	0.59	0.80
Wald Test P-value	0.32	0.23

Notes: This table uses OLS regressions to estimate the effects of divestment on co-firing by the fuel type of the unit divested. Observations are at the unit level and weekly frequency. The dependent variable is equal to one if the unit burned a secondary fuel, while burning coal as the primary fuel. The dummy variable (*Post* $\times$ *Public Buyer*) is equal to one if unit i was divested by a public firm to a public firm, and t is in the post-divestment period and the dummy variable (*Post* $\times$ *Private Buyer*) is equal to one if unit i was divested by a public firm to a private firm, and t is in the post-divestment period. Standard errors are clustered at the plant level and reported in parentheses. Fixed effects are unit fixed effects and interactions of the following variables: state, NERC region, a combined cycle indicator equal to one if the boiler was part of a combined cycle system, primary and secondary fuel type burned, week, year. The table reports also the p-value of a Wald test of the null that the pre-divestment dummies for public buyers and private buyers are equivalent. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

In coal-fired units, divestments to public buyers lead to an average 1.4 percentage point decline in the probability of burning a cleaner secondary fuel; divestments to private buyers lead to no decline in the probability of burning a cleaner secondary fuel. In gas-fired units, divestments to public buyers lead to a 0.4 percentage point increase in the probability of burning a dirtier secondary fuel; divestments to private buyers lead to a 4.4 percentage point decline in the probability of burning a dirtier secondary fuel. None of these point estimates are statistically indistinguishable from zero.

Reporting, Fuel Subtype, and Heat Rate

Empirical Strategy. Here, I study whether divestment leads to changes in the emissions intensity via (a.) changes in the fuel type within the major fuel categories of coal, oil, and gas, and reporting method changes, and/or (b.) changes in the heat rate.

Before proceeding, I briefly touch on how owners can change emissions factors and heat rates.

The reported emissions factor can change by (a.) changing the unit’s methodology for reporting carbon dioxide emissions, (b.) changing parameters used to estimate carbon dioxide emissions, (c.) changing the fuel sub-type. Depending on the fuel type, the EPA decides the methodologies allowed for reporting carbon dioxide emissions. For solid fuels like coal, the EPA requires emissions to be estimated by a continuous emissions monitoring system (CEMS). These systems measure pollutant concentration and stack gas flow at least every 15 minutes to estimate emissions. While owners have some latitude in the choice of the exact CEMS system to install, variation in estimates across CEMS systems is bounded by stringent system guidelines.^{19,20} For oil or natural gas, carbon dioxide emissions may be estimated using CEMS, continuous monitoring of the fuel flow rate and periodic sampling of the fuel characteristics, or fuel-specific default emission rates and hourly heat inputs.

The heat rate can change by improving the technology, or changing dispatch to more frequently operate at levels at which the unit is the most efficient. For example, heat rates at coal plants can be improved by implementing processes such as rank coal drying, intelligent or neural network soot blowing. Heat rates at gas units can be improved by coating turbine blades and conductor components, repairing steam leaks, adding or restoring insulation, and implementing an effective steam-trap maintenance program.

To decompose changes in the emissions intensity, I estimate the baseline equation broken out by fuel type. Equation 1.3.2, where the outcome y_{it} is for each unit i and week t , the vector of the natural log of the emissions intensity, defined as the ratio of emissions to gross generation, the natural log of the emissions factor, defined as the ratio of emissions to the heat input, and the natural log of the heat rate, defined as the ratio of the heat input to gross generation:

$$y_{it} = \begin{bmatrix} \ln(Emissions\ Intensity) \\ \ln(Emissions\ Factor) \\ \ln(Heat\ Rate) \end{bmatrix}_{it} = \begin{bmatrix} \ln(\frac{CO_2}{MW_h}) \\ \ln(\frac{CO_2}{MMBtu}) \\ \ln(\frac{MMBtu}{MW_h}) \end{bmatrix}_{it}. \quad (1.4)$$

All other variables are as before, except X_{it} , which here is the interacted set of the variables included in the baseline specification but with one change: the interaction set does not include the type of fuels the unit can burn, but the actual primary and secondary fuel type burned. Thus, the control are units that actually burned the same primary and secondary

¹⁹CEMs must be certified by the EPA, and undergo daily calibration tests, and annual reference test audits. For compliance, it must be in continuous operation even if no process is on and for valid measurement, must record at least one reading every 15 minutes for 3 out of 4 quarters and the readings are averaged quarterly.

²⁰The EPA provides an CEMS Cost Modeler spreadsheet on its website <https://www.epa.gov/emc/emc-continuous-emission-monitoring-systems>. A single carbon dioxide analyzer would cost \$114,988 initially and \$37,549 a year with capital recovery (10 years, 7% interest rate).

fuel type that week, not units that could have burned the same primary and secondary fuel type. This prevents any effects on the emissions intensity coming from fuel mixing and thus, isolates changes in emissions intensities coming from changes in the emissions factor or heat rate.

Note that in this setting, mechanically, post-divestment effects on the emissions intensity can only come through changes in the emissions factor and the heat rate. The emissions intensity is the product of the emissions factor and the heat rate

$$\underbrace{\left(\frac{CO_2}{Generation}\right)_{it}}_{Emissions\ Intensity} = \underbrace{\left(\frac{CO_2}{Heat\ Input}\right)_{it}}_{Emissions\ Factor} \times \underbrace{\left(\frac{Heat\ Input}{Generation}\right)_{it}}_{Heat\ Rate}.$$

Taking log changes,

$$\Delta \ln(Emissions\ Intensity)_{it} = \Delta \ln(Emissions\ Factor)_{it} + \Delta \ln(Heat\ Rate)_{it}.$$

Thus, the coefficient on the dummy for post-divestment to a buyer of type b in the emissions intensity regression will be the sum of coefficients on the same dummy variable from the emissions factor and heat rate regressions.

Results Table [1.8](#) presents the main results as ATT estimates to measure the short-horizon impact of divestments on the emissions intensity through changes in reporting, fuel subtype and the heat rate. Note that the point estimates of the divestment effect on the emissions intensity will differ from the benchmark specification, which controlled for the type of fuel the unit could burn, not the actual primary and secondary fuel burned as in this specification. The point estimates from the benchmark specification aggregated the effects of changes in reporting, fuel subtype, heat rate, and co-firing. Here, I separate the divestment effects on the emissions intensity through changes in reporting, fuel subtype and heat rate, from divestment effects through changes in co-firing.

Table 1.8: Decomposition of the Change in the Emissions Intensity By Fuel Type

	$\ln(Emissions\ Rate)$ (1)	$\ln(Emissions\ Factor)$ (2)	$\ln(Heat\ Rate)$ (3)
$Post \times Public\ Buyer \times Coal\ Unit$	-0.014** (0.007)	0.001 (0.001)	-0.015** (0.007)
$Post \times Private\ Buyer \times Coal\ Unit$	-0.041** (0.021)	-0.005* (0.003)	-0.036* (0.021)
$Post \times Public\ Buyer \times Gas\ Unit$	-0.014 (0.017)	0.003 (0.002)	-0.017 (0.017)
$Post \times Private\ Buyer \times Gas\ Unit$	0.014 (0.018)	-0.001 (0.001)	0.015 (0.017)
Sample			
Observations	255,210	255,210	255,210
R-squared	0.94	0.99	0.84
Wald Test P-value			
Coal	0.22	0.07	0.35
Gas	0.31	0.14	0.25

Notes: This table uses OLS regressions to test the effect of divestment on starts, capacity factors, and the log of the emissions intensity by the primary fuel burned by the divested unit prior to divestment. Observations are at the unit level and weekly frequency. Starts is a dummy variable equal to one if gross generation is positive; the capacity factor is defined as the ratio of gross generation to nameplate capacity conditional on gross generation being positive; the log of the emission rate is defined as the ratio of carbon dioxide emissions to gross generation conditional on gross generation being positive. Columns (1)-(3) present the effects of divestment by fuel type. The variable $(Post \times b - Buyer)$ is a dummy variable equal to one if the unit was divested by a public firm to a buyer of type b , where b is either a public firm or a private firm, and t is in the post-divestment period. Standard errors are clustered at the plant level and reported in parentheses. Fixed effects are unit fixed effects and interactions of the following variables: state, NERC region, a combined cycle indicator equal to one if the boiler was part of a combined cycle system, the primary and secondary fuel type burned that week, week, year. The table reports also the number of relevant observations used to identify the coefficients reported in that column only, and the p-value of a Wald test of the null that the post-divestment dummies for public buyers and private buyers are equivalent by divested unit fuel type. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

The divestment of coal-fired units through changes in reporting, fuel subtype, and heat rate is associated with an 1.5% and 3.6% average decline in emissions intensities when sold to public and private buyers, respectively. The former, but not the latter, is statistically significant at the 5% level. However, the difference is not statistically significant across buyer types.

Point estimates in Columns (2) and (3) decompose this decline into two components: the change in the emissions factor and the change in the heat rate. I find that 88-100% of the

decline in the emissions intensity of coal-fired units is due to statistically significant declines in the heat rate, which corresponds to an increase in the physical efficiency at which fuel energy is converted into electricity. The remaining decline can be attributed to statistically insignificant changes in the emissions factor. As noted earlier, because all coal-fired units must use CEMS to estimate emissions, the most likely lever by which owners can change the emissions factor is by changing the coal type used. I obtain annual data on plants' primary coal and secondary coal types from S&P Capital IQ Pro. Of the 51 plants that were divested to a public firm, none changed their primary coal type and six changed their secondary fuel types at some point in the post-divestment period. Of the eight plants that were divested to a private firm, none changed their primary coal type and one changed its secondary fuel type at some point in the post-divestment period. In all, this suggests "tuning" type improvements, like sootblowing and turbine upgrades.

As noted earlier, post-divestment declines in the emissions intensity of gas-fired units are statistically indistinguishable from zero.

Another question is whether new owners are pushing the boundary on emissions intensity or picking units that have lagged behind comparable units in efficiency improvements and bringing them in line with the average. To distinguish between the two, I estimate the baseline equation broken out by fuel type Equation 1.3.2, but with pre-divestment dummies and without unit fixed effects:

$$y_{it} = \Sigma_f \beta_1 (Pre \times Public Buyer \times f Unit)_{it} + \quad (1.5)$$

$$+ \Sigma_f \beta_2 (Post \times Public Buyer \times f Unit)_{it} \quad (1.6)$$

$$\Sigma_f \beta_3 (Pre \times Private Buyer \times f Unit)_{it} + \quad (1.7)$$

$$+ \Sigma_f \beta_4 (Post \times Private Buyer \times f Unit)_{it}$$

$$+ X_{it} + \alpha_i + \eta_{it},$$

where the dummy variable $(Pre \times Public Buyer \times f - Unit)_{it}$ is equal to one if unit i is in the pre-divestment period at time t , was divested by a public buyer and of type $f \in \{Coal, Gas\}$; and the dummy $(Pre \times Private Buyer \times f - Unit)_{it}$ is equal to one if unit i is in the pre-divestment period at time t , was divested by a private buyer, and of type f . Let y_{it} denote the vector of the natural log of the emissions intensity and the natural log of the heat rate

$$y_{it} = \left[\ln(Emissions Intensity) \right]_{it}. \quad (1.8)$$

Table [1.9](#) reports the pre-divestment dummies, which parameterizes the average unit fixed effects of divested units in the pre-divestment period, defined as the eighteen month period before divestment. The coefficient on the pre-divestment dummies of coal-fired units divested to public buyers and private buyers in the log emissions intensity regression are 0.007 and 0.048, respectively. The coefficients on the pre-divestment dummies of gas-fired units divested to public and private buyers in the regression of the log of the emissions intensity are similarly positive. This raises the possibility that acquired coal-fired units, which experienced an average 1.4% and 4.2% decline in the emissions intensity when divested to public and private buyers, respectively, were units that had been lagging behind comparable units being brought into line with the average. With the caveat that these are very imprecisely estimated coefficients, this leans against the idea of new owners pushing the boundary on emissions intensity improvements and favors the idea of new owners bringing inefficient units to the standard baseline.

One question the regression results raise, however, is why natural gas units that are acquired do not also experience similar post-divestment declines. One possibility is that the range of heat rate improvement that can be made at coal gas units is larger. Another is that heat rate improvements in coal units are particularly attractive and thus, taken up, because they deliver extra benefits beyond fuel cost savings that are not realized for heat rate improvement in gas units. For instance, coal combustion also generates other pollutants such as sulfur dioxide, nitrogen oxides, and mercury, which are either directly or indirectly taxed; improving the heat rate thus reduces the costs of regulatory compliance.^{[21](#)}

This suggests that divestments have a positive, aggregate climate impact through improvements in the physical efficiency of units, which in this context, directly translates to reduced emissions intensity of generation, but that divestments to private firms are not particularly better than those to public firms.

²¹See the cap-and-trade Acid Rain Program (ARP), mercury emissions limits imposed by the Mercury and Air Toxics Standards (MATS).

Table 1.9: Average Emissions Intensity of Treated Units vis-à-vis Control Units in the Pre-divestment Period, Without Unit Fixed Effects

	<i>ln(Emissions Intensity)</i> (1)
<i>Pre × Public Buyer × Coal Unit</i>	0.007 (0.015)
<i>Pre × Private Buyer × Coal Unit</i>	0.048 (0.058)
<i>Pre × Public Buyer × Gas Unit</i>	-0.037 (0.025)
<i>Pre × Private Buyer × Gas Unit</i>	0.037* (0.022)
Sample	
Observations	265,698
R-squared	0.80
Wald Test P-value	
Coal	0.51
Gas	0.58

Notes: This table uses OLS regressions to test for selection. Observations are at the unit level and weekly frequency. The dependent variable in Column (1) is the natural log of the emissions intensity: the ratio of emissions to gross generation. The dummy variable ($Post \times Public Buyer \times f - Unit$) is equal to one if unit i is in the post-divestment period at time t , was divested by a public buyer, and of type f and the dummy ($Post \times Private Buyer \times f - Unit$) is equal to one if unit i is in the post-divestment period at time t , was divested by a private buyer, and of type f . Two types of units exist: coal and gas units. Standard errors are clustered at the plant level and reported in parentheses. Fixed effects are unit fixed effects and interactions of the following variables: state, NERC region, a combined cycle indicator equal to one if the boiler was part of a combined cycle system, primary and secondary fuel type burned, week, year. The table reports also the R-squared contribution of the pre-divestment dummies, and the p-value of a Wald test of the null that the pre-divestment dummies for public buyers and private buyers are equivalent. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Table 1.10: Average Emissions Intensity of Treated Units vis-à-vis Control Units in the Pre-divestment Period, With Unit Fixed Effects

	$\ln(\text{Emissions Intensity})$	$\ln(\text{Heat Rate})$
	(1)	(2)
$Pre \times Public Buyer \times Coal Unit$	-0.018* (0.010)	-0.014 (0.010)
$Pre \times Private Buyer \times Coal Unit$	-0.014 (0.016)	-0.016 (0.015)
$Pre \times Public Buyer \times Gas Unit$	-0.035 (0.034)	-0.042 (0.034)
$Pre \times Private Buyer \times Gas Unit$	0.053** (0.023)	0.049** (0.023)
Sample		
Observations	265,692	265,692
R-squared	0.93	0.84
R-squared contribution	0.00	0.01
Wald Test P-value		
Coal	0.86	0.86
Gas	0.07	0.61

Notes: This table uses OLS regressions to test for selection. Observations are at the unit level and weekly frequency. The dependent variable in Column (1) is the natural log of the emissions intensity: the ratio of emissions to gross generation. The dependent variable in Column (2) is the natural log of the heat rate: the ratio of the heat input to gross generation. The dummy variable ($Post \times Public Buyer \times f - Unit$) is equal to one if unit i is in the post-divestment period at time t , was divested by a public buyer, and of type f and the dummy ($Post \times Private Buyer \times f - Unit$) is equal to one if unit i is in the post-divestment period at time t , was divested by a private buyer, and of type f . Two types of units exist: coal and gas units. Standard errors are clustered at the plant level and reported in parentheses. Fixed effects are unit fixed effects and interactions of the following variables: state, NERC region, a combined cycle indicator equal to one if the boiler was part of a combined cycle system, primary and secondary fuel type burned, week, year. The table reports also the R-squared contribution of the pre-divestment dummies, and the p-value of a Wald test of the null that the pre-divestment dummies for public buyers and private buyers are equivalent. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

So far, I have said nothing about the social welfare implications of the drop in the emissions intensity. If the social objective is to minimize short-horizon emissions at whatever cost, the drop in the emissions intensity is good for social welfare. The reality is that there are competing objectives both at a particular point in time, and across time. For example, by dropping the emissions intensity via the heat rate of coal-fired units, new owners may be reducing the path of expected, future fuel costs and extending the economic lifespan and thus, expected, future long run emissions of the units. This could reduce social welfare. Or, if emissions intensity improvements were made but the costs were passed onto consumers, the value of the drop in emissions might be too small to compensate for the loss of consumption

value. A welfare analysis would require information about the private net present value of projects implemented to achieve reductions in the emissions intensity and an assumption about the social cost of carbon. If these projects had positive net present values, the emissions intensity reductions would be behavior that I would expect to observe in an equilibrium in which social welfare was being maximized. Because I do not observe projects implemented, their cash flows, I am unable to do this.

1.4 Valuation Implications

If investors expect negative consequences for public firms with high emissions, the announcement of an unanticipated divestiture might generate positive abnormal returns to shareholders. In this section, I measure the effect of power plant divestment announcements on the value of the public seller by conducting an event study of abnormal returns around divestment announcements.

The approach rests on the idea that events that affect a firm's future value are reflected immediately in asset prices because of the forward looking nature of asset prices; thus, the event's economic impact can be measured using asset prices observed over a relatively short time period, whereas direct measures may require many months or years of observation MacKinlay (1997).

1.4.1 Data

To estimate the valuation effects of divestment announcements on the seller, I assemble a dataset of power plant sale announcement dates from S&P Capital IQ Pro, buyer and seller names, and, where available, tickers and listed exchanges. I then pair this with stock price data from WRDS's CRSP. This allows me to consider 153 sales in the sample: 62 that were to a publicly traded firm and 91 that were to a private firm (see Table L.11). The number of deals announcements I can consider is nearly double what I can consider when estimating climate impacts because a.) I am able to include announcements from repeat sales, and b.) I am no longer bound by the restriction that deals occur eighteen months after the start date of emissions data and eighteen months before the end of the emissions data.

Table 1.11: Deal Counts in Study of Announcement Returns

Sale Type	Deal Count
	Asset Purchases Only
Public to Public	62
Public to Private	91
All	153

Notes: This table reports on number of events over which average cumulative abnormal returns were computed, by sale type.

1.4.2 Empirical Strategy

To assess the event's impact, a measure of abnormal returns is required. It is defined as the actual ex-post return of the security over the event window minus the expected return of the firm over the event window:

$$AR_{t-1,t}^i \equiv r_{t-1,t}^i - \mathbb{E}_{t-1}(r_{t-1,t}^i), \quad (1.9)$$

where $AR_{t-1,t}^i$ is stock i 's abnormal return over $t - 1$ to t , $r_{t-1,t}^i$ is stock i 's ex-post realized return over $t - 1$ to t , and $\mathbb{E}_{t-1}(r_{t-1,t}^i)$ is stock i 's expected return at time $t - 1$ over $t - 1$ to t .

Here, the expected return is the Fama-French Plus Momentum Model. This model assumes a stable linear relationship between the security's excess return, the risk-free rate, and four factors, which are the three Fama-French factors: the excess return on the market, size of firms, and book-to-market values; and the momentum factor. I can thus express abnormal returns as

$$AR_{t-1,t}^i \equiv r_{t-1,t}^i - \mathbb{E}_{t-1}(r_{t-1,t}^i) \quad (1.10)$$

$$= (r_{t-1,t}^i - r_{t-1,t}^{Risk-free}) \quad (1.11)$$

$$= (\alpha^i + \beta_1 f_{t-1,t}^{Market} + \beta_2 f_{t-1,t}^{SMB} + \beta_3 f_{t-1,t}^{HML} + \beta_4 f_{t-1,t}^{Mom}), \quad (1.12)$$

where $r_{t-1,t}^{Risk-free}$ is the risk-free rate over $t - 1$ to t estimated by the one-month T-bill rate, the factors $f_{t-1,t}^{Market}$, $f_{t-1,t}^{SMB}$, $f_{t-1,t}^{HML}$, and $f_{t-1,t}^{Mom}$ are the value-weighted aggregate market return on the CRSP US. stock universe, and returns on value-weighted, zero-investment, factor-mimicking portfolios for size, book-to-market equity, and one-year momentum in stock returns, respectively over $t - 1$ to t . The factors $f_{t-1,t}^{SMB}$, $f_{t-1,t}^{HML}$ are obtained from Ken French's website; the factor $f_{t-1,t}^{Mom}$ is the equal-weight average of firms with the highest 30 percent eleven-month returns lagged one month minus the equal-weight average of firms with the

lowest 30 percent eleven-month returns lagged one month. The portfolios include all NYSE, Amex, and Nasdaq stocks and are re-formed monthly.

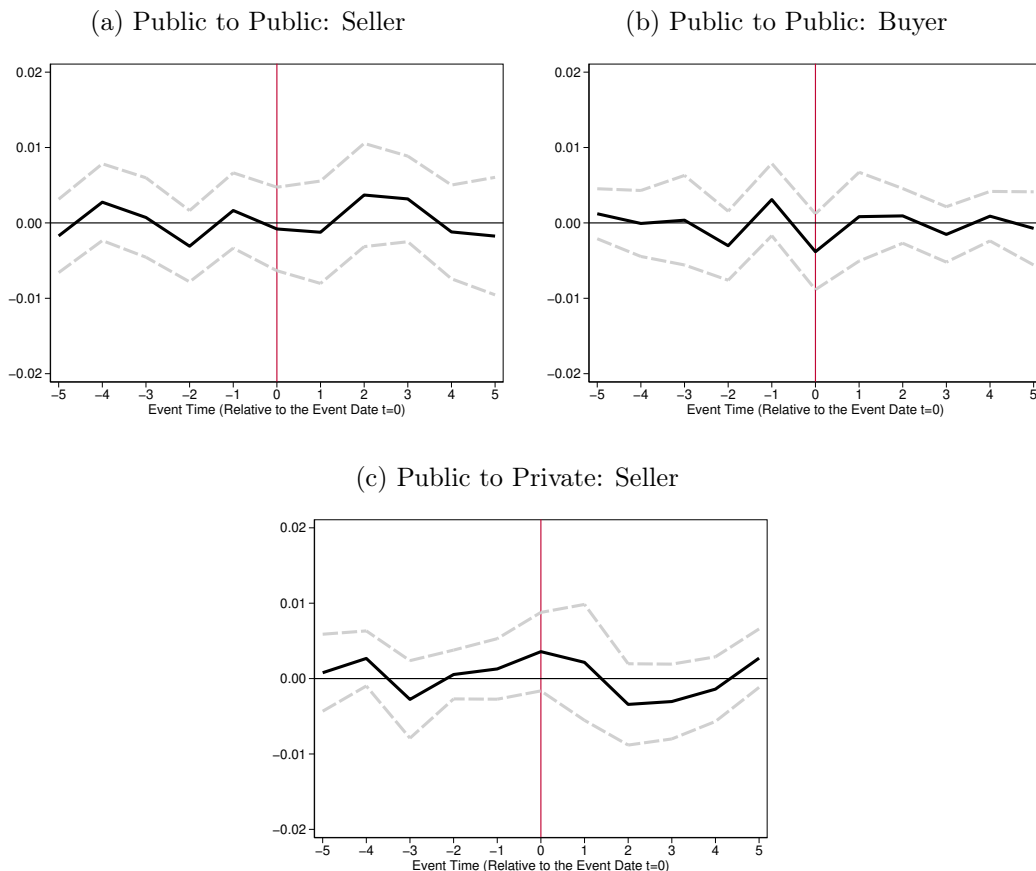
The parameters of the model are estimated using the 100 days of trading data that are 50 days before the start of the event window. Having estimated the parameters, I compute abnormal returns and cumulative abnormal returns over the window. I then aggregate these returns across events on a nameplate weighted basis, where the nameplate for a given event is the total nameplate fossil fuel generation capacity that changed hands.

1.4.3 Result

Average The results suggest that sellers were not rewarded financially for selling to private firms compared to public firms.

Figure [1.1](#) plots the nameplate-weighted average cumulative abnormal returns over the eleven-day symmetric window surrounding the announcement of a power plant divestment by a publicly traded firm, by buyer type. In public to public transactions, sellers' stocks have cumulative abnormal returns of 0.6%; in public to private transactions, 1.4%. The point estimates and their difference are not statistically significant. Thus, I reject the null that publicly traded sellers were rewarded more by the stock market when they announced divestments to privately held firm compared to when they announced divestments to public firms. This suggests that in my period, there was no detectable financial incentive for sellers to divest high greenhouse gas emitting assets to private firms in particular. This result is robust to using smaller and larger windows for the event study, to averaging CARs using nameplate weights, to including mergers and acquisitions (see Appendix).

Figure 1.1: Average Cumulative Abnormal Returns Around Plant Divestment Announcements



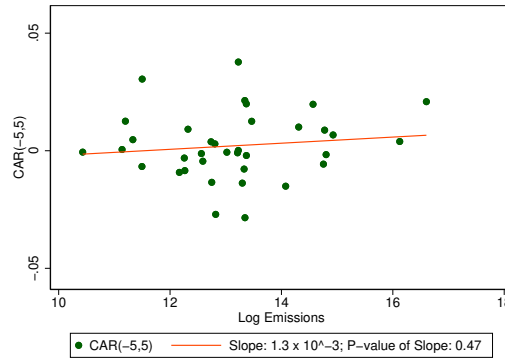
Notes: These figures show the average cumulative abnormal returns (CARs) over an eleven-day window surrounding announcement of a plant divestment. Figures are distinguished by divestment type: Public to Public and Public to Private. The event window is plotted on the x-axis and starts five business days before the event, and ends five business days after the event. The average CAR displayed is nameplate weighted average of firm CARs across firms. The return model is the Fama-French factor plus Momentum model. Dashed lines indicate the 95% confidence interval.

One may wonder why there are statistically significant, and economically meaningful announcement returns in public to private sale announcements in the absence of statistically significant changes in the plant operating behavior along the dimensions this paper considers. One possibility is that private firms do indeed have a comparative advantage in running power plants financially, but not operationally. For example, [Liu \(2021\)](#) finds that when private equity investors, an important class of private firms, buy out a hospital, total healthcare spending for the privately insured in affected markets are driven by higher bargained prices at PE-backed hospitals, with PE investors' superior bargaining skills accounting for 43%

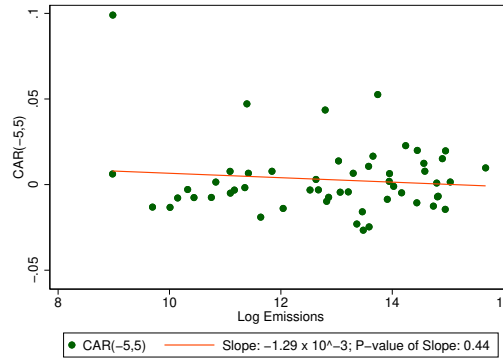
of the price and spending increases. In this setting, the relevant contracts that could be renegotiated are fuel contracts, which determine 80% of plant operating costs, and labor contracts.

Cross-section Figure 1.3 displays CARs around divestment announcements by the annual emissions of the fossil fuel power plants sold in the year prior to sale announcement. Cumulative abnormal returns do not appear to have a statistically robust, linear relation to log emissions. In public to public, and public to private transactions, cumulative abnormal returns are approximately flat in log emissions for the seller.

Figure 1.2: Cumulative Abnormal Returns vs Log Emissions



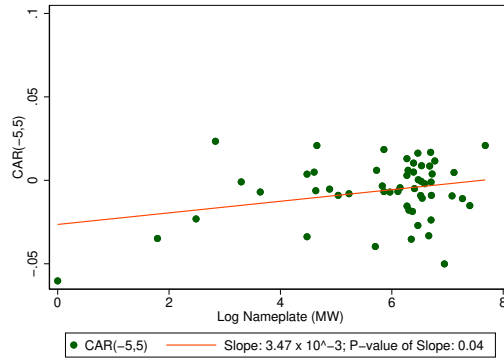
(a) Public to Public: Seller



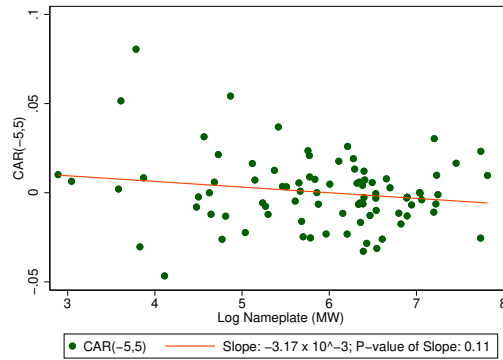
(b) Public to Private: Seller

Notes: These plots plots sellers' average cumulative abnormal returns (CARs) to announcing the sale of a fossil fuel power plant by buyer type against the log of emissions transferred in the transaction. CARs are over an eleven-day symmetric generated and estimated using the Fama-Carhart model.

Figure 1.3: CARS vs. Log Nameplate



(a) Public to Public: Seller

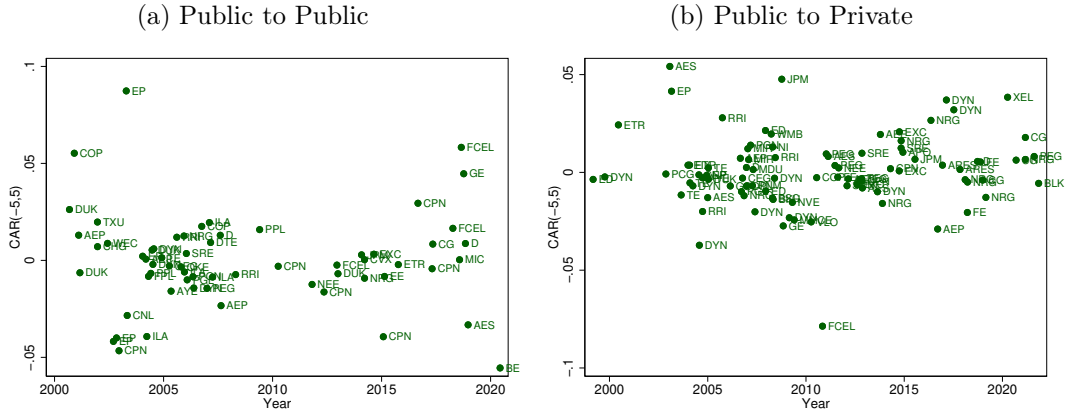


(b) Public to Private: Seller

Figure [1.4](#) displays CARs around divestment announcements by the year of the announcement date. Data points are labeled by seller tickers. Returns largely straddle zero. Positive abnormal returns are not surprising, as they suggest selling firms are taking a positive net present value action. Negative abnormal returns require more explanation. This could be noise associated with the event study - that is, that there was a downward short-term trend in the underlying stock not absorbed by the factors, but orthogonal to the divestment announcement; they could be instances of miscalculation of the benefits and costs of the trade by publicly traded firms or large internal private benefits from divestments not reflected in state prices, which encode preferences of the marginal investors.

Notably, in the plot, there are very large observations for El Paso Corp (NYSE: EP), and Dynegy (NYSE:DYN), which close examination reveals occurred during sharp price trends that spanned many months. All three companies were also penny stocks around the time of these large observations, and large swings are not uncommon for such stocks.

Figure 1.4: Time Series of CARs



Notes: This plot displays the cumulative abnormal returns (CARs) of divestment announcements by the year of the announcement date. CARs are over an eleven-day symmetric generated and estimated using the Fama-Carhart model. Data points are labeled by seller tickers.

1.5 Conclusion

This paper empirically studied whether in my sample of fossil fuel power plant divestments in 2002-2020, shifting asset ownership from the hands of publicly traded firms to those of privately held firms a.) leads assets to be operated in ways that are more emitting in absolute terms and relative to what we might expect if they were sold to another publicly traded firm and b.) rewards publicly traded firms through increased valuations.

Estimated short-horizon, defined here as eighteen months, divestment effects on unit starts, capacity factors, fuel switching and co-firing behaviors, and emissions intensities suggest that in my sample, such ownership transfers did not lead assets to individually be operated in ways that were more emitting, either in absolute or relative terms. This favors the greenwashing narrative that asset divestments are unlikely to have a meaningful effect on emissions.

One question is whether we expect these results to hold in the future. The nature of emissions generation from this sector implies that the change in emissions from ownership changes is likely to be limited, especially on the side of emissions dramatically increasing. This is because unlike other kinds of pollution, like ground pollution or even sulfur dioxide or nitrous oxide pollution from this sector, power plants cannot choose whether or not they will “dump” emissions into the air. This is because carbon capture technology is in its infancy and has generally been implemented only at pilot scale with government support (Howard, 2022). Thus, at existing technological constraints, the only margin along which firms can change emissions is by changing how much they produce, which is bounded above by capacity constraints, and constrained by ordinary economic forces like final goods demand—which is highly inelastic in this sector—and marginal costs.

To study the valuation implications of divestitures by publicly traded firms, I estimated cumulative abnormal returns around divestment announcements and found that divestments to private firms did not command a premium to divestments to public firms, suggesting the absence of large and systematic financial incentives to divest these highly emitting assets to private firms over public firms.

Both the empirics and model contribute to the growing literature suggesting that divestments are likely to have near zero, or negative climate impacts and encouraging engagement through the exercise of rights of control (Broccardo et al., 2022; Berk and van Binsbergen, 2022; Hartzmark and Shue, 2023; Krueger et al., 2020) over exit.

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Appendix

1.5.1 Averages of Dependent Variables by Unit Fuel Type

Table 1.12: Averages of Treated Coal and Gas Units

	Coal	Gas
	(1)	(2)
Weekly Starts		
Mean	0.82	0.75
Std. Dev.	0.38	0.43
Capacity Factor		
Mean	0.72	0.45
Std. Dev.	0.24	0.32
Emissions Intensity (short tons/MWh)		
Mean	1.02	0.59
Std. Dev.	0.13	0.15
Carbon Factor (short tons/MMBtu)		
Mean	0.10	0.06
Std. Dev.	0.00	0.01
Heat Rate (MMbtu/MWh)		
Mean	9.85	9.74
Std. Dev.	1.19	2.12

1.5.2 Climate Effects: Robustness

Longer Window to Five Years

One might be concerned that results may change if regressions are run at higher frequencies.

I re-estimate the results using a 5 year horizon.

Table 1.13: Co-firing

Dependent Variable:	Coal	Gas
Burned Secondary Fuel	(1)	(2)
$Post \times Public Buyer$	0.002 (0.004)	0.008 (0.006)
$Post \times Private Buyer$	0.000 0.000	-0.013 (0.021)
Observations	175,585	541,243
R-squared	0.99	0.99
R-squared contribution	0.00	0.00
Wald Test P-value	0.68	0.41

Notes: This table uses OLS regressions to estimate the effects of divestment on co-firing by the fuel type of the unit divested. Observations are at the unit level and weekly frequency. The dependent variable is equal to one if the unit burned a secondary fuel, while burning coal as the primary fuel. The dummy variable ($Post \times Public Buyer$) is equal to one if unit i was divested by a public firm to a public firm, and t is in the post-divestment period and the dummy variable ($Post \times Private Buyer$) is equal to one if unit i was divested by a public firm to a private firm, and t is in the post-divestment period. Standard errors are clustered at the plant level and reported in parentheses. Fixed effects are unit fixed effects and interactions of the following variables: state, NERC region, a combined cycle indicator equal to one if the boiler was part of a combined cycle system, primary and secondary fuel type burned, week, year. The table reports also the R-squared contribution of the pre-sale dummies, and the p-value of a Wald test of the null that the pre-sale dummies for public buyers and private buyers are equivalent. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Table 1.14: Divestment Effects on Unit Starts, Capacity Factors, and Emissions Intensities

	All		
	Starts	Capacity Factor	$\ln(Emissions\ Intensity)$
	(1)	(2)	(3)
<i>Post × Public Buyer</i>	-0.001 (0.014)	-0.007 (0.013)	0.005 (0.013)
<i>Post × Private Buyer</i>	-0.007 (0.014)	0.009 (0.014)	0.034 (0.022)
Sample			
Observations	621,821	466,012	466,012
R-squared	0.40	0.72	0.92
Wald Test P-value			
All	0.73	0.41	0.30

Notes: This table uses OLS regressions to test the effect of divestment on starts, capacity factors, and the log of the emissions intensity by the primary fuel burned by the divested unit prior to divestment. Observations are at the unit level and weekly frequency. Starts is a dummy variable equal to one if gross generation is positive; the capacity factor is defined as the ratio of gross generation to nameplate capacity conditional on gross generation being positive; the log of the emission rate is defined as the ratio of carbon dioxide emissions to gross generation conditional on gross generation being positive. Columns (1)-(3) present the effects of divestment on all units without distinguishing by unit fuel type. The dummy variable (*Post × Public Buyer*) is equal to one if unit i was divested by a public firm to a public firm, and t is in the post-divestment period and the dummy variable (*Post × Private Buyer*) is equal to one if unit i was divested by a public firm to a private firm, and t is in the post-divestment period. Columns (4)-(6) present the effects of divestment on units by fuel type. The dummy variable (*Post × Public Buyer × f – Unit*) is equal to one if unit i is in the post-divestment period at time t , was divested by a public buyer, and of type f and the dummy (*Post × Private Buyer × f – Unit*) is equal to one if unit i is in the post-divestment period at time t , was divested by a private buyer, and of type f . Two types of units exist: coal and gas units. Standard errors are clustered at the plant level and reported in parentheses. Fixed effects are unit fixed effects and interactions of the following variables: state, NERC region, a combined cycle indicator equal to one if the boiler was part of a combined cycle system, the fuels the unit is able to burn, week, year. The table reports also the number of relevant observations used to identify the coefficients reported in that column only, and the p-value of a Wald test of the null that the post-divestment dummies for public buyers and private buyers are equivalent. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Table 1.15: Divestment Effects on Unit Starts, Capacity Factors, and Emissions Intensities

	By Fuel		
	Starts (1)	Capacity Factor (2)	$\ln(Emissions\ Intensity)$ (3)
$Post \times Public\ Buyer \times Coal\ Unit$	0.033 (0.022)	0 (0.014)	-0.013 (0.011)
$Post \times Private\ Buyer \times Coal\ Unit$	0.039 (0.038)	0.022 (0.026)	-0.034* (0.017)
$Post \times Public\ Buyer \times Gas\ Unit$	-0.019 (0.018)	-0.012 (0.019)	0.017 (0.020)
$Post \times Private\ Buyer \times Gas\ Unit$	-0.014 (0.016)	0.007 (0.015)	0.043* (0.024)
Sample			
Observations	621,821	466,012	466,012
R-squared	0.40	0.72	0.92
Wald Test P-value			
Coal	0.91	0.42	0.32
Gas	0.80	0.45	0.45

Notes: This table uses OLS regressions to test the effect of divestment on starts, capacity factors, and the log of the emissions intensity by the primary fuel burned by the divested unit prior to divestment. Observations are at the unit level and weekly frequency. Starts is a dummy variable equal to one if gross generation is positive; the capacity factor is defined as the ratio of gross generation to nameplate capacity conditional on gross generation being positive; the log of the emission rate is defined as the ratio of carbon dioxide emissions to gross generation conditional on gross generation being positive. Columns (1)-(3) present the effects of divestment on all units without distinguishing by unit fuel type. The dummy variable ($Post \times Public\ Buyer$) is equal to one if unit i was divested by a public firm to a public firm, and t is in the post-divestment period and the dummy variable ($Post \times Private\ Buyer$) is equal to one if unit i was divested by a public firm to a private firm, and t is in the post-divestment period. Columns (4)-(6) present the effects of divestment on units by fuel type. The dummy variable ($Post \times Public\ Buyer \times f - Unit$) is equal to one if unit i is in the post-divestment period at time t , was divested by a public buyer, and of type f and the dummy ($Post \times Private\ Buyer \times f - Unit$) is equal to one if unit i is in the post-divestment period at time t , was divested by a private buyer, and of type f . Two types of units exist: coal and gas units. Standard errors are clustered at the plant level and reported in parentheses. Fixed effects are unit fixed effects and interactions of the following variables: state, NERC region, a combined cycle indicator equal to one if the boiler was part of a combined cycle system, the fuels the unit is able to burn, week, year. The table reports also the number of relevant observations used to identify the coefficients reported in that column only, and the p-value of a Wald test of the null that the post-divestment dummies for public buyers and private buyers are equivalent. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Nameplate-Weighted Results

One might be concerned that results may change if we weight results by nameplate.

Table 1.16: Co-firing

Dependent Variable: Burned Secondary Fuel	Coal (1)	Gas (2)
$Post \times Public Buyer$	0.001 (0.004)	0.005 (0.004)
$Post \times Private Buyer$	0.000 0.000	-0.008 (0.007)
Observations	71,742	272,498
R-squared	0.99	1.00
R-squared contribution	0.00	0.00
Wald Test P-value	0.84	0.22

Notes: This table uses OLS regressions to estimate the effects of divestment on co-firing by the fuel type of the unit divested. Observations are at the unit level and weekly frequency. The dependent variable is equal to one if the unit burned a secondary fuel, while burning coal as the primary fuel. The dummy variable ($Post \times Public Buyer$) is equal to one if unit i was divested by a public firm to a public firm, and t is in the post-divestment period and the dummy variable ($Post \times Private Buyer$) is equal to one if unit i was divested by a public firm to a private firm, and t is in the post-divestment period. Standard errors are clustered at the plant level and reported in parentheses. Fixed effects are unit fixed effects and interactions of the following variables: state, NERC region, a combined cycle indicator equal to one if the boiler was part of a combined cycle system, primary and secondary fuel type burned, week, year. The table reports also the R-squared contribution of the pre-sale dummies, and the p-value of a Wald test of the null that the pre-sale dummies for public buyers and private buyers are equivalent. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Table 1.17: Divestment Effects on Unit Starts, Capacity Factors, and Emissions Intensities, Nameplate-Weighted

	All		
	Starts (1)	Capacity Factor (2)	$\ln(\text{Emissions Intensity})$ (3)
$Post \times Public Buyer$	-0.003 (0.013)	0.000 (0.009)	-0.011 (0.010)
$Post \times Private Buyer$	-0.034** (0.015)	-0.020 (0.015)	-0.006 (0.018)
Sample			
Observations	348,252	265,692	265,692
R-squared	0.349	0.622	0.949
Wald Test P-value			
All	0.11	0.24	0.84

Notes: This table uses OLS regressions to test the effect of divestment on starts, capacity factors, and the log of the emissions intensity by the primary fuel burned by the divested unit prior to divestment. Observations are at the unit level and weekly frequency. Starts is a dummy variable equal to one if gross generation is positive; the capacity factor is defined as the ratio of gross generation to nameplate capacity conditional on gross generation being positive; the log of the emission rate is defined as the ratio of carbon dioxide emissions to gross generation conditional on gross generation being positive. Columns (1)-(3) present the effects of divestment on all units without distinguishing by unit fuel type. The dummy variable ($Post \times Public Buyer$) is equal to one if unit i was divested by a public firm to a public firm, and t is in the post-divestment period and the dummy variable ($Post \times Private Buyer$) is equal to one if unit i was divested by a public firm to a private firm, and t is in the post-divestment period. Standard errors are clustered at the plant level and reported in parentheses. Fixed effects are unit fixed effects and interactions of the following variables: state, NERC region, a combined cycle indicator equal to one if the boiler was part of a combined cycle system, the fuels the unit is able to burn, week, year. The table reports also the number of relevant observations used to identify the coefficients reported in that column only, and the p-value of a Wald test of the null that the post-divestment dummies for public buyers and private buyers are equivalent. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Table 1.18: Divestment Effects on Unit Starts, Capacity Factors, and Emissions Intensities, Nameplate-Weighted

	By Fuel		
	Starts (1)	Capacity Factor (2)	$\ln(Emissions\ Intensity)$ (3)
$Post \times Public\ Buyer \times Coal\ Unit$	0.001 (0.015)	0.004 (0.011)	-0.008 (0.008)
$Post \times Private\ Buyer \times Coal\ Unit$	-0.004 (0.024)	-0.015 (0.031)	-0.049** (0.024)
$Post \times Public\ Buyer \times Gas\ Unit$	-0.012 (0.023)	-0.008 (0.016)	-0.014 (0.023)
$Post \times Private\ Buyer \times Gas\ Unit$	-0.041** (0.018)	-0.022 (0.017)	0.006 (0.021)
Sample			
Observations	348,252	265,692	265,692
R-squared	0.349	0.622	0.949
Wald Test P-value			
Coal	0.85	0.56	0.11
Gas	0.28	0.54	0.58

Notes: This table uses OLS regressions to test the effect of divestment on starts, capacity factors, and the log of the emissions intensity by the primary fuel burned by the divested unit prior to divestment. Observations are at the unit level and weekly frequency. Starts is a dummy variable equal to one if gross generation is positive; the capacity factor is defined as the ratio of gross generation to nameplate capacity conditional on gross generation being positive; the log of the emission rate is defined as the ratio of carbon dioxide emissions to gross generation conditional on gross generation being positive. Columns (1)-(3) present the effects of divestment on units by fuel type. The dummy variable ($Post \times Public\ Buyer \times f - Unit$) is equal to one if unit i is in the post-divestment period at time t , was divested by a public buyer, and of type f and the dummy ($Post \times Private\ Buyer \times f - Unit$) is equal to one if unit i is in the post-divestment period at time t , was divested by a private buyer, and of type f . Two types of units exist: coal and gas units. Standard errors are clustered at the plant level and reported in parentheses. Fixed effects are unit fixed effects and interactions of the following variables: state, NERC region, a combined cycle indicator equal to one if the boiler was part of a combined cycle system, the fuels the unit is able to burn, week, year. The table reports also the number of relevant observations used to identify the coefficients reported in that column only, and the p-value of a Wald test of the null that the post-divestment dummies for public buyers and private buyers are equivalent. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

1.5.3 Climate Implications

Comparison of Treated Units and Controls

Table 1.19 presents the average operations and emissions behavior of treated and control units, as well as their unit characteristics: their nameplate capacities and the ages of associated plants, across the entire sample. Treated units were, on average, 221 MW and slightly larger than the control group by 23MW; they were also likely to be a part of plants that were younger by five years. When means are compared within fuel groups, the difference in nameplate capacities and age are statistically significant for gas-fired units only.

Table 1.19: Comparison of Control and Treated Units, Full Sample

	Treated Units	Control Units	Difference
	(1)	(2)	(3)
Annual Generation (MWh)			
Mean	630,738	684,833	-54,095
Std. Dev./Std. Error	(1,030,292)	(1,158,452)	(31,769)
Annual Emissions (s. tons)			
Mean	476,358	569,902	-93,544
Std. Dev./Std. Error	(964,672)	(1,106,915)	(30,101)
Emissions Intensity ($\frac{s.tons}{MWh}$)			
Mean	0.71	0.78	-0.07***
Std. Dev./Std. Error	(0.23)	(0.30)	(0.01)
Nameplate (MW)			
Mean	177.88	178.57	-0.69
Std. Dev./Std. Error	(172.79)	(192.40)	(5.64)
Age (Years)			
Mean	37.48	46.30	-8.81***
Std. Dev./Std. Error	(22.10)	(24.72)	(0.68)

Notes: Means and standard deviations reported in columns (1)-(2). Means and standard errors reported in column (3).

*, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

In the sample used, the average coal-fired unit acquired by public buyers was smaller, generated less, and emitted less than those acquired by private buyers. They also tended to be a part of older plants. Differences in average emissions intensities are not statistically significant.

The average gas-fired unit acquired by public buyers tended to be a part of older plants than those acquired by private buyers. Differences in average generation levels, emissions, emissions intensity, and nameplate capacity are not statistically significant.

Heat Rate Improvements

In its May 2015 report “Analysis of Heat Rate Improvement Potential at coal-fired Power Plants”, the EIA identified heat rate improvements at existing plants ranging from 0.8% - 10% through such methods. Drying low-rank coal prevents heat energy from being lost to evaporate moisture in coal. “Sub-bituminous and lignite coals contain relatively large amounts of moisture (15 to 40%) compared to bituminous coal (less than 10%). A significant amount of the heat released during combustion of low-rank coals is used to evaporate this moisture, rather than generate steam for the turbine. As a result, boiler efficiency is typically lower for plants burning low-rank coal. The technologies include using waste heat from the flue gas and/or cooling water systems to dry low-rank coal prior to combustion.” Intelligent sootblowing can help maintain boiler efficiency. “Sootblowers intermittently inject high velocity jets of steam or air to clean coal ash deposits from boiler tube surfaces in order to maintain adequate heat transfer. Proper control of the timing and intensity of individual sootblowers is important to maintain steam temperature and boiler efficiency. The identified technologies include intelligent or neural network sootblowing (i.e., sootblowing in response to real-time conditions in the boiler) and detonation sootblowing.”

In its October 2017 report “Improving Heat Rate on Combined Cycle Power Plants” delivered to the Environmental Defense Fund, clean energy consulting firm Andover Technology Partners identified possible heat rate improvements in gas plants through such processes. Coating turbine blades can improve a turbine’s heat resistance, fouling resistance, corrosion resistance, and reduce the turbine’s surface friction. Repairing steam leaks and adding or restoring insulation, and effective steam-trap maintenance program prevents the loss of the heat energy being converted to electricity. Heat rate improvements were estimated to be around 0.5-3%.

Logit

Description: This table reports average marginal effects of a logit model of the probability of being sold by a public firm seller in Column (1) and the conditional probability of a unit being sold to a private buyer in Column (2). *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively. The table reports McFadden’s pseudo R-squared,

defined as one minus the ratio of the log likelihood of the full model to the log likelihood of the model with just the intercept: $1 - \frac{\ln(\mathcal{L}_{Full})}{\ln(\mathcal{L}_{intercept})}$.

Table 1.20: Selection

Average Marginal Effects	$P(\text{Public Seller})$		$P(\text{Private Buyer} \text{Public Seller})$	
	(1)	All (2)	Coal (3)	Gas (4)
<i>Gas unit</i>	0.138*** (0.040)	0.359*** (0.061)		
<i>Combined cycle</i>	-0.142*** (0.029)	0.018 (0.040)		0.038 (0.048)
<i>Nameplate quartile</i>	0.005 (0.014)	0.087*** (0.021)	0.024 (0.026)	0.087*** (0.027)
<i>Age quartile</i>	0.064*** (0.014)	-0.096*** (0.018)	-0.172*** (0.037)	-0.088*** (0.022)
<i>In deregulated retail electricity state</i>	-0.088*** (0.024)	0.142*** (0.034)	0.218*** (0.051)	0.123*** (0.043)
Pseudo R-squared	0.04	0.16	0.30	0.05

Plot of Pre-trends

Figure 1.5: Parallel Trends Assumption, Weekly Starts

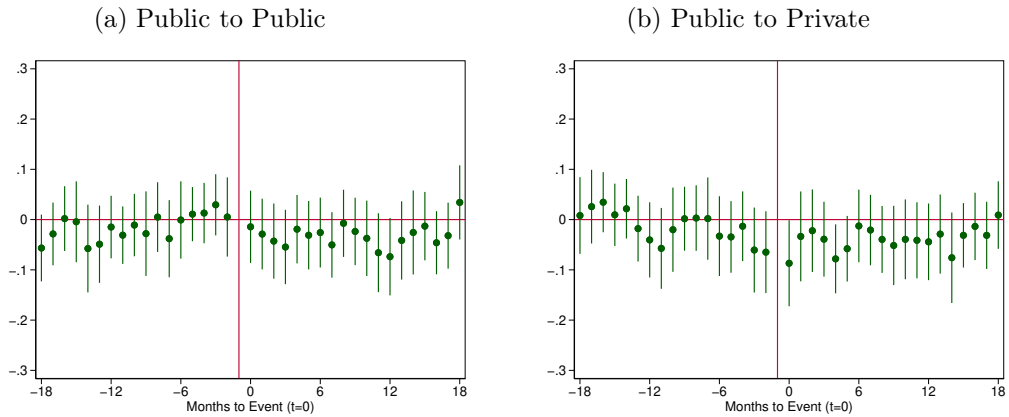


Figure 1.6: Parallel Trends Assumption, Capacity

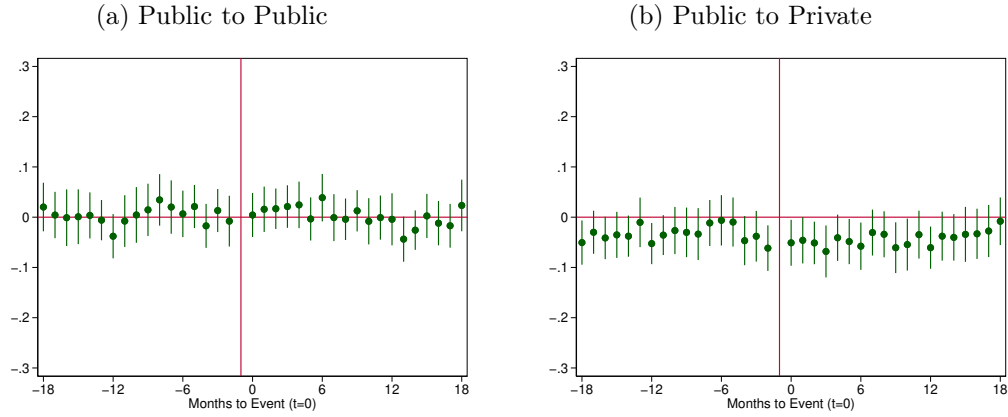
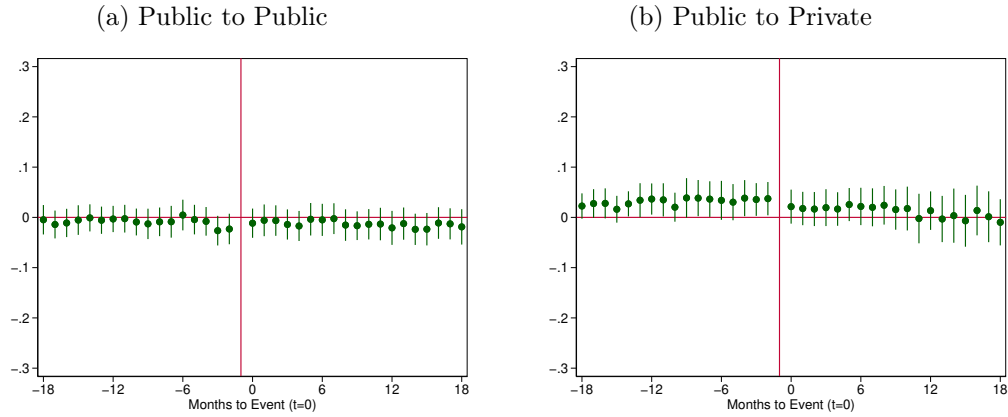


Figure 1.7: Parallel Trends Assumption, Emissions Intensity



Notes: These figures plot pre-trends.

Diff-in-Diff Goodman-Bacon critique

Goodman-Bacon (2021) decomposes the DID estimator into the weighted average of all possible two-group/two-period DID estimators in the data, where the groups are earlier treated, later treated, and never treated. The paper also shows that the coefficient can have the wrong sign if treatment induces a trend shift. Because I use never-treated, or not yet treated units as controls, the coefficients will always be of the right sign as the never/not-yet treated units will never experience a treatment-related trend-shift. Thus, the concerns about identification and interpretation in Goodman-Bacon (2021) are not issues in this setting.

Institutional Background

Balancing Load In the aggregate, electricity demand is short-term inelastic and production simply clears demand. At the plant level, production decisions are more nuanced. A plant's choice to produce is the joint decision between the plant operator and the balancing authority that controls the transmissions system and dispatches power plants within the power control areas (PCAs) in its territory.

Balancing authorities take one of two approaches to dispatch power plants: a “command and control approach”, and a “market dispatch approach.” Under a “command and control approach”, balancing authorities have their own algorithms for calling on plants to meet demand based on the authority's modeling of plants under its control. This was the traditional approach. The Energy Policy Act of 1992, however, sought to make the wholesale market for electricity more competitive by requiring the separation of transmission system owners and power marketers. The Act was codified in 1996 by the Federal Energy Regulatory Commission (FERC) orders 888 and 889, which required open-access, nondiscriminatory tariffs for wholesale electricity transmission. These acts precipitated a wave of deregulation that saw many, though not all, balancing authorities move to a “market dispatch” model in which generators would submit bids to provide capacity the day before, and the day of, the independent system operator calls upon different generators to meet demand based on its bid schedule. Generators could either meet the demand, or buy back one's allocated output at the real time price (Cicala, 2022).

Fossil Fuel Combustion Fossil fuels are formed from the fossilized, buried remains of plants and animals. Over time, layers of these remains were buried under sand, silt, and rock. Pressure and heat caused carbon and hydrogen from these materials to bond together and create fossil fuels. The chemical bonds holding carbon and hydrogen together effectively became stores of energy that are released when fossil fuels could be combusted by contact with heat and oxygen.

Fossil-fuel power plants generate electricity by combusting a primary energy source, such as coal, gas and oil, to generate heat. This heat is converted into the mechanical energy of a rotating turbine that is linked by an axle to a generator. The generator is a large magnet that rotates around coils, and in doing so, induces an alternating current in the coil - a flow of electrons, which I call electricity. During the combustion process, primarily carbon dioxide is released.

Combustion is the rapid combination of a substance with oxygen involving the production of heat and light. In the case of fossil fuel combustion, the heat generation from combustion causes elements in the fossil fuel and air to combine and form greenhouse gases. To illustrate

by example, in the case of most natural gas boilers, natural gas and oxygen in the air combusts on contact with an ignition source, such as an open flame. Natural gas contains many compounds, the largest of which is the carbon compound methane CH_4 . Nearly all the fuel carbon is converted to CO_2 in the process. High temperatures during the combustion process also cause nitrogen, and sulfur to react with oxygen and create sulfur and nitrogen oxides. (Note: Air contains nitrogen. Natural gas contains both nitrogen and sulfur, although in smaller quantities than natural as most of it must be removed before it can be transported, and sold (<https://www.eia.gov/energyexplained/natural-gas/>)). These however, while toxic to human health, are not greenhouse gases and therefore, not the object of our study. In this paper, I will use the term CO_2 emissions and emissions interchangeably.

Regulation Regulation of plant operations and ownership falls into two categories: wholesale sales and interstate electric transmission, and local sales and local electric distribution. The former is regulated by Federal Energy Regulatory Commission (FERC); the latter, by states, and sometimes, self-regulated municipalities and electric co-operatives.

Regulation of plant emissions is largely at the federal and state level.

At the federal level, the EPA regulates greenhouse gas emissions from power plants under Section 111 of the Clean Air Act. Monitoring occurs through its Clean Air Markets Division.

In January, 1993, Congress passed the Part 75 rule, which established a continuous emission monitoring (CEM) program in support of EPA's Acid Rain Program (ARP), which was instituted in 1990 under Title IV of the Clean Air Act. Part 75 requires continuous monitoring and reporting of carbon dioxide (CO_2) mass emissions (among other emissions) of electric generating units (EGUs) that burn fossil fuels such as coal, oil and natural gas and that serve a generator greater than 25 megawatts. The program is run by the EPA's Clean Air Markets Division programs.

However, few federal efforts to regulate the carbon dioxide emissions of all power plants have survived legal review. At the state level, some states have their own, or are members of market-based programs to reduce greenhouse gas emissions. For example, Massachusetts has its own allowance trading program for carbon dioxide emissions from electricity generation, with planned declines in total allowances in the future.²² It is also involved in the Regional Greenhouse Gas Initiative which is an allowance trading program involving several states.²³

²²Allowance at present, however, exceed actual emissions. See <https://www.mass.gov/doc/fact-sheet-massdep-electricity-sector-regulations/download> and <https://www.potomaceconomics.com/wp-content/uploads/2021/06/774-q4report20.pdf>

²³At the state level, twelve states have active carbon-pricing programs: California and the eleven Northeast states — Connecticut, Delaware, Maine, Maryland, Massachusetts, New Hampshire, New Jersey, New York, Rhode Island, Vermont, and Virginia — that make up the Regional Greenhouse Gas Initiative (RGGI). RGGI is the first mandatory cap-and-trade program in the United States to limit carbon dioxide emissions

Concerns that policy variations will bias our empirical results is mitigated by our empirical strategy, which implicitly controls for time variation at the state level.

Efforts to Regulate Emissions Most federal efforts to regulate carbon emissions has met legal challenge. A notable exception is the the August 3, 2015 Final Rule “Carbon Pollution Standard for New Power Plants” (CPP) which contains a rule limiting GHG emissions from new fossil fuel-fired utility boilers and from natural gas-fired stationary combustion turbines that remain in place today. More common are legal challenges. For example, in the recent 2022 Supreme Court Case *West Virginia v. EPA*, the Court ruled that the EPA does not have the authority to substantially restructure the energy industry to limit emissions, and that agency action with large economic and political significance falls under the “major questions doctrine”- that is, Congress cannot defer significant issues of national policy to an administrative authority unless such intent is clearly expressed - and therefore requires clear Congressional authorization. However, the Court still retains significant practical authority to regulate plant emissions at the individual level.

New natural gas power plants can emit no more than 1,000 pounds (lbs) of carbon dioxide per megawatt-hour (MWh) of electricity produced, which is achievable with the latest combined cycle technology. New coal power plants can emit no more than 1,400 lbs CO₂/MWh, which almost certainly requires the use of carbon capture and storage (CCS) technology (<https://www.c2es.org/document/epa-regulation-of-greenhouse-gas-emissions-from-new-power-plants/>).

Recent efforts to regulate carbon dioxide emissions from existing power plants have failed legal review. On October 23, 2015, the EPA Clean Power Plan, which established emission guidelines for states to follow in limiting carbon dioxide emissions from existing power plants, but the agency determined that it exceeds the EPA’s statutory authority under the Clean Air Act and it was repealed. On June 19, 2019, this was replaced by the Affordable Clean Energy rule (ACE), which on January 19, 2021, was vacated by the D.C. Circuit and remanded to the Environmental Protection Agency for further proceedings consistent with its opinion. On June 30, 2022 the Supreme Court ruled in *West Virginia v. EPA* that the Clean Power Plan and its enactment of the replacement ACE rule was unconstitutional. The Court ruled that the EPA does not have the authority to substantially restructure the energy industry to limit emissions, and that agency action with large economic and political significance falls under the “major questions doctrine”- that is, Congress cannot defer significant issues

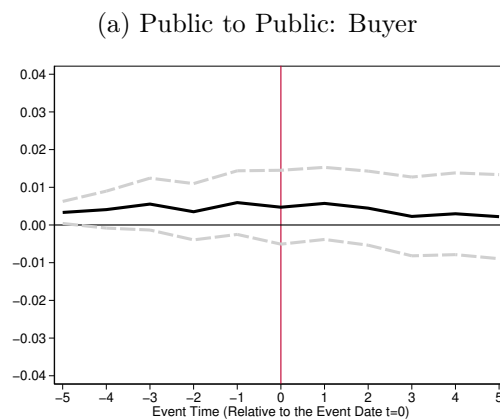
from the power sector. California’s program was the first multi-sector cap-and-trade program in North America. Massachusetts has also implemented regulations to establish an additional cap-and-trade program for its power sector that runs in parallel with RGGI but extends out to 2050. Washington state recently enacted new cap-and-invest legislation to take effect beginning in 2023.

of national policy to an administrative authority unless such intent is clearly expressed - and therefore requires clear Congressional authorization. However, the Court still retains significant practical authority to regulate plant emissions at the individual level by setting standards.

1.5.4 Event Study

Event-study of Buyer

Figure 1.8: Average Cumulative Abnormal Returns Around Plant Sale Announcements, Nameplate-Weighted



Robustness Check: Equally Weighted

Figure 1.9: Average Cumulative Abnormal Returns Around Plant Sale Announcements (Equally-Weighted): Eleven Day Window



Notes: This figure shows the average cumulative returns (CARs) around plant sale announcements by sale type: Public to Public and Public to Private, and party: Buyer and Seller. The event window is plotted on the x-axis and starts five business days before the event, and ends five business days after the event. The CARs averaged across firms are plotted on the y-axis; daily CARs are based on expected return estimates generated using the Fama-French factor plus Momentum model. Dashed lines indicate the 95% confidence interval.

Table 1.21: Table of Average Cumulative Abnormal Returns (-5,5)

		CAR (Percent)	
		Equally-weighted	Nameplate-weighted
Public to Public			
<u>Seller</u>	Mean	0.56	-0.23
	Standard Error	0.82	0.35
	Count	62	62
	Sharpe Ratio	0.09	-0.08
<u>Buyer</u>	Mean	0.22	-0.07
	Standard Error	0.57	0.18
	Count	62	62
	Sharpe Ratio	0.05	-0.05
Public to Private			
<u>Seller</u>	Mean	1.44**	0.27
	Standard Error	0.67	0.17
	Count	91	91
	Sharpe Ratio	0.23	0.17
Comparison of Two Means Test¹			
	P-value	0.41	0.20

Notes: *, **, *** denotes statistical significance at the 1%, 5%, 10% levels, respectively.

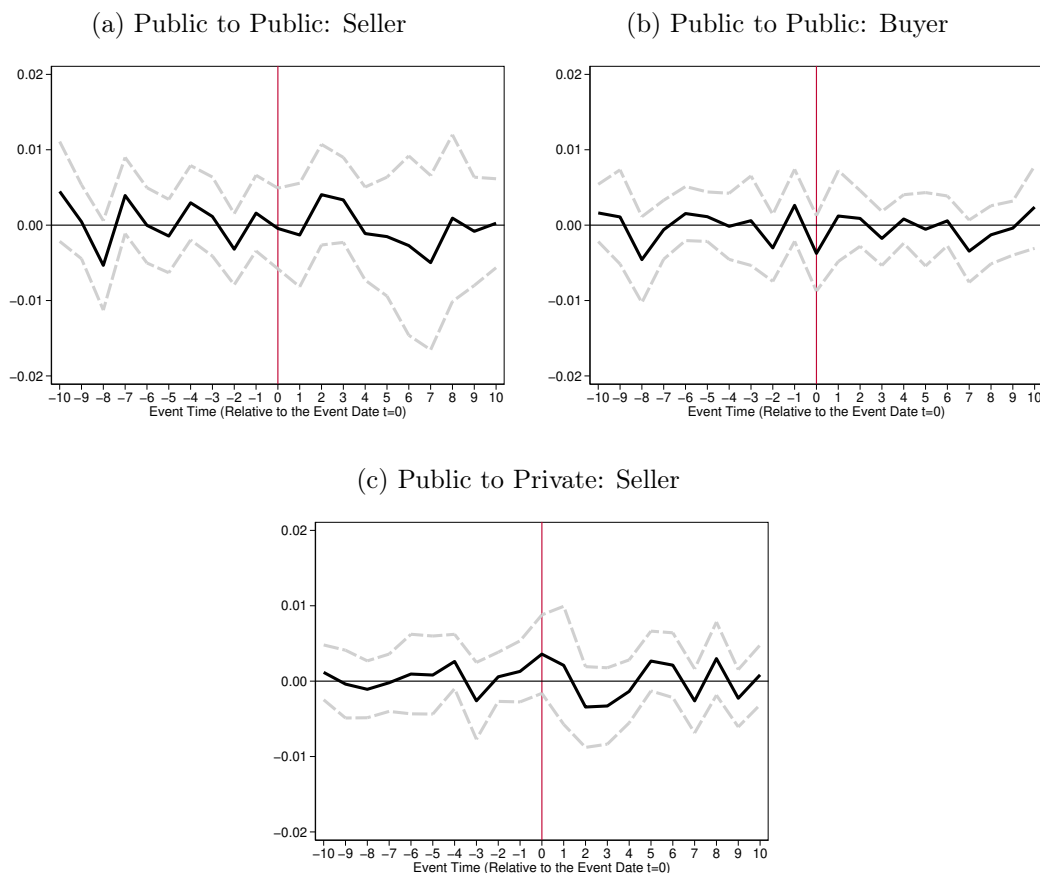
¹ Here, I test the comparison of the mean of the seller CAR in public to private transactions with the mean of the seller CAR in public to public transactions. In the setting in which means and standard deviations are unknown, we can use the two-sample t statistic:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Conservative p-values may be obtained using the $t(k)$ distribution where $k = \min(n_1 - 1, n_2 - 1)$, which is what I report.

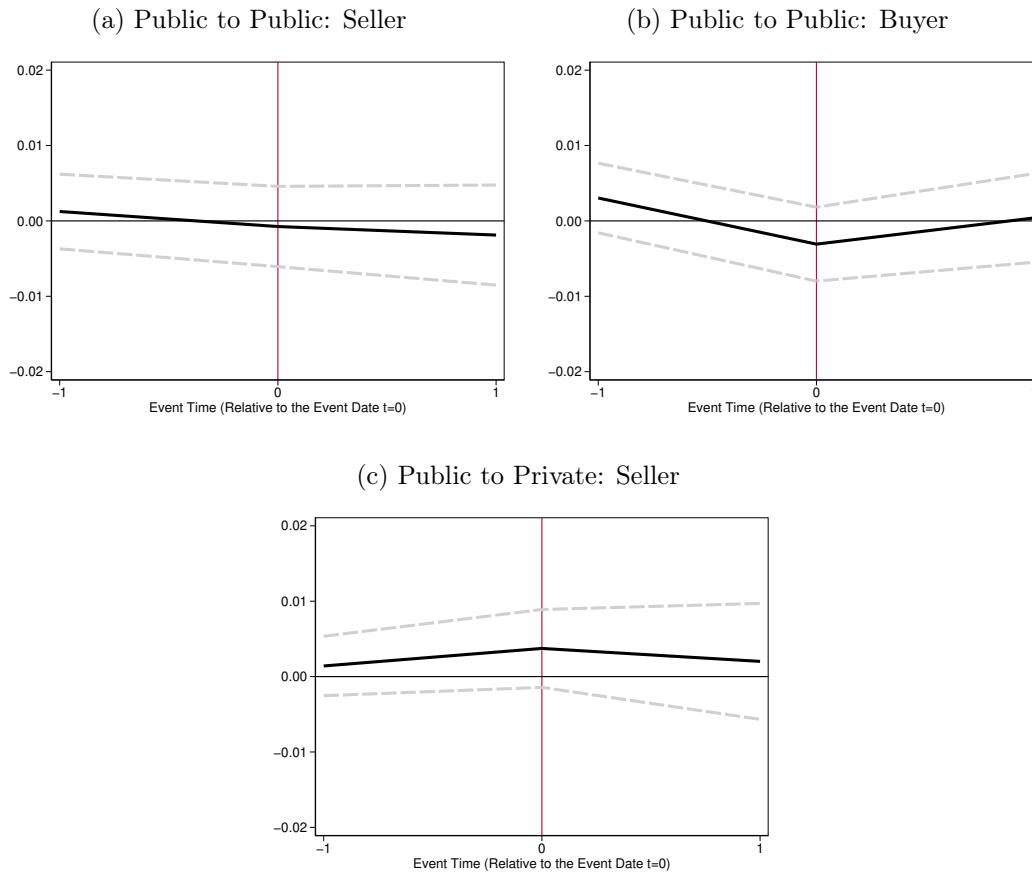
Different Window Sizes: Nameplate-weighted

Figure 1.10: Average Cumulative Abnormal Returns Around Plant Sale Announcements: 21-Day Window



Notes: This figure shows the average cumulative returns (CARs) around plant sale announcements by sale type: Public to Public and Public to Private, and party: Buyer and Seller. The event window is plotted on the x-axis and starts five business days before the event, and ends five business days after the event. The CARs averaged across firms are plotted on the y-axis; daily CARs are based on expected return estimates generated using the Fama-French factor plus Momentum model. Dashed lines indicate the 95% confidence interval.

Figure 1.11: Average Cumulative Abnormal Returns Around Plant Sale Announcements: Three Day Window



Notes: This figure shows the average cumulative returns (CARs) around plant sale announcements by sale type: Public to Public and Public to Private, and party: Buyer and Seller. The event window is plotted on the x-axis and starts five business days before the event, and ends five business days after the event. The CARs averaged across firms are plotted on the y-axis; daily CARs are based on expected return estimates generated using the Fama-French factor plus Momentum model. Dashed lines indicate the 95% confidence interval.

Chapter 2

Environmental Effects of Differential Shocks to Private Costs of Polluting

2.1 Introduction

Growing concerns over climate change have prompted the rise of environmental activist investors that acquire the shares of publicly traded firms that own “dirty”, high greenhouse gas emitting assets in an effort to pressure firms to decarbonize at the firm-level.

Publicly traded firms, henceforth referred to as simply public firms, are a natural target for these investors because their shares, unlike privately held firms, henceforth referred to as private firms, are traded actively, and anonymously on a public exchange, making them easier to acquire. If investors actively target public but not private firms, it is unclear whether their strategy will, ex-ante, be effective at reducing aggregate emissions.

Existing literature has found that the effects of a pollution tax that falls only on a subset of firms can be weakened by emissions leakages. This literature, however, keeps asset ownership fixed and does not allow for trading in assets (van Benthem et al., 2022; Fowle, 2009). This paper fills this gap by introducing trading in assets. I present a general equilibrium model of firm behavior in which firms begin with identical private costs to emitting, and identical assets that produce some final good and emissions as a fraction of inputs. Firms are price-takers in the input and final goods market. A subset of firms, which I will call “public firms”, are shocked with a higher cost to emitting. I find that this shock generates three potential equilibria: two of the equilibria lead to a null effect on aggregate emissions, and one leads to a decline.

The first equilibrium is one of no trade, which highlights that the presence of differential pressure is insufficient to generate trade. It provides one way to think about the fact that though pressure to decarbonize has increased dramatically, changes in the parent ownership of high greenhouse gas emitting facilities have been very low (Im, 2023b).

The second equilibrium is one in which shocked firms sell to unshocked firms. This occurs if the sensitivity of demand to the final goods price is sufficiently high. If shocked firms retain dirty assets, the reduction in output is accompanied by an insufficient rise in the price such that the asset's value to the shocked firm is much lower than the asset's value to the unshocked firms. This is an intuitive equilibrium, and highlights that activist behavior may affect changes in ownership but not production.

The third equilibrium is one in which unshocked firms sell to shocked firms. If shocked firms retain dirty assets, the reduction in output is accompanied by a sharp and large rise in the price such that the value of all assets under exclusive ownership by shocked firms will be larger than if they were held both by the shocked and unshocked firms or held by unshocked firms exclusively. In this way, the cost shock acts like a coordination device that induces all firms that have experienced their shock to jointly reduce their output and achieve higher profits.

The paper is organized as follows. Section 2.2 surveys the existing literature. Section 2.3 describes the model set-up. Section 2.4 concludes.

2.2 Literature Review

This paper is related to the small, but burgeoning literature on the effects of ESG strategies.

This paper is related to the theoretical literature which studies the effectiveness of ESG strategies. Broccardo et al. (2022), which studies the relative effectiveness of exit (divestment and boycott) and voice (engagement) strategies in promoting socially desirable outcomes in companies. In their model, firms choose between not greening and greening their companies and production functions are linear and capacity constrained. This model focuses on the effectiveness of any strategy which would increase costs to emitting on emissions, keeping the technology fixed. It also examines how such costs might affect asset ownership.

This paper is also related to the theoretical and empirical literature on emissions leakage. When pressure is applied to only a subset of facilities contributing to the pollution problem, empirical and theoretical studies have long recognized that this pressure may be muted or wholly ineffective if it prompts a shift in production from pressured to unpressured producers (van Benthem et al., 2022; Fowle, 2009). These studies focus on a version of emissions leakage in which ownership is fixed. This paper seeks to comment on emissions leakage that occurs through a change in ownership.

2.3 Model

In this section, I present a static, general equilibrium model of firm production and emissions in two settings: trading in assets and no trading in assets. The model provides intuition for the mechanisms by which a positive shock to the cost of emissions for public firms can have an emissions impact.

2.3.1 Framework

Firms There are two firm types: public and private firms. Each type exists in unit continuum. The two types have the same endowment of production technologies but differ in their cost of emissions. Denote private firm variables with a tilde; public firm variables without a tilde. Each firm is endowed with two types of assets—a clean asset and a dirty asset, where asset type is denoted by $f \in \{C, D\}$. The dirty asset produces emissions as a byproduct of production; the clean asset does not. There is no firm entry or asset creation. Hence, total productive assets in this industry are fixed.

Asset f takes input x_f and generates y_f of the final good. Its production function is

$$y_f \equiv \min(x_f^{1/2}, \bar{n}p_f) \quad (2.1)$$

where $\bar{n}p_f$ is the asset's capacity constraint. Emissions generated by the dirty asset are

$$e_D = x_D. \quad (2.2)$$

The emissions generating function captures the fact that the emissions level of fossil fuel-burning units is primarily determined by the quantity and carbon content of the fuel burned. The concavity of the production function captures the fact that more efficient assets have lower marginal costs than less efficient ones and thus are used first.¹

¹Fuel costs are approximately 70-80% of electricity generation. More efficient assets use less fuel to generate electricity thus have lower marginal costs and will be dispatched earlier.

I assume no fuel switching² and no opportunity to adopt technologies that will make the asset more or less clean.³ By shutting down these two channels, emissions levels are solely determined by production decisions. It will be true in equilibrium that asset sales between firms are at the unit level; there is no fractional retention of ownership.

Firms take the final goods price p given. Inputs to the clean asset have zero marginal costs (e.g. wind and solar). Inputs to the dirty asset (e.g. gas, oil, coal) are in perfectly elastic supply at exogenous cost c .

The optimization problem of firm i is as follows. Firm i chooses its input demands for its clean and dirty assets x_C^i, x_D^i to maximize the following objective function subject to its individual rationality constraint

$$\begin{aligned} x_C^{i*}, x_D^{i*} &\equiv \arg \max_{\{x_C, x_D\}} py_C^i + py_D^i - (\phi^i + c)x_D^i \\ &s.t. py_C^i + py_D^i - (\phi^i + c)x_D^i \geq 0, \end{aligned} \quad (2.3)$$

where ϕ^i denotes firm i 's cost to emitting.

Representative Consumer Aggregate demand d is linear in the price of the final good. The demand function is

$$d = \bar{n}p_C + a - bp, \quad (2.4)$$

where $a, b > 0$.⁴ There is no short selling.

I assume demand is sufficiently high so that clean assets are always operating at maximum capacities and that the dirty asset is the marginal asset. Since clean assets have zero marginal costs, this assumption is equivalent to the restriction that the equilibrium price of the final

²This is consistent with reality. As of January 2021, 18% of US electric power generating capacity could functionally switch fuels, 27% of utility scale electric generation that could potentially use multiple energy sources could functionally switch fuels; less than 20% of coal capacity could fuel switch; 19% of natural gas-fired combined-cycle applications can fuel switch; 45% of natural gas combustion turbines can fuel switch (Source: [EIA, 2022](#)). Furthermore, transforming the asset to go from burning one fuel to many fuels is so costly that the EPA does not include fuel switching as part of the “best system of emissions reduction.” For example, “although the EPA acknowledged in the GHG Abatement Measures TSD that some coal plant owners are engaging in repowering projects, the agency concluded that this kind of fuel switching will be on average more expensive than other available options, such as constructing a new natural gas combined-cycle unit. Because gas and biomass cofiring options were found to be relatively expensive when national average cost data were used, the EPA declined to include fuel switching as part of the “best system of emissions reduction” in its proposed emissions guidelines” ([NACAA, 2015](#)).

³Making assets less emitting will primarily be through production decisions. Carbon capture technology is in its infancy and has generally been implemented only at pilot scale with government support ([Howard, 2022](#)).

⁴In the electricity sector, this is a common modeling assumption. For example, [Holland et al. \(2022\)](#) assume linear demand for short-term electricity consumption of the form $d(p) = \underline{b} - bp$, where the two parameters are $b = 0.15q_0/p_0$ and $\underline{b} = 1.15q_0$ where p_0 and q_0 are the observed price and demand.

good is greater than zero

$$p^* = \frac{2a}{\frac{1}{\phi+c} + \frac{1}{\bar{\phi}+c} + 2b} > 0.$$

Emissions cost shock Initially, the final goods price is such that supply and demand are in equilibrium. That equilibrium is disrupted when public firms experience an unanticipated, positive shock Φ to their cost of emitting

$$\Phi \equiv \frac{\phi}{\bar{\phi}} > 0.$$

The shock will affect equilibrium output, emissions, and price, and may cause a desired reallocation of dirty assets between public and private firms.

Within a firm type, firms have symmetric beliefs and strategies: each firm believes that if its type finds it individually profitable to buy or sell assets, all firms of its type will do the same. Thus, there are three possible ownership outcomes post-shock: (1) both public and private firms retain ownership of their initial dirty assets; (2) private firms own all dirty assets; and (3) public firms own all dirty assets.

The relative bargaining power of public firms is fixed and parameterized by $\lambda \in (0, 1)$. After any trading has occurred, firms choose inputs demands, production, and emissions. The final goods price clears the final goods market.

2.3.2 Intermediate Results

In this section, I present intermediate results of the pre-shock equilibrium, and the equilibrium post-shock when trading is suppressed.

Pre-shock For the dirty assets, the pre-shock equilibrium output and input quantities by firm type are

$$y_D^{i*} = \frac{p^*}{2(\phi^i + c)}, \tag{2.5}$$

$$x_D^{i*} = \left[\frac{p^*}{2(\phi^i + c)} \right]^2. \tag{2.6}$$

Market clearing in the final goods market

$$y_D^* + \tilde{y}_D^* = a - bp \tag{2.7}$$

pins down the equilibrium output price

$$p^* = \frac{2a}{\left(\frac{1}{\phi+c} + \frac{1}{\tilde{\phi}+c}\right) + 2b}. \quad (2.8)$$

Notice that for larger b , i.e. higher sensitivity of demand to price, the influence of changes in production costs $\left(\frac{1}{\phi+c} + \frac{1}{\tilde{\phi}+c}\right)$ on p^* will diminish. This is a property of the fact that $f(x) = \frac{1}{x}$ is highly convex, especially near $x = 0$.

Substituting Eq. 2.6 into Eq 2.7, we see that the price is a linear function of aggregate production with sensitivity b

$$p^* = \frac{a - (y_D^* + \tilde{y}_D^*)}{b}. \quad (2.9)$$

When b is very small, a small change in aggregate production will produce a very large change in the price.

Post-shock with no asset trading If trading in assets between public and private firms is either not desired or not permitted, Eqs. 2.5, 2.6, 2.8 continue to hold, but with $\tilde{\phi}$ replaced by $\tilde{\phi}' = \Phi \times \phi$.

Post-shock with trading in assets Before characterizing the equilibrium in which there is trading in assets, I define new notation. Let ξ^* denote the private cost of emitting of the owner, in equilibrium, of the dirty asset that initially belonged to the public firm; let $\tilde{\xi}^*$ be the private cost of emitting of the owner, in equilibrium, of the dirty asset that initially belonged to the private firm. Eqs. 2.5, 2.6, 2.8 continue to hold, but $\phi, \tilde{\phi}$ is replaced by $\xi^*, \tilde{\xi}^*$, respectively.

2.3.3 Trading in Assets

When trading in assets is permitted post-shock, there exist three qualitatively distinct, possible equilibria. Which one will prevail depends on b , the sensitivity of demand to price.

Greenwashing Equilibrium The first equilibrium is the “Greenwashing Equilibrium” in which public firms sell their dirty assets to private firms. Private firms then own all dirty assets. They operate those assets more intensively than would public firms that have a higher cost of emissions; hence, aggregate production is higher, emissions are higher, and the final goods price is lower than if no sales had occurred. Note that the increased output from the units previously owned by public firms reduces the value of private firms’ existing assets compared to their value if there were no trade. However, because the market is competitive,

that effect is only indirectly taken into account through price effects in the final goods market, and the dominant effect is the private firms' relatively low cost of emitting. In the range of demand elasticities that support this equilibrium, there is a net increase in the value of dirty assets when they move from public to private ownership. The increase is so large that there are gains to trade and the equilibrium sale of assets from public firms to private firms.

What happens in comparison to the pre-shock equilibrium depends on the pre-shock emissions costs for both types of firms. An interesting case is when pre-shock the private and public cost of emitting is the same and equal to the post-shock cost to private firms of emitting. In that case production and emissions will be unchanged because private owners will operate assets as the public owner would have in the pre-shock equilibrium. Another interesting comparison is the case of public owners initially having a higher cost of emitting than private firms, but not sufficiently higher for them to have already sold the assets. Post-shock, the new, private owners will produce more with the acquired assets, and at a higher, average emissions intensity since the emissions function is convex in the production level. However, because aggregate production increases, the final goods price will decline. Consequently, private firms' existing assets will produce less and at a lower average emissions intensity. On net, the private firm will shift production to be more efficient, which will reduce aggregate emissions. This thought exercise makes the point that shocks like the one described can have highly ambiguous effects on aggregate emissions.

This possible equilibrium outcome highlights the fact that activist pressure for the divestiture of dirty asset may cause changes in ownership that can have no effect on emissions. Lemma 1 formally establishes this result in the special case in which pre-shock emissions costs are identical for both firm types.

Lemma 1. *Greenwashing Equilibrium* - *Given a positive shock to the public firm's private cost to emitting of size Φ , the public firm will sell its assets to the private firm if the demand sensitivity to price b is sufficiently high. Production and emissions of the divested assets, as well as aggregate production and emissions, will be unchanged from levels in the pre-shock equilibrium and had trading in assets been suppressed.*

Proof. See Lemma 10 in the Appendix. □

Impact Equilibrium The second possibility is an "Impact Equilibrium" in which private firms sell their dirty assets to public firms. In this equilibrium, a positive emissions cost shock has "impact" because aggregate emissions go down, but counterintuitively, dirty assets are all owned by public firms. This equilibrium only exists when demand is sufficiently inelastic. The increase in the cost of emitting causes public firms reduce output, and given

very inelastic demand, this generates a large increase in the final goods price and asset values. That output price increase, and hence the value of the assets, is greatest when public rather than private firms hold the dirty assets. Intuitively, in this equilibrium, the shock to the cost of emitting acts to commit all public firms to output reductions when they own all assets. In the range of demand elasticities that support this equilibrium, this substantially raises the final goods price and asset values when they move from private to public ownership.

The existence of this equilibrium requires that the production function be concave; it would not exist for a linear production function. Consider the setting in which the production function is linear. Each firm's marginal cost for producing using the dirty technology is equal to cost of a dirty input plus its private cost of emitting. Since the public firm has a higher cost of emitting, its marginal cost of producing using the dirty technology will be higher and will be the marginal producer if the private firm has exhausted its capacity. In a competitive market, the price will equal the marginal cost of the marginal supplier and the public firm will earn zero profit, but the private firm will earn positive profits. Here, the private firm has no incentive to acquire the asset from the public firm; if it were to do so, the final goods price would equal its marginal cost of producing which has two implications. First, its acquired asset will earn zero profits; second, its existing assets will go from earning positive profits to zero profits.

Lemma 2. *Impact Equilibrium* - *Given a positive shock to the public firm's private cost to emitting of size Φ , the private firm will sell dirty assets to the public firm if the demand sensitivity to price b is sufficiently small. Production and emissions of the divested assets, as well as aggregate production and emissions, will decline relative to levels in the pre-shock equilibrium and had trading in assets been suppressed.*

Proof. See Lemma [11](#) in the Appendix. □

No Trade Equilibrium The third possibility is a “No Trade Equilibrium.” This equilibrium exists in an intermediate range for the sensitivity of demand to price.⁵ This equilibrium exists because when public firms sell to public firms, the increased output from the units previously owned by public firms reduces the value of private firms' existing assets. This creates a wedge in the cut-off point for the Greenwashing Equilibrium and Impact Equilibrium. This is more formally stated in the Appendix. The existence of this equilibrium makes

⁵A more mathematical, mechanical explanation is that the Greenwashing Equilibrium exists when $\{t : 2\tilde{V}(\phi, \tilde{\phi}) - \tilde{V}(\phi, \tilde{\phi}) - V(\phi, \phi) > 0\}$ is non zero. The Impact Equilibrium exists when $\{t : V(\phi, \phi) - \tilde{V}(\phi, \tilde{\phi}) > 0\}$. The wedge exists because the marginal, portfolio value of a new asset to the public firm $2\tilde{V}(\phi, \tilde{\phi}) - \tilde{V}(\phi, \tilde{\phi})$ is not equal to the value of the asset to the public firm if it retains the asset $\tilde{V}(\phi, \tilde{\phi})$.

clear that increased pressure on public firms to reduce emissions is insufficient to generate trade in assets. It provides one way to contextualize my empirical findings that though the pressure to decarbonize has increased dramatically, changes in the parent ownership of high greenhouse gas emitting facilities have been very low.

Lemma 3. *No Trade Equilibrium* - *Given a positive shock to the public firm's cost to emitting of size Φ , there will be no trade in assets if the demand sensitivity to price is neither too small nor too large. The public firms reduce emissions and aggregate emissions decline relative to levels in the pre-shock equilibrium.*

Proof. See Lemma [12](#) in the Appendix. □

The emissions implications are then identical to the case in which no trade was allowed: the assets of the public firms generate less emissions, as does the system as a whole.

2.3.4 No Trading in Assets

When there is no trading in assets, for instance because it is expensive to undertake such transactions, there is no Greenwashing equilibrium. Pressure on public firms has the intended result on their assets' emissions levels: they decline. Without trading in assets, public firms must internalize the positive shock to their emissions through production and emissions decisions.

Lemma 4. *With no trading in assets, there is no greenwashing equilibrium. The public firms reduce emissions and aggregate emissions decline.*

Proof. See Appendix.

The result on aggregate emissions, however, is sensitive to the functional form of the production function assumed. One can easily generate an increase in aggregate emissions by having output reductions by public firms replaced (even if only in part) by output increases by private firms at very high emissions intensities. □

2.3.5 Discussion

In this section, I discuss the rationale and implications of key assumptions made.

No new dirty assets A critical assumption is that there is that the aggregate stock of dirty assets in this industry are fixed. This is a plausible assumption if the technology becomes illegal or its production highly taxed. For example, Green and Vallee (2023) find that size weighted, most banks active in commercial bank lending and bond underwriting are implementing coal exit policies and that bank intermediation of credit to coal companies drops significantly after a bank adopts a coal policy. In particular, the total amount of bank loans and corporate bonds issued by a firm drops when it is more exposed to bank exit policies.⁶

Relaxing this assumption and allowing for new firms with the same cost to emitting as existing private firm to enter freely creates a setting with perfect competition and zero profits. An increase to the cost of emitting by public firms will make operating dirty assets unprofitable, and those assets will be sold to private firms.

Homogeneity of assets With assets that are heterogeneous with regard to how efficiently they convert inputs to the final good and in the relation between their inputs and emissions, the effects of a shock that changes the production of public firms' assets will depend on the technological and emissions profile of these other assets. For example, if the shock causes public firms' assets to decrease production and thus, emissions, the aggregate emissions implications depend on the emissions profile of replacement generation. This point has been repeatedly made in the empirical literature about the aggregate emissions consequences of various policies that affect the production of a subset of emitting facilities (Mansur, 2007).

No trading costs This paper assumed no significant trading costs. As long as trading costs are sufficiently low, all three equilibria will continue to exist. As the cost of trade increases, the set of bs in which there will be a no-trade equilibria will expand.

Concavity of the production function In a given regional electricity market, there are generators that use different fuels. The dispatch curve is then ordered by the relative price of fuels. Within fuels, the curve is ordered by the efficiency of the generators, with the least efficient and thus, more emissions-intense generators having higher marginal costs and therefore being dispatched later. The concavity of the production function can be

⁶It is also reasonable if I am only interested in the short-term climate response, and if adding productive capacity is expensive and requires a lot of time. For example, if I were interested in the consequence of a cost shock like the one described on one to two year horizon environmental outcomes, this assumption is reasonable given that plant construction takes years (e.g. combined cycle plants take 2 years to construct, and coal plants take 3-4 years). However, for larger horizons, I would want to relax this assumption.

rationalized by assuming that marginal assets that are of a single fuel type. Infra-marginal assets are assumed to be supplying at maximum capacity and so who holds them is irrelevant to production, emissions, and emissions intensities unless the asset is shut down. I am assuming that the shock to private costs of emitting is not sufficiently high to cause a shut down.

Fuel switching By assuming homogenous technologies, the only margin along which production can become more/less emissions intensive is if production moves to a more/less efficient level. That is, I do not allow for jumps in emissions intensities through decisions like fuel switching. Suppose, however, that it is technologically feasible for assets to switch fuels. If the cleaner fuel is the more expensive fuel, if the shock is large enough to overcome fuel price differentials this will induce fuel switching by the public firm. The aggregate emissions impact will depend on the relative magnitude of the decline in emissions from this channel versus the increase in emissions from the private firm from increasing production at a higher emissions intensity.

With these caveats in mind, this simple model highlights features of a setting in which we might expect greenwashing to occur. It also makes clear what would be needed to prevent greenwashing: apply pressure on firms while blocking divestments, forcing firms to express increased costs of emitting through production and emissions decisions. There are two major caveats. The first is that if demand is very inelastic, reductions in emissions coming primarily through reductions in production will generate a near symmetric increase in production elsewhere. If that replacement production is more emissions intensive, aggregate emissions will increase on net. The second is that I did not allow firms to alter their assets. If there are differences in firm abilities to make assets more, or less emissions intensive, screening trades to find and allow only those trades that would facilitate the flow of assets towards firms that would make assets less emissions intensive would be good for the climate.

Symmetric beliefs and trading strategies Individual firms will not deviate from symmetric beliefs and trading strategies. To illustrate by example, consider the second equilibrium in which private firms sell to public firms. On the surface, it may appear that individually, public firms might deviate because the private firms will be willing to operate the asset more intensely, and thus have higher valuations for the asset. However, this assumes the private buyer believes it will operate the acquired asset in a world in which all other assets are owned by public firms who operate assets less intensely and thus, support a higher final goods price. Said another way, it assumes the private buyer doesn't think through the equilibrium implications of a single public firm deviating. This paper assumes they do. If

a private firm is offered the opportunity to buy the asset from a public firm, it will assume that all other public firms will make the same offer. Thus, if the private firm makes an offer for the asset, it will do so assuming that in equilibrium, all assets will be owned by private firms. This offer will be lower than the public firm's asking price for the asset and so the public firm will not deviate.

2.4 Conclusion

In the fight against climate change, there have been private activist attempts to align corporate interests with aggregate reductions in greenhouse gas emissions. This paper examines the issue theoretically in a simple model of firm behavior in which firms own identical assets but have firm specific costs to emitting.

It shows that there is a form of greenwashing that can exist in which shocked, public firms sell to unshocked, private firms that operate assets just as they would have been operated pre-shock. The solution in such a case would then be to impose no trade such that a shock to private costs to polluting will induce a decline in emissions.

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Appendix

2.4.1 Theory

Notation Before proceeding, I introduce simplifying notation and auxiliary functions.

Denote the state by s , which is a pair of parameters that specify the equilibrium owners of the asset originally owned by the public firm and the asset originally owned by the private firms, in that order. Owner identity is represented by the owner's private cost of emitting. Since each firm believes that if its type finds it individually profitable to sell all firms of its type will sell, there are three possible states:

$$S \equiv \{(\phi, \tilde{\phi}), (\phi, \phi), (\tilde{\phi}, \tilde{\phi})\}.$$

The first corresponds to the state in which both public and private firms own assets; the second corresponds to the state in which public firms own all assets; the third corresponds to the state in which private firms own all assets.

Denote the ratio of the public and private firms' costs by

$$k \equiv \frac{(\phi + c)}{(\tilde{\phi} + c)}.$$

Note that there is a one-to-one mapping between $\Phi = \frac{\phi}{\tilde{\phi}}$ and k and that as Φ increases, so does k .

Let $V(s), \tilde{V}(s)$ denote the value of an asset owned in state $s \in S$ by a public firm and private firm, respectively. $V(s)$ can take on the following values:

$$V(\phi, \tilde{\phi}) = \frac{1}{4k(\tilde{\phi} + c)} \left(\frac{2a}{\frac{1+k}{k}(\frac{1}{\tilde{\phi}+c}) + 2b} \right)^2, \quad (2.10)$$

$$V(\phi, \phi) = \frac{k}{4(\tilde{\phi} + c)} \left(\frac{a}{\frac{1}{\tilde{\phi}+c} + kb} \right)^2, \quad (2.11)$$

$$V(\tilde{\phi}, \tilde{\phi}) = 0. \quad (2.12)$$

$\tilde{V}(s)$ can take on the following values:

$$\tilde{V}(\phi, \tilde{\phi}) = \frac{1}{4(\tilde{\phi} + c)} \left(\frac{2a}{\frac{1+k}{k}(\frac{1}{\phi+c}) + 2b} \right)^2, \quad (2.13)$$

$$\tilde{V}(\phi, \phi) = 0, \quad (2.14)$$

$$\tilde{V}(\tilde{\phi}, \tilde{\phi}) = \frac{1}{4(\tilde{\phi} + c)} \left(\frac{a}{\frac{1}{\tilde{\phi}+c} + b} \right)^2. \quad (2.15)$$

Denote the total value function, which specifies the total value of all assets in the equilibrium state s by $W(s)$. $W(s)$ can take on the following values:

$$W(\phi, \phi) = \frac{k}{2(\tilde{\phi} + c)} \left(\frac{a}{\frac{1}{\phi+c} + kb} \right)^2, \quad (2.16)$$

$$W(\tilde{\phi}, \tilde{\phi}) = \frac{1}{2(\tilde{\phi} + c)} \left(\frac{a}{\frac{1}{\tilde{\phi}+c} + b} \right)^2, \quad (2.17)$$

$$W(\phi, \tilde{\phi}) = \frac{1}{4} \frac{1+k}{k} \frac{1}{\tilde{\phi} + c} \left(\frac{2a}{\frac{1+k}{k} \frac{1}{\phi+c} + 2b} \right)^2. \quad (2.18)$$

2.4.2 Equilibrium

Given the assumptions that the clean assets will always produce to capacity, a public firm solves the following optimization problem

$$x^* \equiv \arg \max_x py - (\phi + c)x \quad (2.19)$$

$$s.t. py - (\phi + c)x \geq 0,$$

taking p as given. First order conditions give the following equilibrium output and input demands from a public firm⁷

$$p \frac{1}{2} x^{\frac{-1}{2}} - (\phi + c) = 0$$

⁷where I have invertibility of the production function under the particular functional form I assumed. The derivative of f is $0.5x^{-0.5}$ which is strictly monotonic, and therefore injective.

$$y^* = \frac{p^*}{2(\phi + c)}, \quad (2.20)$$

$$x^* = \left[\frac{p^*}{2(\phi + c)} \right]^2. \quad (2.21)$$

By symmetry, the equilibrium output and input demands of a private firm are:

$$\tilde{y}^* = \frac{p^*}{2(\tilde{\phi} + c)}, \quad (2.22)$$

$$\tilde{x}^* = \left[\frac{p^*}{2(\tilde{\phi} + c)} \right]^2. \quad (2.23)$$

Note that because emissions scale one to one with input, a firm's equilibrium input demand is equal to the firm's equilibrium emissions. Market clearing in the final goods market pins down the equilibrium price.

Market clearing in the final goods market pins down the equilibrium final goods price:

$$\begin{aligned} y^* + \tilde{y}^* &= a - bp^* \\ \frac{p^*}{2(\phi + c)} + \frac{p^*}{2(\tilde{\phi} + c)} &= a - bp^* \\ \frac{1}{2} \left(\frac{1}{\phi + c} + \frac{1}{\tilde{\phi} + c} \right) &= \frac{a - bp^*}{p^*} \\ y^* + \tilde{y}^* &= \frac{a}{p^*} - b \end{aligned} \quad (2.24)$$

Solving for the equilibrium final goods price yields

$$p^* = \frac{2a}{\left(\frac{1}{\phi + c} + \frac{1}{\tilde{\phi} + c} \right) + 2b}. \quad (2.25)$$

Note for the future that

$$\begin{aligned} \frac{\partial p^*}{\partial \phi} &= \frac{2a \left(\frac{1}{\phi + c} \right)^2}{\left(\left(\frac{1}{\phi + c} + \frac{1}{\tilde{\phi} + c} \right) + 2b \right)^2} \\ &= p^* \frac{\frac{1}{(\phi + c)^2}}{\left(\frac{1}{\phi + c} + \frac{1}{\tilde{\phi} + c} \right) + 2b} > 0. \end{aligned}$$

2.4.3 Lemmas

Lemma 5. *The equilibrium price satisfies the following inequality*

$$p(\tilde{\phi}, \tilde{\phi}) < p(\phi, \tilde{\phi}) < p(\phi, \phi). \quad (2.26)$$

Proof. By definition

$$\begin{aligned} p(\tilde{\phi}, \tilde{\phi}) &= \frac{2a}{(1+1)\frac{1}{\tilde{\phi}+c} + 2b} \\ p(\phi, \tilde{\phi}) &= \frac{2a}{(1+\frac{1}{k})\frac{1}{\phi+c} + 2b} \\ p(\phi, \phi) &= \frac{2a}{(\frac{1}{k}+\frac{1}{k})\frac{1}{\phi+c} + 2b}. \end{aligned}$$

Since $\frac{1}{k} + \frac{1}{k} < 1 + \frac{1}{k} < 1 + 1$, I have that $p(\tilde{\phi}, \tilde{\phi}) < p(\phi, \tilde{\phi}) < p(\phi, \phi)$. □

Lemma 6. *The value functions satisfy the following inequalities*

$$V(\phi, \tilde{\phi}) < V(\phi, \phi) \quad (2.27)$$

$$\tilde{V}(\tilde{\phi}, \tilde{\phi}) < \tilde{V}(\phi, \tilde{\phi}) \quad (2.28)$$

Proof. By definition,

$$\begin{aligned} V(\phi, \phi) &= \frac{p(\phi, \phi)^2}{4(\phi + c)} \\ V(\phi, \tilde{\phi}) &= \frac{p(\phi, \tilde{\phi})^2}{4(\phi + c)} \end{aligned}$$

Since $p(\phi, \tilde{\phi}) < p(\phi, \phi)$, $V(\phi, \tilde{\phi}) < V(\phi, \phi)$.

By definition,

$$\begin{aligned} \tilde{V}(\tilde{\phi}, \tilde{\phi}) &= \frac{p(\tilde{\phi}, \tilde{\phi})^2}{4(\tilde{\phi} + c)} \\ \tilde{V}(\phi, \tilde{\phi}) &= \frac{p(\phi, \tilde{\phi})^2}{4(\tilde{\phi} + c)} \end{aligned}$$

Since $p(\tilde{\phi}, \tilde{\phi}) < p(\phi, \tilde{\phi})$, $\tilde{V}(\tilde{\phi}, \tilde{\phi}) < \tilde{V}(\phi, \tilde{\phi})$. □

Lemma 7. *If*

$$b > \frac{(k^2 - 2k) + \sqrt{4(k + k^2 - 2)(k^2 - k)}}{k^2 - k} \frac{1}{\tilde{\phi} + c}$$

then

$$\tilde{W}(\tilde{\phi}, \tilde{\phi}) - \tilde{V}(\phi, \tilde{\phi}) - V(\phi, \phi) > 0.$$

Proof. By definition

$$2\tilde{V}(\tilde{\phi}, \tilde{\phi}) - \tilde{V}(\phi, \tilde{\phi}) = \frac{1}{2(\tilde{\phi} + c)} \left(\frac{a}{\frac{1}{\tilde{\phi} + c} + b} \right)^2 - \frac{1}{4(\tilde{\phi} + c)} \left(\frac{2a}{(1 + \frac{1}{k})\frac{1}{\tilde{\phi} + c} + 2b} \right)^2.$$

This quantity is bounded below by

$$\frac{1}{2(\tilde{\phi} + c)} \left(\frac{a}{\frac{1}{\tilde{\phi} + c} + b} \right)^2 - \frac{1}{4(\tilde{\phi} + c)} \left(\frac{a}{\frac{1}{k}\frac{1}{\tilde{\phi} + c} + b} \right)^2.$$

Thus, the set of bs that satisfy the following inequality will also satisfy the inequality $V(\phi, \phi) < 2\tilde{V}(\tilde{\phi}, \tilde{\phi}) - \tilde{V}(\phi, \tilde{\phi})$

$$\begin{aligned} \frac{k}{4(\tilde{\phi} + c)} \left(\frac{a}{\frac{1}{\tilde{\phi} + c} + kb} \right)^2 &< \frac{1}{2(\tilde{\phi} + c)} \left(\frac{a}{\frac{1}{\tilde{\phi} + c} + b} \right)^2 - \frac{1}{4(\tilde{\phi} + c)} \left(\frac{a}{(\frac{1}{k})\frac{1}{\tilde{\phi} + c} + b} \right)^2 \\ k \left(\frac{1}{\frac{1}{\tilde{\phi} + c} + kb} \right)^2 &< 2 \left(\frac{1}{\frac{1}{\tilde{\phi} + c} + b} \right)^2 - k^2 \left(\frac{1}{\frac{1}{\tilde{\phi} + c} + kb} \right)^2 \\ k \left(\frac{1}{1 + kN} \right)^2 &< 2 \left(\frac{1}{1 + N} \right)^2 - k^2 \left(\frac{1}{1 + kN} \right)^2, \text{ where } b \equiv N \frac{1}{\tilde{\phi} + c} \\ 0 &< (k^2 - k)N^2 + (2k - k^2)N + (2 - (k + k^2)). \end{aligned}$$

Since $\forall k$, $(k^2 - k) > 0$ and $(2 - (k + k^2)) < 0$ and $g(x) \equiv (k^2 - k)N^2 + (2k - k^2)N + (2 - (k + k^2))$, $g(x)$ will have only one positive root, which I can bound above

$$\begin{aligned} N_{pos} &= \frac{-(2k - k^2) + \sqrt{(2k - k^2)^2 - 4(k^2 - k)(2 - (k + k^2))}}{2(k^2 - k)} \\ &< \frac{(k^2 - 2k) + \sqrt{4(k + k^2 - 2)(k^2 - k)}}{k^2 - k}. \end{aligned}$$

Since $g(x)$ faces up, this in turn implies that the inequality holds for

$$b > \frac{(k^2 - 2k) + \sqrt{4(k + k^2 - 2)(k^2 - k)}}{k^2 - k} \frac{1}{\tilde{\phi} + c}.$$

□

Lemma 8. *If and only if*

$$b < b_1,$$

where $b_1 \equiv \frac{(\sqrt{k}\frac{1+k}{k}-2)\frac{1}{\tilde{\phi}+c}}{2(k-\sqrt{k})}$, then

$$\tilde{V}(\phi, \tilde{\phi}) < V(\phi, \phi)$$

with inequality if $b = b_1$.

Proof. To see this, note that by definition

$$\begin{aligned} \tilde{V}(\phi, \tilde{\phi}) &< V(\phi, \phi) \\ \frac{1}{4(\tilde{\phi} + c)} \left(\frac{2a}{\frac{1+k}{k}(\frac{1}{\tilde{\phi}+c}) + 2b} \right)^2 &< \frac{k}{4(\tilde{\phi} + c)} \left(\frac{a}{\frac{1}{\tilde{\phi}+c} + kb} \right)^2 \\ \sqrt{4 \left(\frac{1}{\tilde{\phi} + c} + kb \right)^2} &< \sqrt{k \left(\frac{1+k}{k}(\frac{1}{\tilde{\phi} + c}) + 2b \right)^2} \\ 2 \left(\frac{1}{\tilde{\phi} + c} + kb \right) &< \sqrt{k} \left(\frac{1+k}{k}(\frac{1}{\tilde{\phi} + c}) + 2b \right) \\ 2(k - \sqrt{k})b &< (\sqrt{k}\frac{1+k}{k} - 2) \frac{1}{\tilde{\phi} + c} \\ b &< \frac{(\sqrt{k}\frac{1+k}{k} - 2) \frac{1}{\tilde{\phi} + c}}{2(k - \sqrt{k})}. \end{aligned}$$

Where the upper bound is greater than zero since $\sqrt{k}\frac{1+k}{k} > 2$ and $k - \sqrt{k} > 0, \forall k > 1$. The proof in the opposite direction is identically obtained if one replaces the less than sign with a greater than equal sign; the proof for the equality is identically obtained if one replaces the less than sign with an equality. □

2.4.4 Main Propositions

Lemma 9. *With no trading in assets, there is no Greenwashing equilibrium. The public firm reduces emissions, and aggregate emissions decline.*

Proof. Derive partials of the output. I keep things in terms of p^* , which is itself a function of the cost to emitting for the public and private firms types $(\phi, \tilde{\phi})$, to make clear where changes in the partial are arising through changes in the price.

The partial of the final goods output of the private firms with respect to the public firm's cost to emissions is

$$\begin{aligned}
\frac{\partial y^*}{\partial \phi} &= \frac{\partial}{\partial \phi} \frac{p^*}{2(\phi + c)} \\
&= \frac{2(\phi + c) \frac{\partial p^*}{\partial \phi} - 2p^*}{(2(\phi + c))^2} \\
&= \frac{\frac{\partial p^*}{\partial \phi}}{2(\phi + c)} - \frac{p^*}{2(\phi + c)^2} \\
&= \frac{p^* \left[\frac{\frac{1}{\phi+c}}{(\frac{1}{\phi+c} + \frac{1}{\tilde{\phi}+c}) + 2b} - 1 \right]}{2(\phi + c)^2} < 0,
\end{aligned}$$

since $\frac{1}{\phi+c} < (\frac{1}{\phi+c} + \frac{1}{\tilde{\phi}+c}) + 2b$.

The partial of the final goods output of the public firm with respect to its cost to emissions is:

$$\begin{aligned}
\frac{\partial \tilde{y}^*}{\partial \phi} &= \frac{\partial}{\partial \phi} \frac{p^*}{2(\tilde{\phi} + c)} \\
&= \frac{2(\tilde{\phi} + c) \frac{\partial p^*}{\partial \phi}}{(2(\tilde{\phi} + c))^2} \\
&= \frac{\frac{\partial p^*}{\partial \phi}}{2(\tilde{\phi} + c)} \\
&= \frac{p^* \frac{\frac{1}{\phi+c}}{(\frac{1}{\phi+c} + \frac{1}{\tilde{\phi}+c}) + 2b}}{2(\phi + c)(\tilde{\phi} + c)} \\
&> 0,
\end{aligned}$$

since $\frac{\partial p^*}{\partial \phi} > 0$.

Finally, the partial of the total final goods output with respect to its cost to emissions is

$$\begin{aligned}
\frac{\partial Y^*}{\partial \phi} &= \frac{\partial}{\partial \phi}(a - bp^*) \\
&= -b \frac{\partial p^*}{\partial \phi} \\
&< 0.
\end{aligned}$$

Partials for the inputs are

$$\begin{aligned}
\frac{\partial \tilde{x}^*}{\partial \phi} &= \frac{\partial}{\partial \phi}(\tilde{y}^*)^2 \\
&= 2\tilde{y}^* \frac{\partial \tilde{y}^*}{\partial \phi} \\
&= 2\tilde{y}^* \frac{p^* \frac{\frac{1}{(\phi+c)}}{(\frac{1}{\phi+c} + \frac{1}{\phi+c}) + 2b}}{2(\phi+c)(\tilde{\phi}+c)} \\
&> 0 \\
\frac{\partial x^*}{\partial \phi} &= \frac{\partial}{\partial \phi}(y^*)^2 \\
&= 2y^* \frac{\partial y^*}{\partial \phi} \\
&= 2y^* \frac{p^* \left[\frac{\frac{1}{\phi+c}}{(\frac{1}{\phi+c} + \frac{1}{\phi+c}) + 2b} - 1 \right]}{2(\phi+c)(\phi+c)} \\
&< 0 \\
\frac{\partial X^*}{\partial \phi} &= \frac{\partial \tilde{x}^*}{\partial \phi} + \frac{\partial x^*}{\partial \phi} \\
&= 2\tilde{y}^* \frac{p^* \frac{\frac{1}{(\phi+c)}}{(\frac{1}{\phi+c} + \frac{1}{\phi+c}) + 2b}}{2(\phi+c)(\tilde{\phi}+c)} + 2y^* \frac{p^* \left[\frac{\frac{1}{\phi+c}}{(\frac{1}{\phi+c} + \frac{1}{\phi+c}) + 2b} - 1 \right]}{2(\phi+c)(\phi+c)} \\
&= \frac{2y_0^* p^*}{2(\phi_0+c)^2} \left(\frac{\frac{1}{(\phi_0+c)}}{(\frac{1}{\phi_0+c} + \frac{1}{\phi_0+c}) + 2b} + \frac{\frac{1}{\phi_0+c}}{(\frac{1}{\phi_0+c} + \frac{1}{\phi_0+c}) + 2b} - 1 \right) \\
&= \frac{y_0^* p^*}{(\phi_0+c)^2} \left(\frac{\frac{1}{(\phi_0+c)}}{\frac{1}{(\phi_0+c)} + b} - 1 \right) \\
&< 0
\end{aligned}$$

The equilibrium input demands of the public firms $\frac{\partial x^*}{\partial \phi}$ and the equilibrium aggregate input demand $\frac{\partial X^*}{\partial \phi}$ are less than zero. Thus, the assets of the public firm generate less emissions, as does the system as a whole. The result on aggregate emissions however, is sensitive to the functional form of the production function assumed. One can easily generate an increase in aggregate emissions by having output reductions by public firms replaced (even if only in part) by output increases by private firms at very high emissions intensities, or in this case, at a very inefficient, flat region of the production function. $\square$

Lemma 10. *Greenwashing Equilibrium* - *Given a positive shock to the public firm's cost to emitting of size Φ , the public firm will sell its assets to the private firm if the demand sensitivity to price is sufficiently high. Production and emissions of the divested assets as well as aggregate production and emissions will be unchanged vis-à-vis the initial equilibrium.*

Proof. Let t denote the transaction price of the asset. The public firm sells if the payoff from not trading is dominated by the payoff from selling (IR Constraint)

$$V(\phi, \tilde{\phi}) < t$$

and the payoff from buying is dominated by the payoff from selling (IC constraint)

$$\begin{aligned} W(\phi, \phi) - t &< t \\ V(\phi, \phi) &< t. \end{aligned}$$

By Lemma 6, $V(\phi, \tilde{\phi}) < V(\phi, \phi)$, which implies the latter condition is more restrictive and therefore, that the lower bound of the public firm's asking price is the marginal value of an asset in the equilibrium in which the public firms own all assets:

$$\underbrace{V(\phi, \phi)}_{\text{Value of New Asset in } (\phi, \phi)} < t.$$

For trade to occur, the price must also satisfy the conditions required for the private firm to buy. The private firms buy the asset if the payoff from not trading is dominated by the payoff from buying (IR Constraint)

$$\begin{aligned} \tilde{V}(\phi, \tilde{\phi}) &< 2\tilde{V}(\tilde{\phi}, \tilde{\phi}) - t \\ t &< 2\tilde{V}(\tilde{\phi}, \tilde{\phi}) - \tilde{V}(\phi, \tilde{\phi}) \end{aligned}$$

and if the payoff from selling is dominated by the payoff from buying (IC constraint)

$$\begin{aligned} t &< 2\tilde{V}(\tilde{\phi}, \tilde{\phi}) - t \\ t &< \tilde{V}(\tilde{\phi}, \tilde{\phi}). \end{aligned}$$

Note that by Lemma 6, $2\tilde{V}(\tilde{\phi}, \tilde{\phi}) - \tilde{V}(\phi, \tilde{\phi}) - \tilde{V}(\tilde{\phi}, \tilde{\phi}) = \tilde{V}(\tilde{\phi}, \tilde{\phi}) - \tilde{V}(\phi, \tilde{\phi}) < 0$. Thus, the IR constraint is more restrictive and the upper bound of the private firms's bid is the value of an asset under the equilibrium in which it owns all assets plus the decline in the value of the asset it already owns

$$2\tilde{V}(\tilde{\phi}, \tilde{\phi}) - \tilde{V}(\phi, \tilde{\phi}) = \underbrace{\tilde{V}(\tilde{\phi}, \tilde{\phi})}_{\text{Value of New Asset in } (\tilde{\phi}, \tilde{\phi})} - \underbrace{(\tilde{V}(\phi, \tilde{\phi}) - \tilde{V}(\tilde{\phi}, \tilde{\phi}))}_{\downarrow \text{Value of Existing Assets}}.$$

Therefore, the set of prices that support this equilibrium is

$$\{t : V(\phi, \phi) < p < 2\tilde{V}(\tilde{\phi}, \tilde{\phi}) - \tilde{V}(\phi, \tilde{\phi})\}. \quad (2.29)$$

This set is non-zero if

$$2\tilde{V}(\tilde{\phi}, \tilde{\phi}) - \tilde{V}(\phi, \tilde{\phi}) - V(\phi, \phi) > 0,$$

which, by Lemma 7, is satisfied if demand sensitivity to price is sufficiently high

$$b > \frac{(k^2 - 2k) + \sqrt{4(k + k^2 - 2)(k^2 - k)}}{k^2 - k} \frac{1}{\tilde{\phi} + c}.$$

The transaction price t is pinned down by λ , which parameterizes the relative bargaining power of the public firm:

$$t = V(\phi, \phi) + \lambda(2\tilde{V}(\tilde{\phi}, \tilde{\phi}) - \tilde{V}(\phi, \tilde{\phi}) - V(\phi, \phi)).$$

Since all assets will be owned by owners with a cost of emitting equal to ϕ , production and thus, emissions will be unchanged vis-à-vis the initial equilibrium. □

Lemma 11. *Impact Equilibrium* - *Given a positive shock to the public firm's cost to emitting of size Φ , the private firm will sell its assets to the public firm if the demand sensitivity to price is sufficiently small. Production and emissions of the divested assets as well as aggregate production and emissions will decline vis-à-vis the initial equilibrium.*

Proof. Let t denote the transaction price. The private firms sells if the payoff from not trading is dominated by the payoff from selling

$$\tilde{V}(\phi, \tilde{\phi}) < t$$

and if the payoff from buying is dominated by the payoff from selling

$$\begin{aligned} 2\tilde{V}(\tilde{\phi}, \tilde{\phi}) - t &< t \\ \tilde{V}(\tilde{\phi}, \tilde{\phi}) &< t. \end{aligned}$$

By Lemma 6, $\tilde{V}(\tilde{\phi}, \tilde{\phi}) < \tilde{V}(\phi, \tilde{\phi})$. Thus, the former is more restrictive

$$\underbrace{\tilde{V}(\phi, \tilde{\phi})}_{\text{Value of Existing Asset in } (\phi, \tilde{\phi})} < t.$$

When a pressured firm experiences a positive shock to its cost to emitting and retains its asset because there is no buyer for it, aggregate production declines and the equilibrium price increases. The firms that did not experience this shock benefit from an increase in firm profits and asset valuations, since they produce at quantities at least greater than the baseline but sell at a higher price.

The public firms buy if the payoff from not trading is dominated by the payoff from buying:

$$\begin{aligned} V(\phi, \tilde{\phi}) &< 2V(\phi, \phi) - t \\ t &< 2V(\phi, \phi) - V(\phi, \tilde{\phi}) \end{aligned}$$

and if the payoff from selling is dominated by the payoff from buying:

$$\begin{aligned} t &< 2V(\phi, \phi) - t \\ t &< V(\phi, \phi). \end{aligned}$$

By Lemma 6, $V(\phi, \tilde{\phi}) < V(\phi, \phi)$. Thus, $2V(\phi, \phi) - V(\phi, \tilde{\phi}) > V(\phi, \phi)$ and the latter condition is more restrictive

$$t < \underbrace{V(\phi, \phi)}_{\text{Value of New Asset in } (\phi, \phi)}.$$

To see this note that

$$\begin{aligned} V(\phi, \phi) &< 2V(\phi, \phi) - V(\phi, \tilde{\phi}) \\ V(\phi, \tilde{\phi}) &< V(\phi, \phi). \end{aligned}$$

Therefore, the set of prices that support this equilibrium is

$$\{t : \tilde{V}(\phi, \tilde{\phi}) < p < V(\phi, \phi)\}. \quad (2.30)$$

This set is non-zero if

$$V(\phi, \phi) - \tilde{V}(\phi, \tilde{\phi}) > 0.$$

which is satisfied when the demand sensitivity to price is sufficiently small

$$b < \frac{(\sqrt{k}^{\frac{1+k}{k}} - 2) \frac{1}{\phi+c}}{2(k - \sqrt{k})},$$

by Lemma [7](#). The transaction price t is pinned down by $(1 - \lambda)$, which parameterizes the bargaining power of the private firm:

$$t = \tilde{V}(\phi, \tilde{\phi}) + (1 - \lambda)(V(\phi, \phi) - \tilde{V}(\phi, \tilde{\phi})).$$

If the private firms sell their assets to the public firms, aggregate production and emissions will decline vis-à-vis the pre-shock equilibrium. Let post-shock equilibrium variables be denoted by an apostrophe. Since assets are identical and have concave production functions, if a firm owns two assets, it will operate the assets identically. Hence,

$$\begin{aligned} Y^{*'} &= 2y^{*'} & Y^* &= 2y^* \\ X^{*'} &= 2x^{*'} & X^* &= 2x^*. \end{aligned}$$

Since $x^{*'} = (y^{*'})^2$, demonstrating that aggregate production and emissions decline vis-à-vis the pre-shock equilibrium, it is sufficient to show that

$$\begin{aligned} y^{*'} &< y^* \\ \frac{p^*(\tilde{\phi}, \tilde{\phi})}{2k(\tilde{\phi} + c)} &< \frac{\frac{a}{z+b}}{2(\tilde{\phi} + c)} \\ \frac{\frac{a}{\frac{1}{\phi+c} + kb}}{2(\tilde{\phi} + c)} &< \frac{\frac{a}{\frac{1}{\phi+c} + b}}{2(\tilde{\phi} + c)} \end{aligned}$$

which holds since $k > 1$.

□

Lemma 12. No Trade Equilibrium - *Given a positive shock to the public firm's cost to emitting of size Φ , there will be no trade in assets if the demand sensitivity to price is neither too small nor too large.*

Proof. In the no trade equilibrium, the conditions required for the two other equilibria should be jointly violated.

The condition for the Greenwashing Equilibrium is violated if

$$V(\phi, \phi) - 2\tilde{V}(\tilde{\phi}, \tilde{\phi}) + \tilde{V}(\tilde{\phi}, \phi) > 0.$$

Since $\tilde{V}(\tilde{\phi}, \tilde{\phi}) < \tilde{V}(\tilde{\phi}, \phi)$ by Lemma 6, it follows that if

$$\begin{aligned} V(\phi, \phi) - 2\tilde{V}(\tilde{\phi}, \tilde{\phi}) + \tilde{V}(\tilde{\phi}, \tilde{\phi}) &> 0 \\ V(\phi, \phi) - \tilde{V}(\tilde{\phi}, \tilde{\phi}) &> 0, \end{aligned}$$

then the condition for the Greenwashing Equilibrium will also have been violated. The set of b s that satisfy this inequality are

$$\begin{aligned} \tilde{V}(\tilde{\phi}, \tilde{\phi}) &< V(\phi, \phi) \\ \frac{1}{4(\tilde{\phi} + c)} \left(\frac{a}{\frac{1}{(\tilde{\phi} + c)} + b} \right)^2 &< \frac{k}{4(\tilde{\phi} + c)} \left(\frac{a}{\frac{1}{(\tilde{\phi} + c)} + kb} \right)^2 \\ b &< \frac{\sqrt{k} - 1}{k - \sqrt{k}} \frac{1}{\tilde{\phi} + c} \equiv b_2. \end{aligned}$$

The condition for the Impact Equilibrium is violated if

$$b > b_1 \equiv \frac{(\sqrt{k} \frac{1+k}{k} - 2) \frac{1}{\tilde{\phi} + c}}{2(k - \sqrt{k})}$$

which follows directly from the proof of the Impact Equilibrium. Note that $\forall k, b_1 < b_2$. Hence, if $b \in (b_1, b_2]$ there will be no trade.

Note that the reason the no-trade equilibrium exists is because in the greenwashing equilibrium, increased output from the units previously owned by public firms reduces the value of the private firm's existing assets.

In the Greenwashing Equilibrium,

$$\underbrace{\tilde{V}(\tilde{\phi}, \tilde{\phi})}_{\text{Value of New Asset in } (\tilde{\phi}, \tilde{\phi})} - \underbrace{V(\phi, \phi)}_{\text{Value of New Asset in } (\phi, \phi)} > \underbrace{(\tilde{V}(\phi, \tilde{\phi}) - \tilde{V}(\tilde{\phi}, \tilde{\phi}))}_{\text{Change in Value of Existing Assets}} > 0$$

In the Impact Equilibrium,

$$\underbrace{\tilde{V}(\tilde{\phi}, \tilde{\phi})}_{\text{Value of New Asset in } (\tilde{\phi}, \tilde{\phi})} - \underbrace{V(\phi, \phi)}_{\text{Value of New Asset in } (\phi, \phi)} < 0.$$

Note that the wedge in the cut-off point for the Greenwashing Equilibrium and Impact Equilibrium is $\tilde{V}(\phi, \tilde{\phi}) - \tilde{V}(\tilde{\phi}, \tilde{\phi})$, which is the change in the value of existing assets under private ownership. $\square$

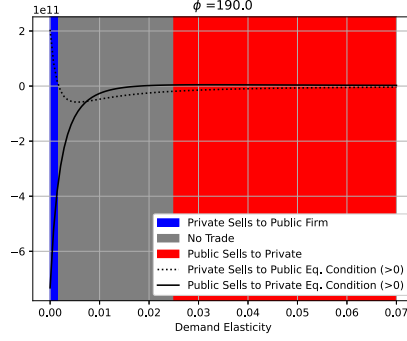
2.4.5 Equilibrium Visualization

To visualize the equilibria, I numerically solve for and present the cut-off points for a set of parameterizations (see Figure 5.1). I let the initial cost of emitting equal 7 for both shocked and unshocked firms, which corresponds to the upper range of the social cost of carbon under the Trump Administration. I examine the equilibria to shocks that make $\varphi = 190$, the EPA's 2023 social cost of carbon.

In Figure 2.1, to visualize the equilibria, I numerically solve for and present the cut-off points for a set of parameterizations. I let the initial cost of emitting equal 7 for both shocked and unshocked firms, which corresponds to the upper range of the social cost of carbon under the Trump Administration. I examine the equilibria to shocks that make $\phi = 190$, the EPA's 2023 social cost of carbon.

Figure 2.1: Equilibria

(a) EPA 2023 Shock



Notes: This figure illustrates how equilibria change as the sensitivity of demand to price b varies. To generate these figures, parameter values are defined as follows: $a = 101,200$, $b = 203$, $y^* = \tilde{y}^* = 44,000 MWh$ and $x^* = \tilde{x}^* = 1,936,000,000$.

I take the demand functions coefficients to be

$$a = 1.15Q_0$$

$$b = 0.15 \frac{Q_0}{P_0}$$

so that the elasticity of demand of 0.15. Let $Q_0 = 88,000 MWh$ and $P_0 = \$65/MWh$, which corresponds to the average load and price in the PJM Interconnection, a regional transmission organization in 2021. Then,

$$a^{Electricity} = 101200$$

$$b^{Electricity} = 203.$$

Assuming competitive generation and homogeneity, the price is the marginal cost of generation c . At the initial symmetric equilibrium, for the value of Q_0 chosen, market clearing implies $y^* = \tilde{y}^* = 44,000 MWh$ and $x^* = \tilde{x}^* = 1,936,000,000$. The derivative of the production function at this point is $\frac{1}{2}x^{-1/2} = 0.00001136363$ - that is, it is nearly flat.

Chapter 3

Long-run Real Bond Parity

3.1 Introduction

Exchange rate determination remains an active puzzle (Meese and Rogoff (1983)). The empirical literature has sought to identify significant correlates to changes in exchange rates (Ferraro, Rossi, and Rogoff (2011), Verdelhan (2018), Jiang, Krishnamurthy, Lustig (2018)). The risk-based view of exchange rate determination has helped explain the cross section of currency portfolio returns and individual exchange rate variation (Lustig, Verdelhan (2009), Verdelhan (2018)), generating explainability of monthly, bilateral exchange rate movements as high as 80%.

This paper adopts the risk-based view of exchange rates. Following in the footsteps of Lustig et al. (2019), I make four assumptions: complete markets, the satisfaction of regularity conditions for a decomposition of the stochastic discount factor into a “permanent”, martingale component and a “transitory” component related to the inverse of long run real bond prices, a common “permanent”, martingale component across currencies, and the validity of the log of ten-year, real bond, monthly holding period returns as an approximation for the log of infinite-maturity, real bond, monthly holding period returns, to derive testable implications for the sign, and magnitude of the relationship between the log of real exchange rate changes and the log of long-maturity (i.e. ten-year) real bond returns. In particular, a positive shock to the log holding period return on the long-run foreign (domestic) real bond should be associated with contemporaneous, and one-to-one depreciation (appreciation) in the log of the foreign real exchange rate; once converted into a common numeraire, log holding period returns on real long-run bonds should be equal across currency baskets. In the style of Lustig et al. (2019) who derived this relation in nominal terms, I call this the “long, real bond return parity.”

The joint hypothesis is rejected in the post-2009 sample of developing and developed market currencies; however, I document a strong relationship between changes in the log real exchange rate and real foreign, and domestic holding period bond returns in the direction of parity.

3.2 Background

Definition 1. Let Assumptions I-IV refer to the following set of assumptions:

- I. Complete markets;
- II. Regularity conditions satisfied to decompose the stochastic discount factor in a permanent, martingale component and a transitory component related to the inverse of long run real bond prices;
- III. Common "permanent" martingale component across currencies;
- IV. Log of ten-year, real bond, monthly holding period returns approximate the log of infinite-maturity, real bond, monthly holding period returns.

Proposition 1. *When the Assumptions I-IV hold,*

$$\Delta q_{t+1} = hpr_{t+1}^{*,(10)} - hpr_{t+1}^{(10)} \quad (3.1)$$

where Δq_{t+1} is the log change in the real exchange rate between the domestic and foreign currency, $hpr_{t+1}^{*,(10)}$ is the log holding period return on the foreign, ten-year real bond, and $hpr_{t+1}^{(10)}$ is the log holding period return on the domestic, ten-year real bond.

Proof. Define a single currency pair as a domestic and foreign currency. To simplify the exposition, I will denote all domestic variables without stars, and foreign variables with stars. Suppose I have the stochastic discount factor M_{t+1} with the following representation in terms of the time $t + 1$ and t pricing kernels Λ_{t+1}, Λ_t

$$M_{t+1} = \frac{\Lambda_{t+1}}{\Lambda_t}. \quad (3.2)$$

Under regularity conditions, [Alvarez and Jermann \(2005\)](#) present a decomposition of the pricing kernel Λ_t into a martingale, permanent component Λ_t^P and a transitory component Λ_t^T using the price of a k maturity long-term bond $P_t^{(k)}$

$$\Lambda_t = \Lambda_t^P \Lambda_t^T, \text{ where } \Lambda_t^T = \lim_{k \rightarrow \infty} \frac{\delta^{t+k}}{P_t^{(k)}}. \quad (3.3)$$

The holding period return from time t to $t + 1$ on the infinite maturity bond, which I denote by $HPR_{t+1}^{(\infty)}$, is the ratio of the transitory components at time t and $t + 1$

$$HPR_{t+1}^{(\infty)} = \lim_{k \rightarrow \infty} HPR_{t+1}^{(k)} = \lim_{k \rightarrow \infty} \frac{P_{t+1}^{(k-1)}}{P_t^{(k)}} = \frac{\Lambda_t^T}{\Lambda_{t+1}^T}. \quad (3.4)$$

The corresponding log holding period return, which I denote by $hpr_{t+1}^{i,(\infty)}$ is

$$hpr_{t+1}^{i,(\infty)} = -\lambda_{t+1}^{i,T} + \lambda_t^{i,T}. \quad (3.5)$$

If markets are complete, there exists a unique, real stochastic discount factor. This factor generates a set of unique, currency-specific factors that differ in the real numeraire that they reference, M_{t+1}, M_{t+1}^* . The Alvarez and Jermann representation of M_{t+1} gives us

$$M_{t+1} = \frac{\Lambda_{t+1}}{\Lambda_t} = \frac{\Lambda_{t+1}^P \Lambda_{t+1}^T}{\Lambda_t^P \Lambda_t^T}, \text{ where } \Lambda_t^T = \lim_{k \rightarrow \infty} \frac{\delta^{t+k}}{P_t^{(k)}}. \quad (3.6)$$

Let Q_{t+1} denote the time $t + 1$ real exchange rate, defined as units of the foreign real goods basket per unit of the domestic real goods basket

$$Q_{t+1} = \frac{\text{Units of foreign real goods baskets}_{t+1}}{\text{Domestic real goods basket}}. \quad (3.7)$$

An accounting identity gives us

$$M_{t+1} = M_{t+1}^* \frac{Q_{t+1}}{Q_t}. \quad (3.8)$$

This is from the Euler equations of the stand-in domestic and foreign investors: for an asset return R^* expressed in foreign currency basket terms $\mathbb{E}_t[M_{t+1} R_{t+1}^* \frac{Q_t}{Q_{t+1}}] = 1$ and $\mathbb{E}_t[M_{t+1}^* R_{t+1}^*] = 1$. State-by-state equality is an accounting identity that holds under no arbitrage. In logs, this corresponds to

$$\Delta q_{t+1} = m_{t+1} - m_{t+1}^*. \quad (3.9)$$

The AJ decomposition gives us

$$\Delta q_{t+1} = [(\lambda_{t+1}^T - \lambda_t^T) + (\lambda_{t+1}^P - \lambda_t^P)] - [(\lambda_{t+1}^{T*} - \lambda_t^{T*}) + (\lambda_{t+1}^{P*} - \lambda_t^{P*})]. \quad (3.10)$$

Lustig et al. (2019) show that average excess returns on the carry trade using long-run bonds and with a three-month horizon, is zero, suggesting that the entropies of the permanent

components are equal.¹ If currency SDFs share a common permanent (martingale) $\lambda_t^{P*} = \lambda_t^P$, this restriction is satisfied and

$$\Delta q_{t+1} = -(\lambda_{t+1}^{T*} - \lambda_t^{T*}) + (\lambda_{t+1}^T - \lambda_t^T) \quad (3.11)$$

$$= hpr_{t+1}^{*,(\infty)} - hpr_{t+1}^{(\infty)}. \quad (3.12)$$

□

This is a statement about realizations not expectations. This is not a novel derivation. Lustig et al. (2019) are the first to derive it in nominal terms in their appendix. This paper rewrote it in real terms, and will test it.

Let t index time and i the foreign currency, this directly motivates a regression of the form

$$\Delta q_{t+1} = \alpha + \beta_1 hpr_{t+1}^{*,(\infty)} + \beta_2 hpr_{t+1}^{(\infty)} + \epsilon_t \quad (3.13)$$

where Δq_{t+1} is the log change in the real exchange rate, $hpr_{t+1}^{*,(\infty)}$ is the log holding period return on the foreign, infinite maturity real bond, and $hpr_{t+1}^{(\infty)}$ is the log holding period return on the domestic, infinite maturity real bond.

However, I face a practical problem. There are no infinite maturity real, risk-free bonds. Therefore, I assume that the log of ten-year bond returns approximate the log of infinite bond returns; that is, that

$$\Delta q_{t+1} = \alpha + \beta_1 hpr_{t+1}^{*,(10)} + \beta_2 hpr_{t+1}^{(10)} + \epsilon_t \quad (3.14)$$

is closely approximated by

$$\Delta q_{t+1} = \alpha + \beta_1 hpr_{t+1}^{*,(10)} + \beta_2 hpr_{t+1}^{(10)} + \epsilon_t \quad (3.15)$$

where $hpr_{t+1}^{*,(10)}$ is the log holding period return on the foreign, ten-year real bond, and $hpr_{t+1}^{(10)}$ is the log holding period return on the domestic, ten-year real bond.

3.3 Data

In this section, I describe the notation and the data.

¹Their result is in nominal terms and on currency portfolios. They argue that the inflation characteristics of portfolios suggest the result holds also in real terms. I find suggestive evidence that this is true when I look at the individual currencies in the sample. Interested readers may find the analysis in the Appendix.

3.3.1 Notation

This section describes notation for exchange rates, bond prices, and currency and bond returns.

Currency and national price levels

Let S_{t+1} denote the time $t + 1$ nominal exchange rate, defined as units of foreign currency per unit of the domestic currency

$$S_{t+1} = \frac{\text{Units of foreign currency}_{t+1}}{\text{Unit of domestic currency}}. \quad (3.16)$$

Let Z_{t+1} denote the time $t + 1$ domestic national price level, defined as the domestic nominal price of one unit of the domestic real goods basket

$$Z_{t+1} = \frac{\text{Units of domestic currency}_{t+1}}{\text{Domestic real goods basket}}. \quad (3.17)$$

Let Π_{t+1} denote gross inflation from time t to time $t + 1$ in the domestic currency

$$\Pi_{t+1} = \frac{Z_{t+1}}{Z_t}. \quad (3.18)$$

National price levels are vis-a-vis currency specific real goods baskets; this is to reflect cross-country differences in what governments consider a basket of real goods when reporting national price levels.

I now define other objects defined vis-a-vis these variables.

Let Q_{t+1} denote the time $t + 1$ real exchange rate, defined as units of the foreign real goods basket per unit of the domestic real goods basket

$$Q_{t+1} = \frac{\text{Units of foreign real goods baskets}_{t+1}}{\text{Domestic real goods basket}} = S_{t+1} \frac{Z_{t+1}}{Z_{t+1}^*}. \quad (3.19)$$

Let Π_{t+1} denote gross inflation from time t to time $t + 1$ in the domestic currency

$$\Pi_{t+1} = \frac{Z_{t+1}}{Z_t}, \quad (3.20)$$

and Π_{t+1}^* denote gross inflation from time t to time $t + 1$ in the foreign currency

$$\Pi_{t+1}^* = \frac{Z_{t+1}^*}{Z_t^*}. \quad (3.21)$$

Following the convention of denoting by the lowercase the logs of variables, let Δq_{t+1} denote the log change in real exchange rates from time t to time $t + 1$

$$\Delta q_{t+1} = \Delta s_{t+1} + \pi_{t+1} - \pi_{t+1}^*. \quad (3.22)$$

Bonds

Let $P_t^{(N)}$ denote the zero-coupon, N -maturity, risk-free real bond which pays a unit of the real basket at maturity.

Let $HPR_{t+1}^{(N)}$ denote the gross, holding period return on the N -maturity zero coupon real bond from time t to time $t + 1$

$$HPR_{t+1}^{(N)} = \frac{P_{t+1}^{(N-1)}}{P_t^{(N)}} \quad (3.23)$$

and $hpr_{t+1}^{(N)}$ denote its log counterpart

$$hpr_{t+1}^{(N)} = \log(HPR_{t+1}^{(N)}). \quad (3.24)$$

For example, the log, monthly holding period return on a ten-year bond, which I denote by $hpr_{t+1}^{(10)}$, is

$$hpr_{t+1}^{(10)} = \log(HPR_{t+1}^{(10)}). \quad (3.25)$$

$$= \log\left(\frac{P_{t+1}^{(9\frac{11}{12})}}{P_t^{(10)}}\right). \quad (3.26)$$

To compute holding period returns, I need zero-coupon real bonds prices at the 10, $9\frac{11}{12}$ -year maturities.

Let M_{t+N} be the nominal stochastic discount factor that prices $t + N$ cash flows in domestic currency. Consider a zero-coupon, N -maturity risk-free nominal bond which pays a unit of currency at maturity. Its price is $B_t^{(N)}$ and its zero rate is $y_t^{(N)}$

$$B_t^{(N)} = E_t[M_{t+N}] \quad (3.27)$$

$$\equiv \frac{1}{(1 + y_t^{(N)})^N}. \quad (3.28)$$

Consider a zero-coupon, N -maturity inflation swap. At maturity, the fixed leg investor will pay the fixed rate $i_t^{(N)}$ on the notional; the floating leg investor will pay the realized change in the referenced inflation index series $\frac{Z_{t+N}}{Z_t}$. Inflation swaps are priced at par: the fixed leg is such that the fixed and floating legs have the same value, and that the contract value is zero. This is expressed as

$$0 = E_t[M_{t+N} \times (\frac{Z_{t+N}}{Z_t} - (1 + i_t^{(N)})^N)]. \quad (3.29)$$

Consider a zero-coupon, N -maturity, risk-free real bond which pays a unit of the real basket at maturity. Its price is $P_t^{(N)}$.

Using the previous expression, I can express $P_t^{(N)}$, which is often not directly observable, as a function of $y_t^{(N)}$, $i_t^{(N)}$, which I often do observe directly:

$$0 = E_t[M_{t+N} \times (\frac{Z_{t+N}}{Z_t} - (1 + i_t^{(N)})^N)] \quad (3.30)$$

$$E_t[M_{t+N} \frac{Z_{t+N}}{Z_t}] = E_t[M_{t+N} \times (1 + i_t^{(N)})^N] \quad (3.31)$$

$$E_t[M_{t+N} \frac{Z_{t+N}}{Z_t}] = E_t[M_{t+N}](1 + i_t^{(N)})^N \quad (3.32)$$

$$P_t^{(N)} = E_t[M_{t+N}](1 + i_t^{(N)})^N, \text{ since } P_t^{(N)} \equiv E_t[M_{t+N} \frac{Z_{t+N}}{Z_t}] \quad (3.33)$$

$$P_t^{(N)} = \left[\frac{(1 + i_t^{(N)})}{(1 + y_t^{(N)})} \right]^N \quad (3.34)$$

3.3.2 Data Sources

I turn to the paper's data. I first describe the dataset, and then discuss the data selection process and features of the data relevant to the interpretation of the results.

Data sources

The data is a panel of monthly national price levels and daily closing prices for zero-coupon nominal overnight index swap (OIS) and zero-coupon inflation swaps, nominal and real bonds, nominal exchange rates, and nominal forward exchange rates for eight currencies. Six of the eight are from developed market economies: the Japanese yen, euro, Swedish krona, British pound, U.S. dollar, Australian dollar; the remaining two are from developing market economies: South African rand, and Brazilian real.²

²I listed the currencies in order of the average level of their three-month, nominal interest rates. I report results in this order as well. This was motivated by the fact that the failure of UIP is a key stylized fact, and so currencies are often thought of in terms of their nominal short interest rate levels.

All data are obtained from Bloomberg at the daily frequency for the period from January 01, 2009 to December 31, 2020. Data are aggregated to the monthly frequency, which corresponds to the most common release frequency of national price level series.

My dataset is constrained by data availability as few currencies have developed markets for long-maturity inflation swaps, real bonds, and overnight index swaps.³

Zero-coupon inflation and OIS rates, and zero-coupon nominal and government bond rates were obtained at the three-month, and one, two, five, nine and ten year maturities. With the exception of inflation swap rates, rates are rates derived by Bloomberg from traded prices.

In an OIS swap, a fixed rate of interest decided at contract birth is exchanged for a floating rate of interest over the swap’s lifetime. The swap rate quote is the rate on the fixed leg of the swap.⁴ Table 3.3.2 lists the series obtained, and their source.

Table 3.1: Unbalanced sample data sources and coverage

Data series	Source
Zero-coupon inflation swap rates	Bloomberg
Zero-coupon OIS rates	Bloomberg
Zero-coupon nominal and inflation-indexed government bond rates	Bloomberg
Nominal exchange rates	Bloomberg
Nominal forward exchange rates	Bloomberg, author’s calculations
National price levels	Bloomberg

Notes: This table describes the sources and coverage of the data. The data covers a small, heterogeneous panel of developed and developing market currencies: Japanese yen, euro, Swedish krona, British pound, U.S. dollar, Australian dollar, South African rand, and Brazilian real. All data are obtained at the daily frequency for the period from January 01, 2009 to December 31, 2020. Data were aggregated to the monthly frequency as this corresponds to the most common release frequency of national price level series. Nominal forward exchange rates for maturities longer than one year was constructed as in Du et al. (2018).

In Table 3.3.2, I report summary statistics for the main data series by currency. Panel A reports the mean, standard error, and standard deviation of the ten-year nominal rate $y_t^{*,(10)}$, the ten-year real rate $r_t^{*,(10)}$, the annual year-over-year inflation rate π_t , and the ten-year

³While inflation indexed bonds have existed before 2009, limited issuance across the maturity structure and embedded options make it difficult to reliably bootstrap the real bond curve for the currency set. Long-maturity inflation swap data exist for 2004 onwards, but long-maturity OIS data do not exist until 2009. The choice of OIS as the benchmark is motivated by the TIPS-Treasury puzzle, which indicates “mispricing” in real and/or nominal government cash bond markets in the U.S. (Fleming et al. (2014)), Japan, U.K., Germany (Kita and Tortice, 2019).

⁴Pricing errors, if they mimic classical measurement error, should minimally affect the cross sectional empirics if the variance of the error is small and the sample large enough; for the time series empirics, they will bias coefficients downwards towards zero.

inflation compensation rate $i_t^{*,(10)}$. Panel B reports the same statistics for the one-year real rate $r_t^{*,(1)}$, and the slope of the real rate curve which is the difference between the ten-year and one-year real rates $r_t^{*,(10)} - r_t^{*,(1)}$.

The table paints a picture of a currency sample with appreciable heterogeneity along the recorded dimensions. The currencies with the lowest and highest average nominal rates at the ten-year maturity are, respectively, the Japanese yen at 0.62% and the Brazilian real at 10.4%, a difference of nearly ten percentage points. The range of average real rates is more narrow, but also appreciable. The currencies with lowest and highest real rates at the ten-year maturity is, respectively, the British pound at -1.12% and the Brazilian real 5.11%. The sharply negative real rate for the British pound, which is below even that for the Japanese yen, underscores the unprecedented nature of the sample's low nominal and real interest rate environment in the context of recorded history.⁵ This warrants cautious interpretation and extrapolation of the empirical results.

On the inflation front, developed market currencies tend to have lower and less variable inflation than developing market currencies. Notably, however, the standard deviation of inflation for the British pound is 1.27%, which is similar in magnitude to the corresponding number for the South African rand. The average level of inflation swap rates generally increase with the level of inflation. The standard deviation on realized annual inflation is larger than the standard deviation on ten-year inflation swaps, which is suggestive of shocks in the inflation process that net out over a ten-year horizon. I emphasize that inflation swap rates are objects that are related to but not equal to either the inflation risk premia, or the holding period return on inflation swaps, the latter of which has a direct relation to the holding period returns on real bonds.

Turning to the yield curve, one-year real yields are negative for all developed market currencies, except the Australian dollar and there is no clear relationship between the level of one-year real yields and the slope of the real yield curve. One possible reason is that statistical relation between these two objects is more obscure when levels of real yields are so low and close across currencies. One-year real yields are positive for the two developing market currencies and are higher the steeper the slope of the real yield curve.

⁵The United Kingdom's decision to leave the European Union in June 2016 precipitated a sharp exchange rate 10% currency depreciation and a concomitant positive shock to inflation.

Table 3.2: Summary statistics for the cross section

		Currency							
(in percentage points)		British pound	Japanese yen	euro	Swedish krona	U.S. dollar	Australian dollar	South African rand	Brazilian real
Panel A: Summary statistics									
$y_t^{*,(10)}$	Mean	2.13	0.62	1.52	1.88	2.41	3.59	8.17	10.40
	s.e.	[0.09]	[0.04]	[0.10]	[0.09]	[0.07]	[0.13]	[0.05]	[0.24]
	s.d.	1.09	0.48	1.19	1.08	0.81	1.61	0.55	2.46
$r_t^{*,(10)}$	Mean	-1.12	0.32	-0.09	-0.14	0.16	1.09	1.58	5.11
	s.e.	[0.09]	[0.07]	[0.07]	[0.09]	[0.05]	[0.10]	[0.06]	[0.12]
	s.d.	1.09	0.83	0.79	1.02	0.63	1.20	0.66	1.22
π_t	Mean	2.87	0.40	1.15	1.04	1.70	1.99	4.87	6.23
	s.e.	[0.11]	[0.09]	[0.08]	[0.09]	[0.08]	[0.07]	[0.10]	[0.44]
	s.d.	1.27	0.99	0.94	0.98	0.87	0.80	1.13	4.14
$i_t^{*,(10)}$	Mean	3.29	0.31	1.61	2.02	2.25	2.47	6.50	5.01
	s.e.	[0.02]	[0.05]	[0.04]	[0.02]	[0.03]	[0.04]	[0.06]	[0.12]
	s.d.	0.22	0.56	0.44	0.27	0.34	0.45	0.73	1.21
Panel B: Yield curve characteristics									
$r_t^{*,(1)}$	Mean	-1.93	-0.04	-0.70	-0.72	-0.48	0.42	0.85	3.65
	s.e.	[0.08]	[0.10]	[0.05]	[0.06]	[0.08]	[0.08]	[0.09]	[0.20]
	s.d.	0.91	1.14	0.64	0.67	0.92	0.96	1.03	2.06
$r_t^{*,(10)} - r_t^{*,(1)}$	Mean	0.81	0.36	0.61	0.58	0.64	0.67	0.72	1.46
	s.e.	[0.10]	[0.05]	[0.05]	[0.06]	[0.08]	[0.04]	[0.05]	[0.12]
	s.d.	1.21	0.61	0.63	0.76	0.91	0.51	0.63	1.16

Notes: This table displays the mean, standard error of the mean, the standard deviation of the ten-year nominal rate $y_t^{*,(10)}$, the ten-year real rate $r_t^{*,(10)}$, the annual inflation rate π_t , and the ten-year inflation compensation rate $i_t^{*,(10)}$, the one-year real rate based $r_t^{*,(1)}$, and the slope of the real rate curve which is the difference between the ten-year and one-year real rates $r_t^{*,(10)} - r_t^{*,(1)}$.

While I make the best use of available data, I pre-emptively emphasize that features of the dataset warrant cautious interpretation of the results. First, the dataset is limited in size. The dataset begin in January 2009, and the first return observation is for February 2009. Currencies have monthly observations that cover anywhere from eight to twelve years, which translates to 101 to 143 observations per currency, tenor pair. Observations are all in the post-crisis period during which I have seen historically unprecedented negative rates and central bank intervention in the long maturity bond market. This suggests the results should be cautiously interpreted within the context of the sample period they are based on. Furthermore, I have a small, and heterogeneous cross section, which restricts the analysis to individual currencies, as opposed to currency portfolios in which I might be able to appeal to diversification to isolate currency factors.

Data Sources

To construct the ten-year holding period returns on real bonds, I require zero-coupon real bond prices/rates. I constructed these using zero coupon, risk-free nominal OIS rates and zero-coupon inflation swap rates, except for Brazilian real for which I used the implied zero-coupon inflation swap rate from real bond prices.

By the law of one price, the price of the N -maturity, zero coupon domestic risk-free real bond $P_t^{(N)}$ can be expressed as a function of the corresponding zero-coupon inflation swap $i_t^{(N)}$ and nominal bond rate $y_t^{(N)}$

$$P_t^{(N)} = \frac{(1 + i_t^{(N)})^N}{(1 + y_t^{(N)})^N}. \quad (3.35)$$

One concern is that this relation may not strictly hold, and therefore, that the real bond price cannot be replicated in this way. In the United States, Fleckstein, Longstaff, and Lustig (2009) find that sovereign bonds are overpriced relative to their inflation indexed counterparts TIPS, by a weighted average of \$2.92, but with significant variation, and a maximum in their sample of nearly \$10. They provide suggestive evidence that this is not due to mispricing in the inflation swaps market: in the corporate bond market, nominal and inflation indexed bond mispricing is minimal, with an average of 1-7 basis points, with standard errors of 60-70 basis points. Even though they do not explicitly account for options, accounting for them would go in the opposite direction—nominal sovereign bonds would be even more overpriced. Kita and Tortorice (2019) document mispricing of similar size and the same direction in the sovereign bond markets for Japan and Germany. Whether the relative mispricing exists because of distortions in the price level of Treasuries, TIPS, or both from

their levels in the envisioned, background no arbitrage world is unclear.

Zero coupon nominal rates As the zero coupon nominal rates I use the discount rates implied by overnight index swaps (OIS). An OIS is a swap with floating payments based on a reference rate for unsecured overnight funding and paid at fixed intervals.⁶ Over the lifetime of a contract, counterparties post collateral to keep the net value of the swap contract at zero. In early 2000, the International Swaps and Derivatives Association (ISDA), a global trade organization of participants in the market for over the counter derivatives, conducted a survey that found more than 65% of plain vanilla derivatives, especially interest rate swaps, were collateralized according to the legal standard for collateralization set out in the Credit Support Annex (CSA), a legal document of recommendations in ISDA’s master agreement which regulates collateral for derivatives transactions; moreover, they were bilateral in that both counterparties, not just the lower rated counterparty posted collateral (Johannes and Sundareshan (2006)). Thus, the OIS is only exposed to overnight credit risk and is broadly considered “risk free.”⁷ Because the OIS contract is approximately, not completely, risk free, the rate itself is influenced by the credit risk of the refreshed panel of participants in unsecured overnight funding markets; however, this effect of credit risk should be limited by the fact that the risk is for the portion of the contract that may be undercollateralized because of overnight moves in the market.⁸ Hull and White (2015) argue that OIS is the best proxy for the risk-free rate; furthermore, post-crisis, practitioners have switched from using the LIBOR zero curve to the OIS zero curve for discounting. The use of OIS constrains the sample size as data on long maturity swaps are only available on Bloomberg after 2009.

In the developed markets, one may be concerned that the “risk-free approximation” error is non-trivial. To alleviate this concern, for the South African rand and the Brazilian real, I construct synthetic OIS rates equal to the OIS rates that would hold if the domestic OIS rate was “risk-free” and covered interest parity held. Let $y_t^{(N)}$ denote the time t , maturity N nominal, domestic OIS rate. Let $\rho_t^{(N)}$ denote the time t , maturity N forward premium for the

⁶The reference rates are in the Appendix.

⁷Arora, Gandhi, and Longstaff (2012) study the effect of counterparty credit risk on the pricing of one type of swaps, credit default swaps. They show that differences in the credit risk of dealers selling credit protection have a very small effect on the pricing of CDS contracts; in particular, an increase in the credit spread of a dealer of about 645 basis points maps into only a one-basis-point decline in the price of credit protection (the value of the credit protection being sold is diminished). Their sample, however is based on a set of fourteen well known dealers in the U.S. and U.K. who may not be as active in the OIS markets of emerging market economies. Furthermore, identification is based on variation among private dealers; the paper does not quantify the fixed effect arising from the credit risk of private dealers as a whole.

⁸Measuring the price of credit risk of the refreshed interbank panel, and thus, the “risk-free approximation” error, however, requires the elusive risk-free benchmark.

foreign currency vis-a-vis the domestic currency; a positive premium implies a risk-neutral expected depreciation of the domestic currency. Covered interest rate parity implies that the time t , maturity N , nominal foreign OIS rate, which I denote by $y_t^{*,CIP,(N)}$ is

$$y_t^{*,CIP,(N)} = y_t^{(N)} - \rho. \quad (3.36)$$

Du, Tepper, Verdelhan document CIP deviations in interbank lending markets that cannot be explained away by credit risk or transaction costs for G10 currencies. They argue that wide deviations on contracts that are on banks' balance sheets at quarter end suggest regulatory capital costs are driving the wedge at quarter end. This would affect developing market currencies as well. The authors estimate that from 2010 to 2016, the average, annualized basis for currencies in their sample ranged from -60 to 30 bps at the five-year horizon. For context, the level of LIBOR ranged from around 25 bps to 400 bps. They also document heterogeneity across currencies and time, which suggests we would not be able to correct our synthetic South African rand, and Brazilian real OIS rates for these deviations by simple subtraction of a level factor from these rates.

Zero coupon real bond rates To back out zero-coupon real bond rates from market data, I privilege the use of inflation swap prices, over traded real bond prices. The reason is two fold. First, while inflation indexed bonds have existed before 2009, limited issuance across the maturity structure makes it difficult to bootstrap an entire curve of zero coupon rates. In contrast, inflation swap markets have quotes at multiple and evenly spaced maturities. Second, many inflation indexed bonds have embedded options that we would have to price and separate out. For example, U.S. real bonds, formally referred to as U.S. Treasury Inflation-Indexed Securities and informally as U.S. Treasury Inflation Protected Securities (TIPS), have deflation floors to protect the principal value in inflation, whereas comparable U.K. real bonds, formally referred to as U.K. Index-Linked Gilts, do not. This is in contrast to inflation swaps for which the floating leg does not have embedded options.

For all currencies, except for the Brazilian real, I used zero-coupon inflation swaps to compute the zero-coupon real bond rate. In the case of the Brazilian real, I used zero rates published on Bloomberg for Brazilian real bonds; because these bonds do not contain options the zeroes can be obtained from standard bootstrapping. In currencies in which both inflation swaps and real bonds are traded, I privileged the use of inflation swaps rates to obtain zero coupon inflation compensation rates because of their standardization across currencies.

One may wonder whether inflation swap markets are sufficiently liquid, justifying the use of prices in that market as empirical measures of an object referenced in the no arbitrage, pricing relationship described earlier. The literature on the liquidity of inflation swap markets tends to focus on those for the United States and the United Kingdom, in which they have found to be reasonably liquid and the prices reliable. For example, Fleming and Sporn (2019) estimate realized bid-ask spreads for customers are on the order of three basis points. As in Fleming, Longstaff, and Lustig (2014), however, to guard against using stale prices in the sample, I only include an inflation swap rate when the rate has changed from the previous day; that I use data aggregated to the monthly frequency in the empirical work should also mitigate concerns of stale prices.

In constructing zero-coupon real rates from nominal, and inflation compensation rates, I implicitly assume that the inflation swap of a given maturity compensates the floating leg investor with the actual, realized inflation of the swap's lifetime. This is not strictly true because of publication lags in the national price index, and the contractual lag built into inflation swap contracts.

One may wonder how publication lags in the national price index affect the empirical strategy. There are publication lags in the national price index because national price level data take time to collect and construct. The publication lag implies that without perfect forecasting abilities, investors cannot know the true real exchange rate today because the national price levels relevant to its computation have yet to be released. However, because publication lags are less than a month, and real exchange rate changes are computed over a month these concerns should be mitigated.

One may wonder how contractual lags in the swaps affect the empirical strategy. Contractual lags exist because the swap contracts specify cash flows to be exchanged based on lagged values of the inflation index. These lags varies across currencies, but with significantly less variation than for real bonds. For example, U.S. swaps (real bonds) are linked to their inflation index series with a three-month (three-month) lag, whereas U.K. swaps are linked to their inflation index series with a two-month (three to eight-month lag). Their existence implies that an inflation swap is compensation for two types of inflation: known, lagged inflation, and unknown, future inflation. Only the latter will carry an inflation premium. It also implies that at time t , the swap rate on an inflation swap contract for year n and with contractual lag l will be compensation for inflation from $t - l$ to $t + n - l$, instead of what it should be, which is from t to $t + n$. However, since contractual lags are small relative to the 10-year maturity being considered, and existing empirical work on the impact of these lags

on yields, contractual lags likely have minimal effects on inflation swap rates.⁹

National price level series Countries release multiple national price level series; in this study, I use the national price level series referenced by the relevant inflation swap or real bond contract.

Across currencies, the relevant national price level series differ in the real baskets they reference and the timeliness of their releases. For example, the Japanese index references a core basket of real goods that is meant to capture the average consumer's real consumption basket excluding perishables, a sizable but relatively volatile component of consumer consumption baskets, whereas the Swedish index references a headline basket of real goods that includes perishables. Meanwhile, the Brazilian general index of market prices (IGP-M) is not simply the consumer price index, but the arithmetic weighted average of the wholesale price index, consumer price index, and national construction cost index. All series are monthly with the exception of Australia's, which is quarterly and is converted into a monthly series by making a given month's observation the last quarterly observation observed.

⁹Perhaps because of how persistent inflation series are. For example, Christensen (2018), who exploits variation in the indexation lags of U.K. real bonds and estimates the yield difference between two real bonds identical except in that one has a three-month indexation lag, and the other an eight-month indexation lag, is around two to three basis points for ten-year bonds. While the study does not directly address the effect of contractual lags on the measured versus true log real bond holding period returns, if the pricing effect of the inflation lag is around the same order of magnitude, it should not have a large effect on the empirical results. I do not explicitly account for publication or contractual lags in the study.

Table 3.3: National price level index series

Currency	National price index series	Freq.	Publication lag (mo.)	Contract	Contract lag (mo.)
British pound	Retail Price Index	M	0.5	Swap	2
Japanese yen	Consumer Price Index ex perishables	M	1	Swap	3
euro	Harmonized Index of Consumer Prices, All items	M	0.5	Swap	3
Swedish krona	Consumer Price Index	M	0.5	Swap	3
U.S. dollar	Consumer Price Index, All urban consumers	M	0.5	Swap	3
Australian dollar	Consumer Price Index, All groups	Q	1	Swap	3
S. African rand	Consumer Price Index, Metropolitan	M	0.5	Swap	4
Brazilian real	FGV Brazil General Prices IGP-M	M	0.5	Bond	1

Notes: This table reports the national price level series referenced by the zero coupon inflation swap contracts, or in the case of the Brazilian real, the real bond contract. Additionally, it gives the release frequency, the publication lag of the national price level series, and the contractual lag specified in the relevant securities contract. Information was obtained from Bloomberg and national statistical offices.

Nominal versus real log holding period returns

To contextualize this paper in relation to [Lustig et al. \(2019\)](#), which runs the nominal version of the real regression I estimate in this paper, I derive one set of testable assumptions that imply that the nominal and real regressions should be equal.

Proposition 2. *The nominal and real regressions will be equal when Assumptions I-IV hold in nominal and real terms and when the difference in inflation equal the difference in the holding period returns of nominal, long-run inflation swaps:*

$$\pi_{t+1} - \pi_{t+1}^* = hpr_{t+1}^{(10),\pi} - hpr_{t+1}^{*(10),\pi}. \quad (3.37)$$

A special case in which this holds is when log inflation is equal to the log holding period return on the ten-year inflation swap

$$\pi_{t+1} = hpr_{t+1}^{(10),\pi} \quad (3.38)$$

$$\pi_{t+1}^* = hpr_{t+1}^{*,(10),\pi}. \quad (3.39)$$

Proof. If Assumptions I-IV hold in nominal terms, the derivation of the real regression [3.15](#) leads to the derivation of the following nominal regression when all real variables are swapped for nominal variables:

$$\Delta s_{t+1} = hpr_{t+1}^{*,nom,(10)} - hpr_{t+1}^{nom,(10)} + \epsilon_{t+1}, \quad (3.40)$$

where $hpr_{t+1}^{(10),nom}$ is the log holding period return on the long run nominal bond, $hpr_{t+1}^{(10),\pi}$ is the log holding period return on the long run inflation swap and the corresponding variables with stars are the foreign counterparts of the domestic variables just defined.

Recall that

$$\Delta q_{t+1} = hpr_{t+1}^{*,(10)} - hpr_{t+1}^{(10)} + \epsilon_{t+1} \quad (3.41)$$

$$\Delta s_{t+1} + (\pi_{t+1} - \pi_{t+1}^*) = hpr_{t+1}^{*,(10)} - hpr_{t+1}^{(10)} + \epsilon_{t+1} \quad (3.42)$$

where the second line flows from the definition of q_{t+1} . By definition, the log holding period return on the long-run real bond $hpr_{t+1}^{(10)}$ is related to the log holding period return on the long-run nominal bond $hpr_{t+1}^{(10),\pi}$ and the log holding period return on the long-run inflation swap $hpr_{t+1}^{(10),\pi}$:

$$hpr_{t+1}^{(10)} \equiv \log\left(\frac{\left[\frac{(1+i_t^{(9\frac{11}{12})})}{(1+y_t^{(9\frac{11}{12})})}\right]^{9\frac{11}{12}}}{\left[\frac{(1+i_t^{(10)})}{(1+y_t^{(10)})}\right]^{10}}\right) \quad (3.43)$$

$$= \log\left(\frac{\left[\frac{1}{(1+y_t^{(9\frac{11}{12})})}\right]^{9\frac{11}{12}}}{\left[\frac{1}{(1+y_t^{(10)})}\right]^{10}}\right) - \log\left(\frac{\left[\frac{1}{(1+i_t^{(9\frac{11}{12})})}\right]^{9\frac{11}{12}}}{\left[\frac{1}{(1+i_t^{(10)})}\right]^{10}}\right) \quad (3.44)$$

$$= hpr_{t+1}^{(10),nom} - hpr_{t+1}^{(10),\pi}. \quad (3.45)$$

This relationship also holds for the foreign currency. Thus, Eq [3.42](#) can be rewritten as

$$\Delta s_{t+1} + (\pi_{t+1} - \pi_{t+1}^*) = hpr_{t+1}^{*,(10)} - hpr_{t+1}^{(10)} + \epsilon_{t+1} \quad (3.46)$$

$$\Delta s_{t+1} + (\pi_{t+1} - \pi_{t+1}^*) = hpr_{t+1}^{*,(10),nom} - hpr_{t+1}^{*,(10),\pi} - hpr_{t+1}^{(10),nom} + hpr_{t+1}^{(10),\pi} + \epsilon_{t+1} \quad (3.47)$$

$$\Delta s_{t+1} + (\pi_{t+1} - \pi_{t+1}^*) = (hpr_{t+1}^{*,(10),nom} - hpr_{t+1}^{(10),nom}) + (hpr_{t+1}^{(10),\pi} - hpr_{t+1}^{*,(10),\pi}) + \epsilon_{t+1}. \quad (3.48)$$

When the difference in inflation equal the difference in the holding period returns of nominal, long-run inflation swaps:

$$\pi_{t+1} - \pi_{t+1}^* = hpr_{t+1}^{(10),\pi} - hpr_{t+1}^{*,(10),\pi} \quad (3.49)$$

we can further reduce the equation to

$$\Delta s_{t+1} = (hpr_{t+1}^{*,(10),nom} - hpr_{t+1}^{(10),nom}) + \epsilon_{t+1}. \quad (3.50)$$

□

In the data, inflation differentials and ten-year inflation swap holding period return differentials are weakly correlated. In Table [3.3.2](#), I report the regression results of regressing the former on the latter; regression coefficients are near zero, and R^2 are near zero. Thus, the empirics reject a simple testable assumption that would imply that the results of the nominal regression would be identical to the results of the real regression and thus, motivate this paper's empirical exercise.

Table 3.4: Inflation differentials and ten-year inflation swap holding period return differentials

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	$(\pi_{t+1} - \pi_{t+1}^*)$	$(\pi_{t+1} - \pi_{t+1}^*)$	$(\pi_{t+1} - \pi_{t+1}^*)$	$(\pi_{t+1} - \pi_{t+1}^*)$	$(\pi_{t+1} - \pi_{t+1}^*)$	$(\pi_{t+1} - \pi_{t+1}^*)$	$(\pi_{t+1} - \pi_{t+1}^*)$
$(hpr_{t+1}^{(10),\pi} - hpr_{t+1}^{*(10),\pi})$	0.00	-0.06**	-0.02	-0.02	-0.02	-0.01	-0.02
	(0.02)	(0.03)	(0.02)	(0.02)	(0.03)	(0.01)	(0.03)
Constant	0.00***	0.00***	0.00***	0.00***	0.00***	-0.00***	-0.00***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
	143	143	143	143	143	143	101
R^2	0.00	0.04	0.00	0.01	0.00	0.01	0.00
	Japanese	euro	Swedish	U.S.	Australian	South African	Brazilian
	yen		krona	dollar	dollar	rand	real
Maturity	10y	10y	10y	10y	10y	10y	10y
Freq.	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly

Notes: Panel A reports the results of a regression of inflation differentials on ten-year inflation swap holding period return differentials.

3.4 Empirical Strategy and Results

3.4.1 Real

Let t index time. Earlier, I showed that Assumption I-IV imply coefficient estimates for the following regression

$$\Delta q_{t+1} = \alpha + \beta_1 hpr_{t+1}^{*,(10)} + \beta_2 hpr_{t+1}^{(10)} + \epsilon_t \quad (3.51)$$

where Δq_{t+1} is the log change in the real exchange rate defined as units of foreign real goods baskets per unit of the domestic real goods basket, $hpr_{t+1}^{*,(10)}$ is the log holding period return on the foreign, ten-year real bond, and $hpr_{t+1}^{(10)}$ is the log holding period return on the domestic, ten-year real bond. If all assumptions, hold, the constant term should equal zero, and the coefficients on the foreign and domestic holding period returns should equal 1 and -1, respectively. This motivates the following hypothesis tests:

$$\text{Test I: } H_0 : \alpha = 0; \beta_1 = 1; \beta_2 = -1 \quad (3.52)$$

$$H_a : \alpha \neq 0; \beta_1 \neq 1; \beta_2 \neq -1. \quad (3.53)$$

I denote this test the “Coefficient Test” and report a rejection if the parameters are jointly insignificant at the 5% level. The Coefficient Test suggests that if the log holding period return on the foreign long-run, real bond increases by 1%, the foreign currency should depreciate contemporaneously by 1%; also, if the holding period return on the domestic, long-run, real bond increases by 1%, the domestic currency should appreciate contemporaneously by 1% appreciation. Thus, using long-run bonds, carry trade returns should be close to zero.

The second testable implication is that the R^2 on the regression is equal to one.

$$\text{Test II: } H_0 : R^2 = 1 \quad (3.54)$$

$$H_a : R^2 < 1. \quad (3.55)$$

I denote this test the “ R^2 Test” and report a rejection if the parameter under the null is not in the confidence set at the 5% level and an acceptance otherwise. Since the parameter under the null is on the boundary of the parameter space it has a non-standard distribution and traditional bootstrap will generate inconsistent estimators. To conduct inference on estimated parameters, I generate confidence intervals by drawing pseudo time series by subsampling (Politis and Romano (1994)). The Appendix contains the algorithm description. The value of this test and of generating confidence intervals based on subsampling in this

setting is questionable because of the small sample size; however, it is the only method that has a chance of being asymptotically consistent.¹⁰

The regression is valid at all frequencies. Since test power increases with sample size, I maximize the number of observations by running the regression at the monthly frequency, which corresponds to the most common release frequency of the national price level series used to construct real exchange rates.

I report the regression point estimates and test outcomes in Table 3.4.1. I describe the results and then discuss their theoretical implications.

Columns 1-5 report results from estimating the regression on the five developed market currencies vis-a-vis the British pound: Japanese yen, euro, Swedish krona, U.S. dollar, Australian dollar. For this set of currencies, the point estimates are of the correct sign: the coefficient on the foreign log holding period return is positive, and the coefficient on the domestic log holding period return is negative. This implies that when there is positive shock to foreign log holding period returns in local currency terms, the foreign currency will tend to contemporaneously depreciate; when there is positive shock to domestic log holding period returns, the foreign currency will tend to contemporaneously appreciate. However, while the coefficient on the domestic return is significant across developed market currencies at the 1% level, the coefficient on the foreign return is only significant at the 1% level for the euro, U.S. dollar, and the Australian dollar.

Columns 6-7 report results from estimating the regression on the two market currencies vis-a-vis the British pound: South African rand and the Brazilian real; Columns 8-9 repeat the regression using holding period returns on the foreign, real bonds that are implied by CIP. For this set of currencies, the point estimates are jointly not of the correct sign when using OIS-based risk-free real bond rates, but are of the correct sign when using CIP-implied risk-free real bond rates. When using OIS-based risk-free real bond rates, the point estimates

¹⁰The value in this context is unclear for two reasons.

The first reason is mechanical: the sample is short, and the block length, which was set to twelve to balance the need for a reasonably sized set of subsamples with the need to keep the subsample length large enough to capture data dependence across time; proofs of subsample estimator consistency, however, rely on double asymptotics: both the sample and block size tend to infinity. As with all proofs that rely on asymptotics, the application of the proof in a setting in which all estimators are derived from finite samples is an act of faith, which may be misplaced.

The second reason is a repeat of Bayesian criticism of frequentist inference that seems particularly relevant in this case. The frequentist 95% confidence interval contains the true parameter with probability 95% before one has seen the data. I have seen the data, and the probability that it is in the confidence interval is exactly zero. The theory is about point-by-point realizations. If the data are all precisely, and correctly measured, a finite sample R^2 less than one is automatic rejection of the hypothesis of point-by-point realizations. Allowing for the possibility that a finite sample R^2 of less than one does not imply automatic rejection requires some idea of why the former might hold, such as measurement error. Suppose the countable, data generating process is such that the null holds true except occasionally. Since the data generating process is countable, in any finite sample, as I take sample size to infinity, the R^2 will approach, but never equal one.

on the domestic log holding period returns are of the correct sign; however, foreign log holding period returns are of the wrong sign: they are negative. This implies that when there is positive shock to foreign log holding period returns, the foreign currency will tend to contemporaneously appreciate. It is possible the sign of the true coefficients are the same as the sign of the estimated coefficients; alternatively, it may be that OIS swaps in developing market economies have sizable and variable risk premia compared to developed market economies that are generating the result.

Adversely towards the null, the coefficients are not one and vary in magnitude across currencies; furthermore, the R^2 on the regressions are far from 1. In particular, the coefficient on foreign returns range from 0.16 to 0.76; the coefficient on domestic returns range from -0.49 to -1.00. The R^2 on the regressions range from 8% for the Swedish krona to 33% for the Brazilian real, with standard errors 18% and 14%, respectively. Thus, at the 5% level, I reject the null hypotheses that the coefficients on the foreign and domestic holding period returns equal to 1 and -1, respectively, as well as the null hypothesis that the R^2 is one. If the domestic OIS rate is truly risk-free, and covered interest parity held, then $hpr_{t+1}^{*,CIP,(10)}$, not $hpr_{t+1}^{*,(10)}$ is the correct empirical counterpart to the ten-year real bond holding period returns in the derivation. Notably, the sign of the coefficient on the foreign holding period bond return reverses signs, and becomes positive (i.e. in the direction of long real bond return parity) and significant, changing from -0.20 to 0.37 for the South African rand, and -0.76 to 0.28 for the Brazilian real. They are, however, still far from the null of 1. The coefficient on the domestic holding period bond return becomes more negative and closer to the null of -1, changing from -0.36 to -0.69 for the South African rand, and from -0.75 to -1.13 for the British real.

Taken together, the results of the real regression estimation uniformly reject the joint hypothesis.

Table 3.5: Benchmark regression using private OIS rates (real)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Δq_{t+1}	Δq_{t+1}	Δq_{t+1}	Δq_{t+1}	Δq_{t+1}	Δq_{t+1}	Δq_{t+1}	Δq_{t+1}	Δq_{t+1}
Panel A: Point estimates									
$hpr_{t+1}^{*,(10)}$	0.16	0.59***	0.21	0.40***	0.61***	-0.20	-0.76***		
	(0.19)	(0.12)	(0.13)	(0.11)	(0.14)	(0.18)	(0.14)		
$hpr_{t+1}^{*,CIP,(10)}$								0.37***	0.28***
								(0.05)	(0.04)
$hpr_{t+1}^{(10)}$	-1.00***	-0.61***	-0.49***	-0.95***	-0.88***	-0.36*	-0.75***	-0.69***	-1.13***
	(0.15)	(0.11)	(0.14)	(0.12)	(0.16)	(0.21)	(0.23)	(0.18)	(0.22)
Constant	0.00*	0.00	0.00	0.00	-0.00	0.00	0.01**	0.00	0.01*
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
N. Obs	143	143	143	143	143	143	101	143	101
R^2	0.24	0.19	0.08	0.29	0.19	0.03	0.33	0.27	0.37
R^2 std. dev.	0.22	0.19	0.18	0.20	0.19	0.16	0.14	0.22	0.24
Currency	Japanese	euro	Swedish	U.S.	Australian	South A.	Brazilian	South A.	Brazilian
	yen		krona	dollar	dollar	rand	real	rand	real
Maturity	10y	10y	10y	10y	10y	10y	10y	10y	10y
Freq.	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly
Panel B: Test results									
Coefficient Test	Reject	Reject	Reject	Reject	Reject	Reject	Reject	Reject	Reject
R^2 Test	Reject	Reject	Reject	Reject	Reject	Reject	Reject	Reject	Reject

Notes: Panel A, Columns 1-7 report the results of a regression of the monthly change in the log of real exchange rates denoted by Δq_{t+1} on the contemporaneous holding period return on the real, ten-year foreign bond in foreign currency numeraire $hpr_{t+1}^{*,(10)}$, and the real ten-year domestic bond in domestic currency numeraire $hpr_{t+1}^{(10)}$. Columns 8-9 report the results of a regression of the monthly change in real exchange rates denoted by Δq_{t+1} on the contemporaneous holding period return on the real, CIP implied ten-year foreign bond in foreign currency numeraire $hpr_{t+1}^{*,CIP,(10)}$, and the real ten-year domestic bond in domestic currency numeraire $hpr_{t+1}^{(10)}$ for the South African rand and the Brazilian real. Panel B reports the results of the hypothesis tests. The row Coefficient Test reports on whether I reject or accept the null hypothesis that the coefficient on foreign holding period returns is +1, and the coefficient on domestic holding period returns is -1 at the 5% level. The row R^2 Test reports on whether I reject or accept the null hypothesis that the R^2 is equal to 1 at the 5% level.

3.4.2 Nominal

If the assumptions hold in nominal terms, I can run the regression in nominal terms as well.

Let t index time and i the foreign currency, this directly motivates a regression of the form

$$\Delta s_{t+1} = \alpha + \beta_1 hpr_{t+1}^{*,nom,(10)} + \beta_2 hpr_{t+1}^{nom,(10)} + \epsilon_t \quad (3.56)$$

where Δs_{t+1} is the log change in the nominal exchange rate, $hpr_{t+1}^{*,nom,(10)}$ is the log holding period return on the foreign, ten-year nominal bond, and $hpr_{t+1}^{nom,(10)}$ is the log holding period return on the domestic, ten-year nominal bond. The testable implications are as before.

Table [3.4.2](#) reports regression estimates.

Columns 1-5 reports point estimates from the regression on the five developed market currencies vis-a-vis the British pound: Japanese yen, euro, Swedish krona, U.S. dollar, Australian dollar. For the set of developed market currencies, the point estimates, with the exception of the Japanese yen, are of the correct sign: the coefficient on the foreign log holding period return is positive and the coefficient on the domestic log holding period return is negative.

Comparing these nominal results with the real results, I find that with the exception of the euro and U.S. dollar, the R^2 are higher in the nominal regressions than in the real regressions for the set of currencies. This makes sense because of the weak time series relation between log inflation differentials and log inflation swap holding period return differentials. The two series look as if generated by a process centered around a mean, and subject to weakly correlated mean zero noise innovations. Thus, innovations to inflation differentials will tend to decrease the R^2 in the real regressions vis-a-vis the nominal regressions since it will add variation to the dependent variable not explained by inflation swap holding period return differentials, and this effect appears to dominate for most currencies. The null hypothesis that the coefficients on the foreign and domestic holding period returns equal to 1 and -1, respectively, is accepted for the Swedish krona and the Australian dollar at the 5% level, and rejected for all other currencies. The null hypothesis that the R^2 is one is rejected for all currencies.

Columns 6-7 report results from estimating the regression on the two market currencies vis-a-vis the British pound: South African rand and the Brazilian real;

For the set of developing market currencies, coefficients are of the wrong sign: they are negative for foreign bond returns, and positive for domestic bond returns, implying that a positive shock to foreign (domestic) bond returns are, in domestic currency terms, amplified by foreign currency appreciation (depreciation).

Columns 8-9 re-estimates the regression for developing market currencies using holding period returns on the foreign, real bonds that are implied by CIP. If the domestic OIS rate is truly risk-free, and covered interest parity held, then $hpr_{t+1}^{*,CIP,(10)}$, not $hpr_{t+1}^{*,(10)}$ is the correct empirical counterpart to the ten-year real bond holding period returns in the derivation. Notably, the sign of the coefficient on the foreign holding period bond return reverses signs, and becomes positive (i.e. in the direction of long-run, real bond return parity), changing from -0.69 to 0.58 for the South African rand, and -0.47 to 0.44 for the Brazilian real. They are, however, still far from the null of 1. The coefficient on the domestic bond return also reverses signs, and becomes negative (i.e. in the direction of long real bond return parity) and closer to the null of -1, changing from 0.08 to -0.99 for the South African rand, and from 0.44 to -1.08 for the British real.

Taken together, the results of the real regression estimation uniformly reject the joint hypothesis.

Table 3.6: Benchmark regression using private OIS rates (nominal)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Δs_{t+1}	Δs_{t+1}	Δs_{t+1}	Δs_{t+1}	Δs_{t+1}	Δs_{t+1}	Δs_{t+1}	Δs_{t+1}	Δs_{t+1}
Panel A: Point estimates									
$hpr_{t+1}^{*,nom,(10)}$	-0.96** (0.39)	0.66*** (0.16)	0.91*** (0.17)	0.20 (0.14)	0.87*** (0.14)	-0.69*** (0.10)	-0.47*** (0.06)		
$hpr_{t+1}^{*,nom,CIP,(10)}$								0.58*** (0.10)	0.44*** (0.07)
$hpr_{t+1}^{nom,(10)}$	-0.88*** (0.15)	-0.59*** (0.14)	-0.83*** (0.17)	-0.79*** (0.17)	-1.03*** (0.17)	0.08 (0.18)	-0.25 (0.20)	-0.99*** (0.20)	-1.08*** (0.23)
Constant	0.01** (0.00)	-0.00 (0.00)	-0.00 (0.00)	0.00 (0.00)	-0.00 (0.00)	0.01** (0.00)	0.01*** (0.00)	0.01** (0.00)	0.01*** (0.00)
N. Obs	143	143	143	143	143	143	101	143	101
R^2	0.33	0.11	0.18	0.21	0.23	0.27	0.38	0.22	0.30
std. dev. of R^2	0.20	0.16	0.17	0.24	0.17	0.22	0.16	0.22	0.20
Currency	Japanese yen	euro	Swedish krona	U.S. dollar	Australian dollar	South African rand	Brazilian real	South African rand	Brazilian real
Maturity	10y	10y	10y	10y	10y	10y	10y	10y	10y
Freq.	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly
Panel B: Test results									
Coefficient Test	Reject	Reject	Accept	Reject	Accept	Reject	Reject	Reject	Reject
R^2 Test	Reject	Reject	Reject	Reject	Reject	Reject	Reject	Reject	Reject

Notes: Panel A Columns 1-7 report the results of a regression of the monthly change in the log of nominal exchange rates denoted by Δs_{t+1} on the contemporaneous log holding period return on the nominal, ten-year foreign bond in foreign currency $hpr_{t+1}^{*,nom,(10)}$ and the nominal, ten-year domestic bond in domestic currency $hpr_{t+1}^{nom,(10)}$. Columns 8-9 report the results of the regression of the monthly log change in nominal exchange rates denoted by Δs_{t+1} on the contemporaneous log holding period return on the nominal, CIP implied ten-year foreign bond in foreign currency $hpr_{t+1}^{*,nom,CIP,(10)}$ and the nominal, ten-year domestic bond in domestic currency $hpr_{t+1}^{nom,(10)}$. Panel B reports the results of the hypothesis tests. The row Coefficient Test reports on whether I reject or accept the null hypothesis that the coefficient on foreign holding period returns is +1, and the coefficient on domestic holding period returns is -1 at the 5% level. The row R^2 Test reports on whether I reject or accept the null hypothesis that the R^2 is equal to 1 at the 5% level.

3.5 Conclusion

In this paper, I considered a specification for real stochastic discount factors in which permanent, martingale components are common across currencies, and tested this assumption jointly with the assumptions that the regularity conditions for the decomposition are satisfied, markets are complete, and ten-year bond returns approximate infinite bond returns. Under these four assumptions, I showed that log real exchange rate changes should be positively correlated with contemporaneous log holding period returns on ten-year foreign real bonds, and negatively correlated with contemporaneous log holding period returns on ten-year domestic bonds; moreover, these relationships should be one-to-one. When using OIS zero rates, the sign of the estimated relationship is correct for developed market currencies, they are not for developing market currencies. Adversely to the joint hypothesis, the magnitude of the relationship, does not align with the theory. A test of the one-to-one relationship and of an R^2 of one is rejected for all currencies by the data. Therefore, I reject the joint hypothesis that all four assumptions jointly hold. An open question is the following: the failure of which assumptions or set of assumptions is primarily responsible for the magnitude of the empirical violation? I leave this for future research.

At the same time, in the post-2009 sample, I document a strong relationship between changes in the log real exchange rate and real holding period bond returns in the direction of parity, contributing to the general question of what explains exchange rate changes (Ferraro, Rossi, and Rogoff (2011), Verdelhan (2018), Jiang, Krishnamurthy, Lustig (2018)).

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Appendix

3.5.1 Alvarez and Jermann decomposition regularity conditions

The regularity conditions in Alvarez and Jermann (2005) are described below.

First, there exists some δ such that

$$0 < \lim_{k \rightarrow \infty} \frac{P_t^{(k)}}{\delta^k} < \infty, \forall t \quad (3.57)$$

which requires the average yield $-1/kP_t^{(k)}$ converges fast enough to a finite, positive value. Second, for each $t+1$ there exist a random variable with finite expected value $E_t(x_{t+1})$ such that $\forall k$ almost surely

$$\frac{\Lambda_{t+1}}{\delta^{t+1}} \frac{P_{t+1}^{(k)}}{\delta^k} \leq x_{t+1} \quad (3.58)$$

which says the marginal valuation multiplied by $\frac{P_{t+1}^{(k)}}{\delta^k}$ is bounded by a variable that has a finite conditional expectation.

They also maintain the background assumption that stochastic discount factors are strictly stationary. That is, the random vectors

$$\left\{ \frac{\Lambda_{t+1+k}}{\Lambda_{t+k}} \right\}_{k=i}^{k=i+j}, \left\{ \frac{\Lambda_{t+1+k}}{\Lambda_{t+k}} \right\}_{k=l}^{k=l+j} \quad (3.59)$$

should have the same joint distribution $\forall i, j, l$. This implies that all bond returns are stationary.

3.5.2 Drawing pseudo time series

Subsampling

Unlike bootstrap, the resample size is smaller than the sample size and resampling is done without replacement. In the simple i.i.d. case, I generate q equals n choose b blocks of size b from the data, where n is data size I generate the bootstrapped statistic with each block of data. I then use the empirical distribution over the q pseudo estimate, where the empirical estimate is treated as if the true mean, to approximate the distribution of the empirical estimate. With time series, one must respect the ordering of the data to preserve time series dependence, so I draw blocks with contiguous observations.

The results about estimates from subsampling come from proofs that take $b, n \rightarrow \infty, \frac{b}{n} \rightarrow \infty$. In implementation, I am faced with a finite sample, and thus, constrained to a finite b . The asymptotic conditions provide little guidance on how to choose b in this case; as Politis, Romano, and Wolf note, the choice depends on the purpose. I wish in subsampling to preserve time series dependence, and based on autocorrelation charts, choose a block length of fourteen.

Han Hong's primer on subsampling is available here:

<https://web.stanford.edu/~doubleh/lecturenotes/lecture13.pdf>

Monte-Carlo and stationary block bootstrap

The Politis, Romano (1992) algorithm for the stationary block bootstrap is a way for sampling the data in a way that preserves any time dependence in the data, and stationary. It works as follows. Suppose one has a dataset of length N , and wishes to generate nB pseudo time series. First, one splices the initial data into a series of N blocks. In constructing these blocks, the data are treated as circular so that the next observation after the “last” observation is the “first observation.” Two sets of indices of length nB are then randomly generated. The first set of indices is drawn from a uniform distribution. The second set of indices is drawn from a geometric distribution. To generate a pseudo time series, you “choose” and then “stitch together.” The choosing is as follows: the first index determines the block number to choose, the second index determines the length of continuous observations to take from that block number.

Both the Monte Carlo algorithm and the Politis, Romano (1992) algorithm can be implemented by installing the R package `tseries`, and using the function `tsbootstrap`. The first corresponds to the block type bootstrap where the length of the block is equal to one. The second corresponds to the block type stationary where the length of the mean block is equal to what is desired.

Cross section

The motivation for testing the assumption of common permanent components in real currency stochastic discount factors was the finding by Lustig et al. (2019) that average excess returns on the carry trade using long-run bonds and with a three-month horizon, is zero, suggesting that the entropies of the permanent components are equal. Their result is in nominal terms and on currency portfolios. They argue that the inflation characteristics of portfolios suggest the result holds also in real terms. I find suggestive evidence that this is true when I look at the individual currencies in the sample.

3.5.3 Estimated average excess returns by currency

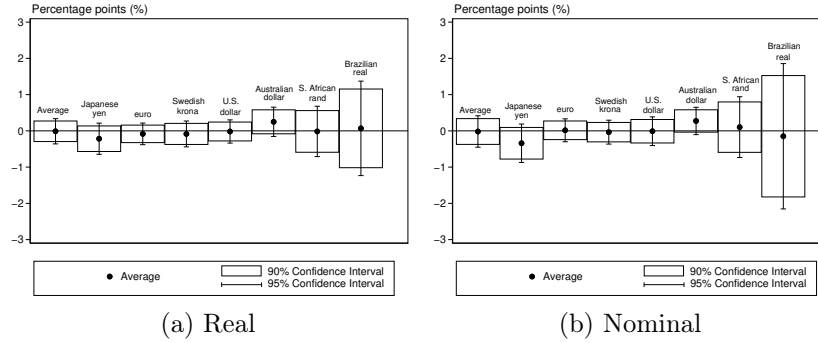
For each currency i , I estimate the average log excess return $\bar{x}r_{t+1}^{i,(10)}$ of going long the ten-year foreign bond and short the ten-year domestic bond expressed in the domestic numeraire (i.e. the British pound basket):

$$\bar{x}r_{t+1}^{i,(10)} \equiv \hat{\mathbb{E}}[hpr_{t+1}^{i,(10)} - \Delta q_{t+1} - hpr_{t+1}^{(10)}]. \quad (3.60)$$

I do this on both a real and nominal basis, and plot the estimated averaged by currency in Figure 3.5.3. The dots denote the point estimates, the bars denote the 90% confidence interval, and the whiskers denote the 95% confidence interval. Confidence intervals were constructed using standard errors that take into account heteroskedasticity and autocorrelation.

Real, average excess returns range from -22 basis points to 32 basis points, which translates in annualized terms to 2.64 and 3.84 percentage points. Symmetric, 90% and 95% confidence intervals around those point estimates, which are denoted by bars and whiskers respectively, include zero; that is, the estimates are not statistically different from zero. The nominal, average excess returns paint a similar picture: estimates of the average are not statistically different from zero. Confidence intervals on estimates are slightly wider than their real counterparts for the Swedish krona and the Australian dollar.

Figure 3.1: Average excess returns on long-maturity bonds



Note: This plot displays the average real and nominal log monthly excess returns on long maturity (i.e. ten-year) bonds. The domestic currency is the British pound, and the real numeraire good is the British pound basket. The dots denote the average, the bars denote the 90% confidence interval, and the whiskers denote the 95% confidence interval. Confidence intervals were constructed using standard errors that take into account heteroskedasticity and autocorrelation. Returns are monthly, holding period returns (i.e. not annualized), reported in percentage points (i.e. $\log(1.05) \approx 5\%$ is reported as 5, not as .05), and cover February 2009 to December 2020 with the exception of the Brazilian real, whose series begins July 2012.

3.5.4 Decomposition

Before I turn to the time series, I present a decomposition of the average real excess return in Table 3.5.4. When I decompose the average excess returns on long-maturity bonds, I see that the change in real exchange rates is offsetting the change in the difference in local currency holding period returns. That is, average excess returns are not zero simply because real exchange rate changes, and local currency holding period returns are zero in expectation.

Table 3.7: Decomposition of the average real excess return on long-maturity bonds

	Currency				
	Average	Japanese yen	euro	Swedish krona	U.S. dollar
(real returns, in percentage points)					
Foreign holding period bond return (A)	0.29	0.24	0.25	0.25	0.22
Domestic holding period bond return (B)	0.20	0.20	0.20	0.20	0.20
Foreign minus domestic holding period bond return (C=A-B)	0.09	0.04	0.05	0.05	0.02
FX return (D)	0.09	0.26	0.13	0.09	0.03
Excess returns in domestic currency (E= C – D)	0.00	-0.22	-0.08	-0.05	-0.02
HAC standard errors (F)	0.18	0.22	0.15	0.18	0.16
No. of monthly return observations	143	143	143	143	143
Sample start	2009-Feb	2009-Feb	2009-Feb	2009-Feb	2009-Feb
Sample end	2020-Dec	2020-Dec	2020-Dec	2020-Dec	2020-Dec

Notes: This table reports the decomposition of the average real excess return on long-maturity bonds. The domestic currency is the British pound, and the real numeraire good is the British pound basket.

Table 3.8: Decomposition of the average real excess return on long-maturity bonds, continued

	Currency		
(real returns, in percentage points)	Australian dollar	South African rand	Brazilian real
Foreign holding period bond return (A)	0.34	0.22	0.56
Domestic holding period bond return (B)	0.20	0.20	0.13
Foreign minus domestic holding period bond return (C=A-B)	0.14	0.02	0.43
FX return (D)	-0.11	0.03	0.36
Excess returns in domestic currency (E= C - D)	0.25	-0.01	0.07
HAC standard errors (F)	0.21	0.36	0.66
No. of monthly return observations	143	143	101
Sample start	2009-Feb	2009-Feb	2012-Jul
Sample end	2020-Dec	2020-Dec	2020-Dec

Notes: This table reports the decomposition of the average real excess return on long-maturity bonds. The domestic currency is the British pound, and the real numeraire good is the British pound basket.

3.5.5 Cross sectional result

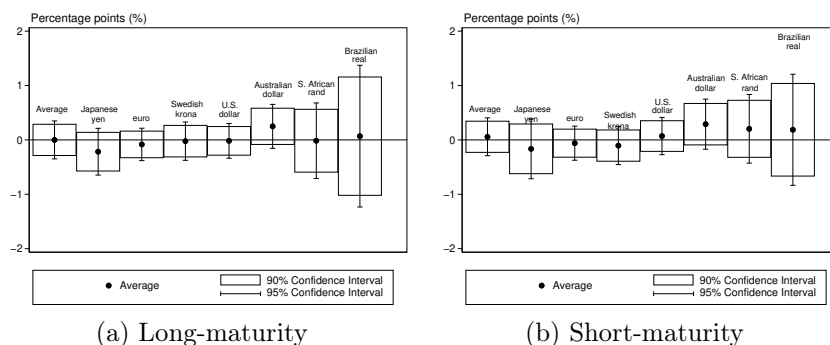
Term structure

To put the long-maturity result in context, I present the corresponding short-maturity result.

Inflation swap data exist only for maturities greater than one year, limiting us to the study of returns on bonds of maturities greater than a year. Therefore, I am unable to estimate the unconditional exchange rate premium as conventionally measured, which is the average monthly bond excess returns on short-run bonds where short-run is the horizon of the trade, in this case one-month. Instead, I estimate the average monthly bond excess returns on short-maturity bonds, defined as two-year bonds, which I plot in Figure 3.5.5. I find that excess returns on short-maturity range from -39 basis points, and 36 basis points, but as before, they are not statistically different from zero.

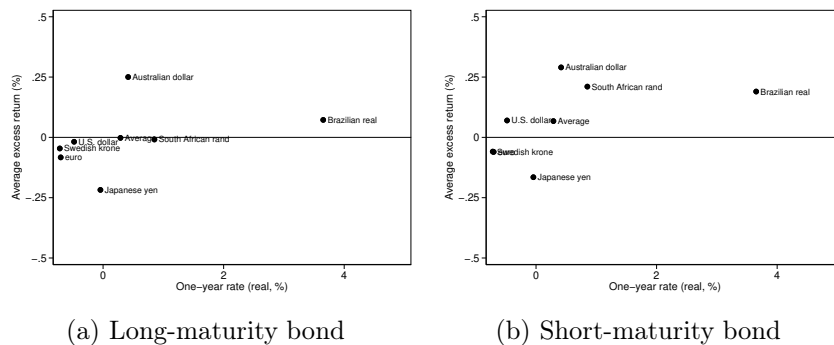
The inability to statistically distinguish from zero is due both to the small magnitude of the point estimates, and large standard errors that reflect the short length of the time series. The point estimates of average excess returns, however, do display a linear relationship vis-a-vis the level of interest rates and the yield curve slopes. Figure 3.5.5 plots the average excess return on the long-maturity and short-maturity bonds vis-a-vis the level of real interest rates. This relationship gets slightly stronger for short-maturity bonds, principally because of the observations for the South African rand and Brazilian real.

Figure 3.2: Average real excess returns on long and short-maturity bonds



Note: This plot displays the average real log monthly excess returns on long maturity (i.e. ten-year) bonds versus short maturity (i.e. two-year) bonds. The domestic currency is the British pound, and the real numeraire good is the British pound basket. Returns are monthly, log holding period returns, reported in percentage points, and cover February 2009 to December 2020 with the exception of the Brazilian real, whose series begins July 2012. Returns for the currency Average was constructed by taking the equally weighted return of currencies with return observations.

Figure 3.3: Average excess returns with the long and short-maturity bond versus the one-year rate



Note: This plot displays the average real log monthly excess returns on long maturity (i.e. ten-year) bonds versus the one-year rate, and the the yield curve slope of the foreign currency. The one-year rate is the one-year, real rate in local currency. The slope is the ten-year minus the one-year real rate in local currency. The domestic currency is the British pound, and the real numeraire good is the British pound basket. Returns are monthly, log holding period returns, reported in percentage points, and cover February 2009 to December 2020 with the exception of the Brazilian real, whose series begins July 2012. Returns for the currency Average was constructed by taking the equally weighted return of currencies with return observations.

3.5.6 Reference interbank rates on plain vanilla OIS contracts by currency

The reference interbank rate on the OIS is typically the average of interest rates at which banks offer to lend unsecured funds to other banks in the interbank market, and is often but not always, an indicative and not market rate; for the Australian dollar the corresponding the rate is the interbank overnight cash rate, and is calculated as the weighted average of the interest rate at which overnight unsecured funds are transacted in the domestic interbank market; for the Brazilian real is the Brazil Interbank Deposit Rate Annualized also known as the CDI, which is the average overnight dollar deposit rate published by the central OTC and private securities and derivatives despoitory CETIP; for the euro is the euro overnight interbank interest rate also known as EONIA; for the British pound is the sterling overnight index average also known as SONIA; for the Japanese yen is the overnight call rate, for the Swedish krona is the tonight next (T/N) Swedish interbank offered rate also known as T/N Stibor; for the U.S. dollar is the interbank interest rate for satisfaction of reserve requirements also known as the fed funds rates.

Table 3.9: Inflation and ten-year inflation swap holding period returns

	(1)	(2)	(3)	(4)	(5)
	π_{t+1}	π_{t+1}^*	π_{t+1}^*	π_{t+1}^*	π_{t+1}^*
$hpr_{t+1}^{(10),\pi}$	0.00 (0.02)				
$hpr_{t+1}^{*(10),\pi}$		-0.05*** (0.02)	-0.07 (0.05)	-0.03 (0.02)	-0.02 (0.02)
Constant	0.00*** (0.00)	0.00 (0.00)	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)
	143	143	143	143	143
R^2	0.00	0.06	0.02	0.01	0.01
0.01	British pound	Japanese yen	euro	Swedish krona	U.S. dollar
Maturity	10y	10y	10y	10y	10y
Freq.	Monthly	Monthly	Monthly	Monthly	Monthly

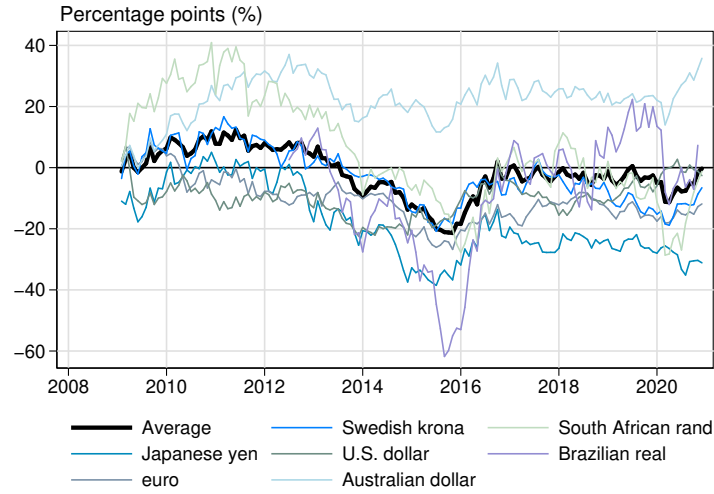
Notes: Panel A reports the results of a regression of inflation on ten-year inflation swap holding period returns.

Table 3.10: Inflation and ten-year inflation swap holding period returns, continued

	(6)	(7)	(8)
	π_{t+1}^*	π_{t+1}^*	π_{t+1}^*
$hpr_{t+1}^{(10),\pi}$			
$hpr_{t+1}^{*(10),\pi}$	0.03	-0.02	-0.03
	(0.03)	(0.01)	(0.03)
Constant	0.00***	0.00***	0.01***
	(0.00)	(0.00)	(0.00)
	143	143	101
R^2	0.01	0.01	0.01
0.01	Australian dollar	South African rand	Brazilian real
Maturity	10y	10y	10y
Freq.	Monthly	Monthly	Monthly

Notes: Panel A reports the results of a regression of inflation on ten-year inflation swap holding period returns.

Figure 3.4: Cumulative real excess returns on the foreign long-maturity bond



Notes: This plot displays the cumulative, log monthly real excess returns of going long the long-maturity (i.e. ten-year) foreign bond and short the domestic bond. The domestic currency is the British pound, and the real numeraire good is the British pound basket. Standard errors take into account heteroskedasticity and autocorrelation. Returns are monthly, log holding period returns, reported in percentage points, and cover February 2009 to December 2020 with the exception of the Brazilian real, whose series begins July 2012. Returns for the currency Average was constructed by taking the equally weighted return of currencies with return observations.